Realising the potential of the Outermost Regions for sustainable blue growth

ANNEX 11 TO THE FINAL REPORT

THE BLUE ECONOMY IN THE SOUTH WEST INDIAN OCEAN SEA BASIN

Written by COGEA srl,
in partnership with AND International, Fundación AZTI and Poseidon Aquatic Resources Management Ltd
September – 2017

EUROPEAN COMMISSION

Executive Agency for Small and Medium-sized Enterprises (EASME)

Department A – COSME, H2020 SME and EMFF

Unit A3 - EMFF

E-mail: EASME-EMFF@ec.europa.eu

European Commission
B-1049 Brussels

Realising the potential of the Outermost Regions for sustainable blue growth

ANNEX 11 TO THE FINAL REPORT

THE BLUE ECONOMY IN THE SOUTH WEST INDIAN OCEAN SEA BASIN

Executive Agency for Small and Medium-sized Enterprises (EASME)

European Maritime and Fisheries Fund (EMFF)

2017

Europe Direct is a service to help you find answers to your questions about the European Union.

Freephone number (*)**:**

00 800 6 7 8 9 10 11

(*) The information given is free, as are most calls (though some operators, phone boxes or hotels may charge you).

LEGAL NOTICE

This document has been prepared for the European Commission however it reflects the views only of the authors, and the Commission cannot be held responsible for any use which may be made of the information contained therein.

More information on the European Union is available on the Internet (http://www.europa.eu).

Luxembourg: Publications Office of the European Union, 2017

ISBN 978-92-9202-282-2

doi: 10.2826/941122

© European Union, 2017

Contents

Mapping economic activities 5

1. Identification and mapping of economic activities 5
2. Synergies and tensions between maritime activities 15
3. Best practices and flagship projects 20
4. Gap analysis 24

Mapping actors and maritime strategies 29

5. Mapping public authorities in the South West Indian Ocean sea basin 29
6. Summary of existing maritime strategies 32
7. Mapping main stakeholders in the South West Indian Ocean sea basin 45

Mapping Blue Growth and assessing the potential 47

8. Growth drivers and barriers 47
9. Future potential for mature and pre-developing activities 57
10. Investments opportunities 69
11. Most relevant maritime activities at sea basin level 69

Identifying public and private intervention 73

12. Potential financial, legislative and non-legislative measures for stimulating growth 73
13. Position of the EU outermost regions in the sea-basin 81
14. Conclusions on concrete measures by type and timeframe 82

List of interviewees 99

List of acronyms

AFD	Agence française de développement – French Development Agency
AfDB	African Development Bank
AIM	African Integrated Maritime [Strategy]
AMD	African Maritime Domain
AU	African Union
AUC	AU Commission
AU-IBAR	African Union — Interafrican Bureau for Animal Resources
AV	added value (acronym generally used in the fiche: VA)
BG	Blue growth
CAADP	Comprehensive Africa Agriculture Development Programme
CAGR	Compound annual growth rate
CAMFA	Conference of African Ministers of Fisheries and Aquaculture
CEMZA	Combined Exclusive Maritime Zone of Africa
cf.	Latin 'confer' meaning in the text 'see [reference]'
CFP	Common Fisheries Policy
CMF	Cluster Maritime Français - French maritime cluster
CMUB	Conseil maritime ultramarin du bassin Sud océan Indien - Ultramarine Maritime Council of the Southern Indian Ocean basin
COMESA	Common Market for Eastern and Southern Africa
COSME	Competitiveness of Enterprises and Small and Medium-sized Enterprises
DCF	(EU) Data Collection Framework
DMSOI	Direction de la mer sud océan Indien - Southern Indian Ocean Directorate
DROM	Département et région d'outre-mer (formerly DOM : département d'outre-mer) — French outermost regions for the EU
EAC	East African Community
EAFRD	European Agricultural Fund for Rural Development
EDF	European Development Fund (FED — fonds européen de développement in French)
EEZ	Exclusive Economic Zone
EFF	European Fisheries Fund
e.g.	'For example', in latin 'exempli gratia'
EMFF	European Maritime and Fisheries Fund (FEAMP in French)
ERDF	European Regional Development Fund
ESA-IO	Eastern Africa, Southern Africa and the *Indian Ocean*
ESF	European Social Fund
ESI	European Structural and Investment
EU	European Union
EU -AITF	EU-Africa Infrastructure Trust Fund (EU-AITF)
EVP	see TUE
FAD	fish aggregating devices
FAO	Food and Agriculture Organisation of the United Nations
FEDER	Fonds européen de développement régional - see ERDF
FIS	Fisheries information system
FSE	Fonds social européen - see ESF
GPMR	Grand Port Maritime de la Réunion -Great Maritime Harbour of Réunion
GVA	Gross Value added
H2020	Horizon 2020 — EU Research and Innovation programme over 7 years (2014 to 2020)
IEDOM	Institut d'Émission des Départements d'Outre-mer - Overseas Departments Note-Issuing Bank, delegation branch of the French Central Bank
IGAD	Intergovernmental Authority on Development
i.e.	'Id est' in Latin meaning 'that is'
IEJ	initiative pour l'emploi des jeunes
Ifremer	Institut français de recherche pour l'exploration de la mer - French marine research institute
IMP	Integrated Maritime Policy

INSEE	Institut national de la statistique et des études économiques - National Institute for Statistics and Economic Studies
IO	Indian Ocean
IOC	Indian Ocean Commission
IOTC	Indian Ocean Tuna Commission
IRT	Île de la Réunion Tourisme - Reunion Island Tourism
IUT	Institut universitaire de technologie - University Institute of Technology
MEA	Maritime economic activity
MoU	Memorandum of understanding
MPA	Marine protected area
MRC	Mauritius Research Council
MRE	Marine Renewable Energy/Energies
MW	Megawatt
NACE	Nomenclature statistique des activités économiques dans la Communauté européenne - Statistical classification of economic activities in the European Community
NEPAD	New Partnership for Africa's Development
OAU	Organization of African Unity
ONML	Observatoire national de la mer et du littoral - National Observatory of the Sea and the Coast
OP	Operational programme
OR	Outermost region
OT	Overseas territory
p.a	Per annum (by year)
PSM	Port State Measures
R&D	Research and development
RAPPICC	Regional Anti-Piracy Prosecution and Intelligence Co-ordination Centre
REFLECS	Regional Fusion and Law Enforcement Centre for Safety and Security at Sea
RDI	Research, development and innovation
RFMO	Regional fisheries management organisation
SADC	South African Development Community
SFPA	Sustainable Fisheries Partnership Agreement (between the EU and a non-EU country)
SIH	Système d'information halieutique - see FIS
SME	Small and medium entreprises
SNML	Stratégie nationale pour la mer et le littoral - National Strategy for the Sea and the Coast
SPREP	Secretariat of the Pacific Regional Environment Programme
SRDAM	Schémas régionaux de développement de l'aquaculture marine - Regional schemes for the development of marine aquaculture
SWAC	Sea-water air conditioning
TAAF	Terres australes et antarctiques françaises - French Southern and Antarctic Territories
TUE	Twenty-foot Equivalent Unit (EVP — équivalent Vingt Pieds in French)
UN	United Nations
UNECA	United nations Economic Commission for Africa
US(A)	United States of America
USD	US dollar
VA	value added (see AV)
VFR	Visiting friends and relatives
WTO	World Trade Organisation
YEI	Youth Employment Initiative

Introduction

The Indian Ocean is the world's third largest, covering around 70 560 000 km² (approximately 20 % of the water on the Earth's surface). It is bounded by Asia on the north, on the west by Africa, on the east by Australia, and on the south by the Southern Ocean. The Western Indian Ocean (WIO) includes a number of coastal states along the south-east, east and north-east of Africa, as well as the Indian sub-continent. It also includes a number of small-island states, including (for north to south), the Maldives, Seychelles, Comoros, Mayotte, Madagascar, Mauritius and Reunion. All four countries have a colourful history of colonisation, immigration, trade and more recently independence. Mauritius and the Seychelles have both developed tourism as a core economic activity, although fisheries and in Mauritius' case, sugar cane and more latterly IT and banking have become increasingly important. Madagascar, by far the largest island of the four in terms of size, is more of an agricultural economy, although it exports considerable volumes of unrefined ores and other natural material. At 1 660 km² (excluding the contested island of Mayotte) the Comoros is the third-smallest African nation by area and is also mainly agriculture-based.

The proposed geographical area of the Indian Ocean sea basin is globally from Somalia to South Africa from the North to the South and from the Eastern African coasts up to Mauritian waters from West to East. This sea basin would then be under broader influences of blue economies of other countries in the Indian Ocean such as Iran and India for instance (two countries influencing the policy of tuna management in the Western Indian Ocean as an example). Our proposed name is the **South West Indian Ocean sea basin.** Note that the Basin we aim to define is named 'the Southern Indian Ocean sea basin' ([1]) in France.

The EU set a specific regional cooperation strategy, including implementing an Integrated maritime policy (IMP), consisting of several specific measures and financial supports to develop the region and derogations to the outermost regions to compensate their handicaps for instance in favour of:

- the fisheries and agriculture sectors;
- the transport sector;
- the taxes and customs sector;
- the rural development.

Several EU funds contain specific financial support and measures to develop the outermost regions in the Indian Ocean such as the ERDF, the EAFRD, the EMFF, COSME financial instruments and H2020 Fund and the EU territorial cooperation ([2]). The EU aims to foster blue growth in the Indian Ocean sea basin mostly through the Indian Ocean Commission (IOC) by regional cooperation towards sustainable and integrated uses of the maritime sectors (IOC regional conference on blue economy — conceptual note, April 2015). The regional context is analysed by considering the development of the blue economy in the sea basin most particularly by:

1) Evaluating a selection of blue economies developed by neighbour countries of La Reunion and Mayotte, which are Comoros, Madagascar, Seychelles and Mauritius, to identify blue growth synergies and competition between the EU outermost regions and the other countries (the French Antarctic and Austral Territories being French overseas territories but not EU territories are mentioned when relevant to the scope of this study); and
2) Assessing the strategy of the Indian Ocean Commission and, to a lesser extent, other international organisations such as the United Nations Economic Commission for Africa.

Dedicated annexes 5 and 6 on Mayotte and Réunion detail the maritime activities in these two EU outermost regions: their situation, gaps, drivers, barriers to growth, potential and proposed actions.

([1]) Bassin Sud océan Indien: http://www.reunion.gouv.fr/installation-du-conseil-maritime-ultramarin-du-a1374.html, access: 29 August 2016.
([2]) Europe en France – example La Réunion, access: 21 April 2016. Other useful website : EU regional policy — outermost regions DG MARE related web-page on outermost regions

Mapping economic activities

1. Identification and mapping of economic activities

Table 1 next page provides an overview of the presence of each activity in each territory of the sea-basin. Maritime sectors are listed in this Chapter in the order of presentation of the existing sectors in the table next page; except for the maritime sectors being marginal but existent in the sea basin (for instance blue biotechnology, extraction of oil and gas, etc.). Maritime sectors for Mayotte and Réunion are analysed in detail in their specific annexes.

Table 1 — Overview of each activity in the different territories of the South West Indian Ocean sea-basin

Group	Sectors	Réunion	Mayotte	Madagascar	Seychelles	Mauritius	Comoros
Living resources	Fisheries and aquaculture	XX	XX	XX	XXX	XXX	XX
	Blue biotechnology	X (m)				X	
Non-living resources	Extraction of oil and gas			X			
	Extraction of salt	X (m)	X (m)	X		X (m)	X (m)
	Desalination		X (m)		XX	XX	
Shipping	Maritime transport	XXX	XX	XX	XX	XXX	X
	Ports (including dredging)	XXX	XX	XX	XXX	XXX	X
Shipbuilding	Shipbuilding			X		XX	
	Ship repair	X (m)		X		XX	
Tourism	Coastal tourism	XX	X	X	XX	XXX	X
	Cruise tourism	X	X	X	X	X	
Marine renewable energy (MRE)	Marine wind energy				X	X (m)	
	Other MREs	X (m)					

Source: Consultant own elaboration

Legend		Activity not identified on the territory
	X (m)	Activity existing but marginal, that is a) disappearing at commercial level or b) still at a research and development
	X	Low development of the activity on the territory
	XX	Medium development of the activity on the territory
	XXX	Strong development of the activity on the territory

1.1 Fisheries and aquaculture

1.1.1 Overview

The Western Indian Ocean is an important fishing area for migratory tuna, with the Somali basin to the north and the Mozambique Channel being key catching areas.

1.1.2 Geographical distribution

The **Seychelles** is well positioned as a base to catch the yellowfin tuna, skipjack and bigeye tuna in the Somali basin and the other parts of the Western Indian Ocean. As a result, it is the main port of landing for a large EU fleet of purse seiners (80 % of purse seine landings), with fishing being canned locally or sent to Mauritius by reefer (70 % of landings in the Seychelles) for further processing and canning. **Mauritius** is further from the fishing grounds, but has developed into an important regional 'seafood hub', with large companies such as Princes Tuna based there. It is also an important transhipment hub for the largely Asian longline fleet based out of Port Louis. **Madagascar** Is not such an important port hub, but does have a larger seasonal fishing ground in the Mozambique Channel, and some fish landed to the canneries at Diego Suarez at the north of the island. Madagascar also has an important wild shrimp fishery, and had considerable levels of shrimp aquaculture, often with economic connections to France. The **Comoros** also licenses access to overseas fishing interests, including the EU (**Réunion** and **Mayotte**), although does not have any associated shore-facilities (based on the recent evaluations of the EU sustainable fisheries partnership agreements in the region ([3]).

Also, since July 2016, France (**Réunion**) has been chosen to host the executive secretary of the Southern Indian Ocean fisheries agreement, which will become a regional fisheries management organisation (RFMO) to set fisheries management and conservation measures for non-tuna fisheries resources (tuna and associated species are managed separately by the Indian Ocean Tuna Commission). Its establishment in Réunion recognizes French fishing activity importance in the area and French role in regional cooperation in accordance with EU policy. Several fishing vessel owners catching toothfish and based in Réunion have shown interest in fishing other species in the Indian Ocean than toothfish and tuna (DMSOI) ([4]).

1.1.3 Main socio-economic trends

The global sale value of tuna from the Western Indian Ocean in 2014 was estimated at USD 5.43 million (EUR 3,97 million) (Poseidon — Macfadyen et al., 2015). Catches have been steadily increasing, but it is now recognised that of the main species, yellowfin tuna, is now unsustainable and harvest control limits are gradually being introduced by the Indian Ocean Tuna Commission (IOTC). Given the upper limits on catch levels, there has been a shift towards use of ultra-low temperature freezing techniques on the European purse seiners ([5]), as well as an intention to decrease the use of fish aggregating devices (FADs) in order to increase average fish sizes and value, as well as to decrease bycatch levels. This is likely to mean some shift way from just canning, focusing on high quality loins and the sashimi market.

1.1.4 Preliminary findings on growth potential

The tuna market remains strong, as so long as fish landings recognise the limits of stock sustainability, all four countries are well positioned to continue to benefit from tuna fishing. The **Seychelles** is expanding its port-side operations, introducing new ultra-low temperature cold storage, and intends to shift some of its vessel operations away from

([3]) For instance: Comoros – report January 2016: https://ec.europa.eu/fisheries/documentation/studies/comoros-2016_en , latest access: 24 May 2017. See other publications under the EU publication website
([4]) http://www.reunion.gouv.fr/IMG/pdf/press_2016-07-12_DMSOI_APSOI_bilan.pdf
([5]) One French vessels owner, Sapmer based in Réunion, has been building its fleet on this technique however the tuna price dropped which put its business plan to be revised by reducing the salary of its crew (Le Marin Hors Série, 10 July 2015 – page 28).

Mauritius to the Seychelles which is closer to the tuna grounds. **Mauritius** is also focused on maintaining landings into Port Louis to support the large processing operation inside the port and at Princes Tuna. It is also expanding the extent of marine cage fish farming, with a number of major investment from South Africa at advanced stages in planning.

1.2 Maritime Transport

1.2.1 Overview

Marine transport is key sector in all the Indian Ocean basin. All of which require connectivity, both with other island nations, other continents (Middle East and Asia), as well as within islands in their own territorial waters. These island states are also adjacent to the main southern shipping route for vessels travelling to Europe to the USA via Southern Africa.

1.2.2 Geographical distribution

The **Seychelles** has a larger inter-island ferry port in the north of Port Victoria, with regular ferry services to Praslin and La Digue islands. **Mauritius** has ferry or cargo ship passenger services with Reunion and Madagascar, as well as with Rodrigues Island to the east. **Madagascar** has shipping lines with South Africa and Mozambique to the west, and Reunion and Mauritius to the east. **Réunion** also has a regional hub operating for one important shipping company. Passengers traffic is also running between **Comoros** (Anjouan) and **Mayotte.**

Prior to the acquisition in 2015 of cranes, stacker and specialized trailers for moving containers in port areas, almost all boats were dedicated for transhipment at/ from Port Louis (**Mauritius**) or, more recently Djibouti, then **Mayotte** was served by feeders (small container). Now, large ships may wish to stop in Mayotte.

1.2.3 Main socio-economic trends

Before the Suez Canal was built, SE Africa was on the main shipping route from Asia to Europe. With most vessels now routed north, this Is not so much the case. Air transport has also taken over the international passenger travel demand. Both **Mauritius** and the **Seychelles** have vibrant economics, with considerable import needs, especially to support their important tourist sectors. Exports — mainly of fish for Seychelles and Mauritius, but also of bulk minerals from Madagascar now form the mainstay of maritime transport demands.

In 2015 ([6]), **Mayotte** has imported goods from ten countries in the Indian Ocean area for a total amount of EUR 32.7 million (24.0 %), 6,5 % of the total. **Reunion, The Mauritius and South Africa** are historically the three main suppliers with respective shares of 31.4 %; 23.9 % and 21.9 %. However, the South African leader in 2009, losing market share and now finds himself third behind Reunion, Mayotte first country importer to the region in 2015, and Mauritius. Madagascar came fourth and 12.4 % in 2015. These four countries suppliers concentrate 89.6 % of imports in the region.

In 2015 ([7]) in Réunion from Outside the European Union, Singapore still reinforces its first country supplier of Reunion, despite a sharp decline of 22.9 % of its inputs, fully composed of oil and petroleum products. It must be noticed that this decline followed already two consecutive declines (-4.2 % in 2014 and -15.1 % in 2013). China is always positioned in 2nd place (4.5 %). Mauritius is the second regional provider Island (-12.3 % to € 32.1 million) after South Africa (main coal supplier, for thermal plants Gol and Bois Rouge) and to Madagascar (+ 19.3 % to € 19 million).

1.2.4 Preliminary findings on growth potential

As suggested above, international maritime transport is mainly of cargo traffic. Recent studies e.g. the 2015 master plan for **Port Louis** suggest that demand will continue to

([6]) http://www.iedom.fr/IMG/pdf/ra_2015_iedom_may.pdf , access: October 2016.
([7]) http://www.iedom.fr/IMG/pdf/ra_2015_iedom_reu_.pdf , access: October 2016.

grow, albeit rather slowly, to support overall economic development. Passenger traffic is manly restricted to inter-island connectivity, which is especially important in the **Seychelles**. Since 2015, the influence of the development of the CMA-CGM Hub in **Réunion** (Port Réunion) and the development of transhipment and port capacities in **Mayotte** (Longoni port) must be taken in account in the area for global import and export goods flow.

1.3 Ports

1.3.1 Overview

As island nations, ports are a key node for both trade and personal connectivity. Thus, the ports tend to have evolved into the principal cities and commercial hubs, with the associated industrial and commercial development.

1.3.2 Geographical distribution

In the **Seychelles,** Victoria is the country's main urban centre with important fish and commercial quays and a separate inter-island harbour. The fish quay is also home to Indian Ocean Tuna, a regionally important cannery owned by Thai Union. In **Mauritius** Port Louis is developed to be a regionally important port that is also the capital of the country. It is home to a large Asian longliner fleet, as well as a container terminal and bunkering services. It is also the centre of the countries' 'seafood hub', with a number of fish landing and processing facilities. The region's largest fish cannery is at Terre Riche, 5 km from the port. Considerable effort is made to ensure that Port Louis continue to expand and take advantage of new opportunities, with a major long-term planning exercises undertaken very five years. **Madagascar** has five container terminal hubs and six other major seaports at strategic ports around the country. The most important seaport in Madagascar is located on the east coast at Toamasina. Ports at Mahajanga and Antsiranana are significantly less used due to their remoteness. The island's newest port at Ehoala, constructed in 2008 and privately managed by Rio Tinto, will come under state control upon completion of the company's mining project near Tôlanaro around 2038. **Comores** has two container ports, one in Maroni and the other in Mutsamudu on Nzwani. Fomboni, the Comores' third largest city also has a small port on the island of Mohéli.

Early in 2016, **Réunion** port handling capacities were put in operational services in order to foster its role as a regional goods exchanges platform to support CMA CGM **new regional hub, a concrete tool for regional services and exports**. EUR 80 million were invested in these new equipment (640 m long dock — 3 'super over panama' cranes) in order to handle 9 000 Twenty-foot Equivalent Unit (TUE) container carrier.

Prior to the acquisition in 2015 of mobile cranes, gantry cranes, stacker and five specialized trailers for moving containers in port areas, almost all boats were dedicated for transhipment at/ from Port Louis (Mauritius) or, more recently Djibouti, then Mayotte was served by feeders (small container).

1.3.3 Preliminary findings on growth potential

Between 2007-2015, the **Mauritius** economy grew at an average rate of about 4 % p.a. (per annum), consistently outperforming the world average (between 2 and 3 % p.a.; World Bank data ([8])) and it is likely that Port Louis will continue to expand to support this growth. The Freeport area is still developing, and fish processing is diversifying to newer, high value products, and is likely to support the growing, export-orientated aquaculture sector. The port area is also expanding to take advantage of less developed land both to the north and south. In the **Seychelles** Victoria is also expanding, partially to counter Mauritius' recent successes, esp. with its seafood hub concept, and will be adding new facilities to retain purse seiner landings and maintenance operations. However, there is limited land to develop, unless further land reclamation takes place. Praslin, the

([8]) Comparison of GDP annual growth rate between Mauritius and the World: http://data.worldbank.org/indicator/ NY.GDP.MKTP.KD.ZG?end=2015&locations=1W-MU&start=2007&view=chart , access: 22 May 2007.

Seychelles' second largest island, has also reclaimed a considerable amount of land to develop a port and associated industrial area near Baie St Ann in the SE of the island. In **Madagascar**, port development has been mostly in the private sector to support mineral export ventures.

Since 2015, the development of the CMA CGM Hub in **Réunion** (Port Réunion) and the development of transhipment and new port capacities in **Mayotte** (Longoni) are to influence the global import and export of goods in the sea basin.

Adapting port infrastructure to cruise tourism in the different Islands is to be investigated (see cruise tourism).

1.4 Ship building and repair

1.4.1 Overview

None of the four countries can compete with South Africa on a regional level, nor Asian ship builder globally for ship-building. However, there is some capacity to service the regional fishing industry, both for the EU purse seiners and the Asian longliners fleet.

1.4.2 Geographical distribution

Only **Mauritius** has a major shipbuilding and repair capacity, mainly at CNOI and Taylor Smith in Port Louis. The shipyard CNOI mainly focused on EU and Reunion purse seiners and longliners, and has a small building capacity. The company has recently built a double-ended vessel to transport passengers and cars between Petite-Terre and Grande-Terre for STM in Mayotte co-funded by the EU and the Conseil général in Mayotte ([9]) (see the annex 'Mayotte'), a fisheries patrol vessel for Madagascar, skiffs for the EU purse sine fleet, a 21 m longliner for a client and have enlarged toothfish longliners fishing in Antarctic waters (fish are often landed in Port Louis).

No dedicated major shipbuilding capacity is available in Réunion or Mayotte. All type of repair could be provided as all requested jobs are available in **Réunion** thanks to the French Navy basis in the EU outermost region. Nevertheless, it is challenged by a regional competition coming from South Africa, Mauritius or Madagascar, for which labour costs are lower. Note that local ship repair companies in the EU outermost regions of the Indian Ocean have signed a collaboration agreement with a Malagasy repair company in 2014.

1.4.3 Preliminary findings on growth potential

Given the relatively high cost of labour, ship building and repair is likely to focus on regionally important fleets such as the tuna purse seiners and longliners. Madagascar has the raw materials and relevantly cheap labour costs to support ship building and repair, but currently lacks the capacity to take advantage of this.

1.5 Coastal tourism

1.5.1 Overview

Tourism is a key economic activity in both the **Seychelles** and **Mauritius** and to a lesser extent in **Madagasca**r and in **Réunion**. It is essentially a coastal activity (except for **Réunion**), drawing on the attractive local landscapes, recreational opportunities from lagoons and reefs, and are positioned high in the market as luxury destination for Europeans and Asian nations.

1.5.2 Geographical distribution

In **Mauritius** tourism development had mostly focused on the western coast, from Le Morne in the south and around Grand Baie in the North. Tourism is less developed in

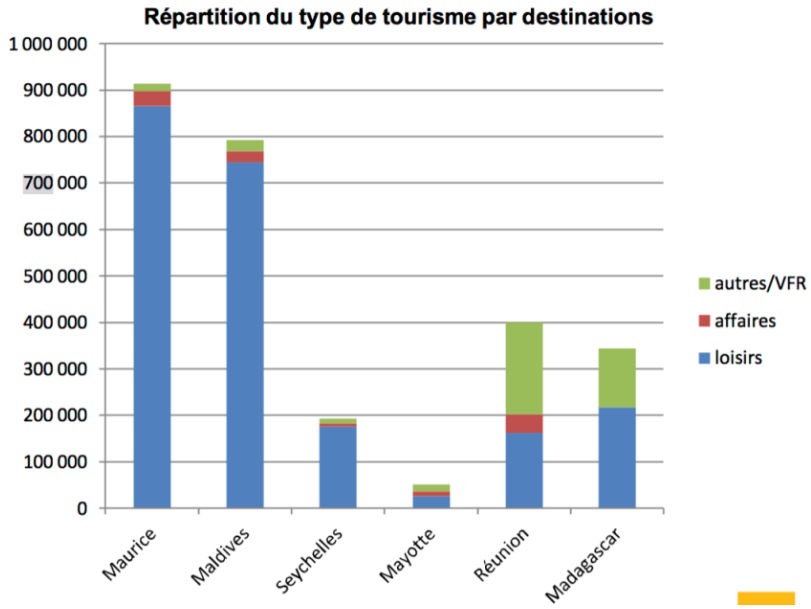
([9]).A second vessel is expected in May 2017 (http://www.lexpress.mu/article/290807/chantier-naval-cnoi-construit-plus-gros-amphidrome-francais , October 2015).

Madagascar, which has potential for a more diverse tourism sector that includes its wildlife and forests as well as the beaches. In **Réunion,** coastal tourism is mainly on its Western side. Although the **Comoros** has many natural resources for tourism, such as its beaches and marine environment, it does not have as strong a tourist industry as its regional competitors Réunion, Mauritius, and Seychelles. The main tourist attractions in the **Comoros** and **Mayotte** are their beaches, underwater fishing and, for **Mayotte**, its marine biodiversity with its lagoon.

1.5.3 Main socio-economic trends

Two typologies of tourism can be observed in the area, friends and family tourism (**Réunion, Mayotte and Comoros**), and leisure tourism (**Mauritius, Seychelles**, Maldives) ([10]). Over 2015 the **Seychelles** had 276 233 visitor arrivals, of which 62 % were from Europe, mainly France (14 %) and Germany (13 %). Around 8 900 people were employed in tourism in 2014, the largest sector at 18 % of the non-governmental work force. In **Mauritius** visitor arrivals have increased from 76 209 from 2003 to an all-time high of 132 735 in December 2015, with nearly 1 million visitors per year, mainly from France (27 %), Reunion (14 %) and South Africa (9 %). Gross tourism receipts have increased from Rs 35 693 million (EUR 903 million based on InforEuro exchange rate at the date of writing ([11])) in 2009 to Rs 44 378 million (EUR 1 124 million) in 2012. Over 28 000 Mauritians are directly employed in tourism. In **Madagascar** travel provides around 577 000 jobs annually and contributes around 14.9 % of the country's GDP with EUR 0.4 billion. Visitor numbers have been variable, often reflecting the political uncertainty that has dogged the country, with a low of 62 000 in 2002 and recent figures of 222 000 in 2014. **Comoros** welcomes around 20 000 visitors a year representing around EUR 34 million a year of receipts. 426 200 non-resident tourists visited **Réunion** in 2015. Visiting friends and relatives (VFR — tourisme affinitaire in French) is the first form of external tourism in Réunion. According to INSEE ([12]), 52 200 tourists were registered in 2013 in **Mayotte**.

Figure 1 — Tourist type in the Indian Ocean



Source: (IRT 2012) Legend: blue, leisure tourism; red, business tourism; green, other and VFR tourism

([10]) https://observatoire.reunion.fr/etudes/marketing-clientele/la-diversification-des-clienteles-touristiques-des-iles-de-locean-indien , access : 7 Sept. 2017.
([11]) http://ec.europa.eu/budget/contracts_grants/info_contracts/inforeuro/index_en.cfm ,latest access: 17 March 2017.
([12]) http://www.insee.fr/fr/themes/document.asp?reg_id=27&ref_id=21362 , access: September 2016.

1.5.4 Preliminary findings on growth potential

Tourism in the **Seychelles** and **Mauritius** continues to grow slowly, with growth limited by the lack of available coastal sites. **Madagascar** — which is relatively a very much larger country — has considerable potential for developing coastal tourism, but is hampered by poor infrastructure, economic and political uncertainty. **Comoros** also suffers from poor infrastructure and inter-island connectivity, as well as occasional political instability. For **Réunion** and **Mayotte**, global trend is to be constant with, to some extent, a switch in visitor topologies and origins and as long as the 'touristic model' is based on VFR. Diversification in tourist target[10], currently based on European and French speaking visitors could change the game. Mayotte is prone to develop niche tourism based on the valorisation of its natural and cultural assets (its MPA for instance) due to its constraints (environment protection, space use, travel cost, lack of high end infrastructures, etc.).

1.6 Cruise tourism

1.6.1 Overview

A global offer in cruise is currently available with a basin tour or islands connection (Mauritius, Réunion, Mayotte, Madagascar).

1.6.2 Geographical distribution

The coastal fringes of Seychelles and Mauritius are both key destinations. In the **Seychelles,** this is mainly concentrated around the north of the main island Mahé, but also on the smaller islands of Praslin and La Digue, with specialist resorts on other islands.

1.6.3 Main socio-economic trends

Drivers of world cruise tourism in number of tourists is mostly originating from the Caribbean however there is a trend of cruise tourism increase in the Indian Ocean sustained by tourists on world cruises and Indian Ocean cruises.

1.6.4 Preliminary findings on growth potential

Cruise tourism could be increased by Indian Ocean tour proposal to discover the sea basin by stops in each island (**Seychelles, Mayotte, Mauritius and Réunion**).

1.7 Other maritime economic activities

Living resources: blue biotechnology. There is some potential for developing new products from marine raw materials, such as high value fish oils from fisheries by-products as currently being started in Port Louis in Mauritius.

Non-living resources: Extraction of oil and gas, aggregates, salt, seabed mining and desalination. Although **Madagascar** has some inland hydrocarbon extraction, offshore oil and gas are not currently a maritime economic activity (MEA) of significance in the area even if it is an issue in diplomatic relationships in Mozambican channel between France and Madagascar (Îles Éparses and EEZ associated). Also, must be taken into account the French geographical position in the Indian ocean and its diplomatic relationships with Madagascar (co-management expected on oil and gas resources). Another co-management agreement with Mauritius for Ile de Tromelin was signed with France in 2010. With current low oil prices, it is doubtful that offshore exploration in the deep oceanic waters of this region will be of importance for some time. O&G prospection licences have been allocated by French government ([13]) in Mozambican canal (Juan de Nova). Logistic support could use Mayotte for logistic support.

([13]) http://lejournaldemayotte.com/une/du-petrole-aux-eparses-mayotte-aux-premieres-loges-de-la-bataille-de-lor-noir-qui-sannonce/ , access: July 2017

Salt extraction for coastal ponds has traditionally been of importance, but its high land footprint in areas often preferred for coastal tourist development means that salt ponds are rapidly disappearing, especially in Mauritius.

Main salt extraction in the sea basin and beyond:

- Madagascar at commercial level in Tulear ([14]) and Diego Suarez ([15]);
- Mauritius with a renown commercial and touristic reputation however it is to be sold for house operations. The owner of the site is Mont Calme Ltd (La Gazette) ([16]); and
- Réunion (St Leu) but not a commercial level anymore: marginal and mostly maintained with public support for eco-tourism (see extended fiche Réunion for further details);
- Mayotte but artisanal: maintained with public support for eco-tourism (see extended fiche Mayotte for further details);
- Anjouan (Comoros), a small production but not responding to the demand of salt for the Island.

Note a large commercial but artisanal production in India too (Chenai — Marakkanam) ([17]).

Seabed mining is still in its infancy with considerable uncertainties about seabed tenure and environmental impact, and is unlikely to become established in the near future in this region.

Renewable energy: wind and other renewable energies. With all countries, largely dependent upon imported fuel oil for power generation, there has been considerable interest in both wind power development and solar photo-voltaic energy including marine wind farms. However past projects have suffered from cyclone damage and on-going maintenance issues.

In the **Seychelles,** there is already an 8-kWh installation of eight turbines in Victoria that supplies 2.2 % of this island nation's power. There is a target to produce 15 % of power from renewable sources by 2030. There is some interest in renewable energy in **Mauritius** which is part of the Maurice Ile Durable (Mauritius Durable Island) long-term energy strategy drawn up by the Ministry of Environment and Sustainable Development. The government says that by 2025 wind energy will account for eight per cent of total electricity generation Within this approach, in March 2017, the country through its research and development council, the MRC, launched an expression of interest to recruit a consultant company to carry out the preliminary studies and supervising an off-shore wind farm development. The preliminary work of the consultant will consist of understanding cyclonic conditions, tsunamis and the potential impacts of the site on reefs (Lexpress.mu, 13 March 2017 ([18])). Mauritius is also looking at relying on locally designed wind turbine technology. **Madagascar** has set targets of 75 % of electricity generation and 54 % of final energy from renewables by 2020, although this will largely be from hydro power. In general, the northern part of Madagascar (around Antsiranana) and the southern part (around Taolagnaro) have wind speeds more than 7m/s (50 m high) favourable to electricity production.

Reunion has developed research and development activities in MRE since 2009. Sea-water air conditioning (SWAC) is to be used to produce heat and cold production for

([14]) http://voyageforum.com/discussion/madagascar-salines-tulear-france-3-19-septembre-d2013776/ , access: September 2016.
([15]) http://latribune.cyber-diego.com/economie/249-la-compagnie-saliniere-lor-blanc-du-nord-de-madagascar-.html , access: Sept. 2016.
([16]) http://www.lagazette.mu/patrimoine-les-salines-tamarin-abandon/ , access: 7 Sept. 2017.
([17]) Picture of salt pans: https://www.flickr.com/photos/dalbera/13957426396 , access: 7 September 2017.
([18]) https://www.lexpress.mu/article/302133/energies-renouvelables-mauritius-research-council-veut-lancer-une-ferme-eolienne (see also: Mauritius Research Council — MRC, http://www.mrc.org.mu/English/News/Pages/Expressions-of-Interest-for-the-Development-of-Offshore-Wind-Farms-for-the-Republic-of-Mauritius.aspx , access: 14 March 2017.

public infrastructures. **Réunion** could be a pilot to develop MRE R&D in extreme weather conditions in the area following lessons learned by MRE projects in the last ten years in the sea basin and beyond. However, Mauritius and Seychelles may be relevant candidates to meet the same objective (cf. above).

1.8 Non-economic (public) activity

1.8.1 Maritime surveillance

Maritime surveillance is an important public maritime activity in the South West Indian Ocean sea basin mostly to fight illegal fishing and piracy. It enables sustainable fishing and cruise and coastal tourism in the sea basin.

1.8.1.1 Overview

Maritime surveillance capability has long been developed in the region to fight IUU fishing in the Western Indian Ocean under the umbrella of Indian Ocean Commission with SmartFish programme (2nd phase of Smartfish running up to mid-2017). It is one of the largest regional programmes for fisheries in Africa covering 20 beneficiary countries in the Eastern, Southern Africa and the Indian Ocean (ESA-IO) area. It is funded by the European Union and Implemented by the Indian Ocean Commission (IOC) jointly with the Food and Agricultural Organization of the United Nations (FAO).

Nevertheless, maritime surveillance, with a key actor being Réunion in the regional surveillance, rose rapidly in prominence to combat the largely Somalia-based piracy crisis in the last decade. The SADC Maritime Security Strategy was adopted in 2013 largely in response to the Somali situation, which has since declined as a regional threat.

1.8.1.2 Geographical distribution

The Somali piracy crisis was mainly restricted to the north of the region, so mainly affected the Seychelles and a number of piracy incidents occurred with their EEZ. The other main threat to maritime economy, IUU fishing, is a much wider problem and has been addressed both at national and regional levels to cover territorial waters, EEZs and the high seas.

1.8.1.3 Preliminary findings on growth potential

Whilst piracy and IUU rates are being reduced through increased catch documentation and traceability, as well as port state measures (both Mauritius and Seychelles ratified the FAO Port State Measures Agreement to eradicate illegal fishing), the combating of IUU fishing and piracy through maritime surveillance will be maintained (or increased). Réunion is playing a key role in maritime surveillance in the sea basin by providing regional maritime surveillance and safety at sea assets (patrol vessels).

1.9 Possible impacts of climate change in the Indian Ocean sea basin ([19])

The EU Climate Change Adaptation Strategy ([20]) was adopted in 2013. It recognises that the EU Outermost Regions (OR) are likely to be impacted by climate change. At the sea basin level, the Indian Ocean Commission (IOC) developed a regional adaptation plan in 2013. The Plan provides guidelines to the actions of IOC cooperation on adaptation until 2020. A plan of actions to implement the strategy up to 2020 followed ([21]).

Small Islands developing States are likely to be more impacted by climate changes due to their high dependency to natural resources and external energy. Adaptation actions are likely to be required in maritime activities, most particularly in tourism (including eco-tourism), port and coastal zone management by a better knowledge of the climate change in the sea basin, managing urban planning and minimising the impacts of climate change on the biodiversity.

2. Synergies and tensions between maritime activities

The table below reports synergies and tensions of each activity with other maritime activities and which are the contribution to sustainable Blue growth.

Group	Sector	Synergies with other maritime activities	Conflicts with other maritime activities	Contribution to sustainable growth
Living resources	Fisheries and aquaculture	A key driver for port development and onshore processing Also, the fisheries sector provides opportunities for ship-building and repair Raw material for biotechnology products Small-scale fishing can be attractive to tourists (by protecting the biodiversity), and hotels welcome fresh fish supplies	Inshore aquaculture can be unattractive to some tourists Competition between neighbour countries in development of fishing and catch landing	Living renewable resource if used responsibly. Move to higher-value, lower waste processing provides growth opportunities
	Blue biotechnology	Can utilise raw material from fishing and aquaculture wastes	None	Potentially very high value-adding opportunities, esp. for fish waste
Non-living resources	Extraction of oil and gas	Opportunities for port development, engineering and maintenance services Also, opportunities for fishing vessels to support offshore installations	In case of pollution, the sector can severely impact both tourism, fisheries and aquaculture (e g major oil spills)	Potential source of foreign exchange earnings Note: an historic sector of importance in the EU blue economy on overall, however, suggested by the European Commission to be a sector outside of

[19] European Commission, 2014, The economic impact of climate change and adaptation in the Outermost Regions, http://ec.europa.eu/regional_policy/sources/activity/outermost/doc/impact_climate_change_en.pdf , access: July 2017
([20]) http://climate-adapt.eea.europa.eu/eu-adaptation-policy/strategy , access: July 2017
([21]) IOC Press release : http://commissionoceanindien.org/activites/changement-climatique/la-coi-developpe-son-plan-daction-climat-regional-2016-2020-29-mars-16-ebene-maurice/#c2067 , access: July 2017.

Group	Sector	Synergies with other maritime activities	Conflicts with other maritime activities	Contribution to sustainable growth
				the EU focus areas for sustainable blue growth ([22])
	Extraction of aggregates	Port and reception area development	Sea area conflict with fishing and aquaculture Short-term environmental impacts e g sediment suspension	Reduction in use of sand mining and other terrestrial extraction
	Extraction of salt	Can be a tourist attraction and provide local sales opportunities	Competes for land with coastal tourism development	Not significant
	Seabed mining	Needs port and inland infrastructure development.	Possible environmental damage	Potential source of foreign exchange earnings Note: an historic sector of importance in the EU blue economy on overall, however, suggested by the European Commission to be a sector outside of the EU focus areas for sustainable blue growth[22]
	Desalination	Supports urban development in areas of scarce freshwater	Hypersaline effluent can affect local coastal fisheries and environment	Supports development in water-stressed areas
Shipping	Maritime transport	Needs port and inland infrastructure development	Space conflict with wind farms in pinch points.	Improves regional trade opportunities
	Ports (inc. dredging)	Key asset for maritime trade and navigation	Can impinge on tourist areas. Competition between neighbour countries in the MEA development in the region	Supports trade
Shipbuilding	Shipbuilding	Supports maritime transport and fishing industries	Noise and other pollution (e.g. on tourism)	Supports trade
	Ship repair	As above	As above	Supports trade
Tourism	Coastal tourism	Small-scale fishing (ecotourism) and coastal environmental protection	Conflict with traditional coastal communities & activities	Provides significant benefits for local employment and economy
	Cruise tourism	Port and infrastructure development needs	None or some with port space exploitation — to balance with other port activities	Economic benefits of port visits

([22]) See section 6 p. 52 of the Commission staff working document — Report on the Blue Growth Strategy towards more sustainable growth and jobs in the blue economy. SWD(2017) 128 final and sections 2 and 5 of the Communication from the Commission to the European parliament, the Council, the European economic and social committee and the Committee of the regions — Blue Growth opportunities for marine and maritime sustainable growth. COM(2012) 494 final. 12 pages. Access to links: July 2017.

Group	Sector	Synergies with other maritime activities	Conflicts with other maritime activities	Contribution to sustainable growth
			Competition between neighbour countries in the MEA development in the region (complementarity required)	
Renewable energy	Wind energy	Also, opportunities for fishing vessels to support offshore installations Ship repair for maintenance and logistic supply Multi-activity (aquaculture)	Potential negative visual impact with coastal tourism. restricted effects regarding the possibility to install wind turbine (high depth) and to take in account cyclonic threat in the sea basin	Low carbon energy production Diversification of historical maritime sector (ship building and repair)
	Other renewable	As above Blue biotechnologies: deep water pumped from SWAC Tourism with green energy production	Conflicts may occur in maritime activities located near or in the area where marine renewable energy installations are installed (mitigation measures and specific technology such as in the mineral and oil extraction interacting with trawl fishing would need to be developed to minimise conflicts between activities near and within the same area)	As above.

2.1.1 Synoptic view of legislative, non-legislative and financial synergies and tensions among maritime activities

The following table reports a synoptic view of main synergies and tensions grouping them under legislative, non-legislative and financial frames.

Synergies with other maritime activities			
ACTIVITY	LEGISLATIVE	NON-LEGISLATIVE	FINANCIAL
Fisheries and aquaculture	Tourism by environmental protection measures	Key driver for port development and onshore processing including ship building and ship repair services in the sea basin especially in Mauritius and Seychelles for tuna-fishing	
Blue biotechnology		Potential for synergies with the fisheries and aquaculture sector and marine renewable energies (SWAC) although blue biotechnology is still a pre-mature sector in the basin	
Extraction of oil and gas (may exist in the future in the		Synergies by developing economic activities in port used by oil and gas super tankers (to	

Synergies with other maritime activities			
ACTIVITY	LEGISLATIVE	NON-LEGISLATIVE	FINANCIAL
Mozambique Channel) ([23])		investigate in Mayotte for instance — see 'Mayotte' annex)	
Extraction of aggregates		Marginal sector or not existing	
Extraction of salt		Synergy with eco-tourism (green tourism) — see Réunion and Mayotte annexes for instance	
Seabed mining[23]		Marginal sector or not existing	
Desalination		Coastal tourism by participating to the reduction of the risk of freshwater draught (see 'Mayotte' annex)	
Maritime transport		Inter-dependence with the port activities	
Ports (inc. dredging)		Inter-dependence with maritime sectors using the ports: fisheries, cruise tourism, maritime transport, etc.	
Shipbuilding		Synergy with the development of port activities in Seychelles, Mauritius and Madagascar	
Ship repair		Synergy with the development of port activities in general	
Coastal tourism		Synergy with fisheries (ecotourism and cruise tourism)	
Cruise tourism		Synergy with coastal tourism	
Marine wind energy		Sector almost not existing: no current synergy identified in the sea basin	
Other marine renewable energies		Blue biotechnologies for instance by use of deep water in SWAC commercial applications (see 'Réunion' annex)	

Conflicts with other maritime activities			
ACTIVITY	LEGISLATIVE	NON-LEGISLATIVE	FINANCIAL
Fisheries and aquaculture	Costal tourism by weak maritime spatial planning	Inshore aquaculture can be unattractive to some tourists	
Blue biotechnology		No major conflict identified	
Extraction of oil and gas[23] (sector not existing but exploration for potential future exploitation in the Mozambique Channel)	Weak management measures can impact other maritime sectors	In case of pollution, the sector can severely impact both tourism, fisheries and aquaculture (e g major oil spills)	
Extraction of aggregates		Sector not existing or marginal	
Extraction of salt		Competes for land with coastal tourism development	
Seabed mining[23]		Sector not existing or marginal	
Desalination	Weak environmental management:		

([23]) Although an important sector of the EU blue economy on overall, the European Commission suggests it to be outside of the EU blue growth focus areas to deliver sustainable blue growth (see footnote **Error! Bookmark not defined.**).

Conflicts with other maritime activities			
ACTIVITY	LEGISLATIVE	NON-LEGISLATIVE	FINANCIAL
	hypersaline effluent can affect local coastal fisheries and the environment		
Maritime transport		No major conflict identified (maritime space conflict to manage in case of marine wind farm development)	
Ports (inc. dredging)		Can impinge on tourist areas	
Shipbuilding		Can impinge on tourist areas	
Ship repair		As above	
Coastal tourism		Can conflict with traditional coastal communities & activities	
Cruise tourism		Conflicts between different port users to manage (fisheries, ship building, ship repair, etc.)	
Marine wind energy		Sector almost absent in the basin (potential conflict to manage by maritime spatial planning)	
Other marine renewable energies		Conflicts may occur in maritime activities located near or in the area where marine renewable energy installations are installed (for instance with trawl fisheries) however the sector is not developed in the basin for the moment	

3. Best practices and flagship projects

1. **Name of the practice**: regional conference on blue growth organised by the Indian Ocean Commission (under the Présidency of France in 2016).
   Objectives: development of a regional cooperation for blue growth in the Indian Ocean sea basin.
   Results achieved: preparation of the conference and organisation of the conference on 12-13 December 2016 in St Denis (Réunion) by the Préfecture and the Conseil Régional of Réunion which enabled the drafting of 30 recommendations as background to a regional strategy expected to be submitted to the IOC by the end of the first semester 2017. Recommendations consisted among others:
   - The collaboration between the IOC and the UN Economic Commission for Africa (UNECA) to use the initiative of the IOC Blue Economy / ocean as an emblematic programme of sub-regional coordination for the UN agency;
   - Dissemination to Member States of the practical guide on the blue economy published by UNECA in April 2016 (see 6.1 for further details);
   - Support to the project 'whale road'' coordinated by the Regional Council of Réunion and as well as actions to enhance the ecotourism potential of the sea basin, rich with an exceptional biodiversity although threatened and requiring to be protected;
   - The exchange of data, expertise and lessons learned in the fields of renewable energy and ecosystem management;
   - The strengthening of port capacity for the improvement of trade and the development of new activities, including the growing cruise tourism;
   - The continuation and amplification of IOC actions for regional connectivity, including sea and air plans; and
   - The effective involvement of IOC and its member countries in the project against trafficking of natural assets (flora and fauna) with funding from the 11th European Development Fund.

   Reasons for considering it as best practice/flagship project: triggered regional cooperation on blue growth development in the Indian Ocean sea basin.

2. **Name of the programme**: Programme to Promote Regional Maritime Security (MASE).
   Objectives: Overall Objective: enhanced maritime security in the ESA-IO region to create a favourable environment for the economic development of the ESA-IO region and beyond.
   The Specific objective: strengthen the capacity of the ESA-IO region in the implementation of the Regional Strategy and Action Plan against Piracy and for Maritime Security.
   Results achieved:
   1) Somalia Inland Action Plan implemented;
   2) National/regional legal, legislative and infrastructural capability for Arrest, Transfer, Detention and Prosecution of Pirates is developed and/or strengthened;
   3) Regional capacity to disrupt the financial networks of pirate leaders and their financiers is strengthened;
   4) National and regional capacity for maritime tasks and support functions enhanced; and
   5) Regional coordination and information exchange improved.

   Reasons for considering it as flagship project: successful strengthening of maritime surveillance and security at sea in the region.

3. **Name of the project/ practice**: Port Victoria Wind Farm, Seychelles (marine renewable energy development), 2012 — 2014.
   Objectives: move towards 15 % renewable energy by 2030 in Seychelles.
   Results achieved: 2.2 % renewable energy contribution.
   Reasons for considering it as flagship project: wind farm on reclaimed land.

4. **Name of the project**: extension of the container terminal at Mer Rouge, Mauritius — works and investments expected to be finalised by mid-2017.
   Objectives: to extend the Mauritius Container Terminal — MCT to accommodate longer vessels with a larger draught given the overall growth in the size of container ships on long-distance, high-sea-traffic lines to generate economies of scale.
   Results achieved: extension running up to 2017.
   Reasons for considering it as best practice/flagship project: example of development of port facilities in the sea basin with EU and French cooperation funds (see details in the table below).

5. **Name of the project**: Mauritius Seafood Hub development, 2008 — ongoing.
   Objectives: development of 'an efficient and attractive environment for the supply of value added processes and services for the export of seafood products' (Ministry of Ocean Economy, Marine Resources, Fisheries and Shipping, Mauritius).
   Results achieved: improved cross-sectoral cooperation, reduced administrative burden.
   Reasons for considering it as flagship project: development of a seafood hub with public and private investments.

3.1 Best practices and flagship projects details

Table 2 — Best practices and flagship projects at regional or InterReg level

Name of the project/ practice	Leading actor(s)	Other subjects/ stakeholders involved	Timing	Maritime sectors involved	Budget	Programming document	Website
Blue economy regional conference, held on 12-13 December 2016, St Denis (Réunion)	Indian Ocean Commission Conference organised by Réunion	Member States of the IOC	2016	all	-	Adopted conclusions and recommendations of the conference	Press releases with regards to its organisation : http://commissionoceanindien.org/activites/leconomie-bleue/economie-bleue-oceanique-les-bases-du-plan-dactions-regional-de-la-coi-jetees-saint-denis-de-la-reunion-13-dec-16/ http://commissionoceanindien.org/fileadmin/resources/COPL012015/ and http://www.reunion.gouv.fr/communique-de-la-commission-de-l-ocean-i-indien-a2382.html (see associated documents on the second hyperlink) See also, a note from the IOC to organise a regional conference on blue growth (Jan. 2015) : http://commissionoceanindien.org/fileadmin/resources/COPL012015/Point%203.2%20Annexe%201%20Note%20conceptuelle%20sur%20l'economie%20bleue.pdf
Flagship programme: programme to Promote Regional Maritime Security (MASE)	EU, IOC, ESA-IO Regional Organisations	SADC, East-African Force (EASF) and EAC military cooperation	2014 — 2018	Maritime transport, maritime surveillance, ports	EUR 37 million	See footnote ([24])	http://commissionoceanindien.org/activites/securite-maritimemase/contexte/

([24]) http://ec.europa.eu/europeaid/documents/aap/2013/af_aap-spe_2013_intra-acp_p3.pdf , access: March 2017.

Table 3 — Best practices and flagship projects at national level

Name of the project/ practice	Leading actor(s)	Other subjects/ stakeholders involved	Timing	Maritime sectors involved	Budget	Programming document	Website
Port Victoria Wind Farm (Seychelles)	Public Utilities Corporation (PUC)	Port Victoria	2012 — 2014	Ports, renewable energy	Grant from UAE: USD 28 million	Not identified	http://www.masdar.ae/en/energy/detail/seychelles-wind-farm
Extension of the container terminal at Mer Rouge (Mauritius)	Mauritius Port Authority	MPA	2012 — 2017	Port and maritime transport	EUR 108 million (estimate in 2012) AFD (French development agency): EUR 40 million (loan — interest rate 3,3 %) EU: EUR 3 million	None identified	EU-Africa Infrastructure Trust Fund (EU-AITF) ([25]) Newspaper : Le Mauricien, 27 October 2012 ([26])
Mauritius Seafood Hub development	Ministry of Fisheries	MPA, Freeport	2008 — ongoing	Ports, fisheries, maritime cargo, ship repairs	Public and private investments Private investments by companies in the fisheries and aquaculture sector total: MUR 2,405 million (equivalent to EUR 61 million)	Available (see hyperlink on the right)	http://fisheries.govmu.org/English/Pages/Fisheries---SEA-FOOD-HUB.aspx

([25]) http://www.eu-africa-infrastructure-tf.net/activities/grants/mauritius-container-terminal-extension-project.htm (latest update June 2016), access: 23 November 2016.
([26]) http://www.lemauricien.com - article: PRÊT DE L'AFD: Rs 1,3 md à la MPA pour le terminal à conteneurs , access: 23 November 2016.

4. Gap analysis

In this section, each maritime activity has been analysed in order to identify 'what is missing for a sustainable Blue Growth' in terms of 'Human resources skills', 'Infrastructures', 'Research' and 'Data'.

- ***Coastal tourism***

HR skills	Tourism requires a mixture of skill levels, from unskilled to hospitality specialists. Given that this region caters particularly for high-end tourism, skills development is an issue. English speaking HR is a limitation in French native territories (Réunion, Mayotte, Madagascar). The Seychelles has a Seychelles Hotel and Tourism Training College (SHTTC), but it is recognised that other training resources are required to address the specialist skills required in running modern, sustainable hotels and food establishments catering of well-paying but discerning guests.
Infrastructure	Tourism development is highly diverse, varying from small guest houses to large beach resorts. The supporting infrastructure in the Seychelles and Mauritius is generally good, with good paid roads, stable power supplies and modern airports. The rural areas of Madagascar have greater needs for good infrastructure development, which is constraining tourism development in some areas. Infrastructure in the Comores and Mayotte is also a constraint, with roads often in poor condition, power interruptions and data links (e.g. for online payment systems) often non-functional. There is still a weak integrated approach of the tourism development in Réunion (in term of 'supply chain' of services and facilities from the port to the airport to accommodation to inland and coastal tourism activities)
Research	There is relatively little need for R&D to support tourism development per se. this said, the development of niche, high end opportunities such as ecotourism may require some degree of site-specific R&D e.g. to provide water / electricity supplies and responsible sewage treatment in remote, off-grid areas.
Data	The Seychelles, Mauritius, Réunion and Madagascar all have well-resourced tourist boards to support hoteliers and other tourist facilities in terms of forecasting occupancy levels and guiding development. Mayotte also has a tourism observatory that may require continuous or additional support to collect data in collaboration with INSEE, the French National Statistical Institute.

- ***Cruise tourism***

HR skills	See coastal tourism above
Infrastructure	Port in the sea basin would require to evolve to respond to cruise demand in terms of quays mostly (for instance in Réunion) See coastal tourism too
Research	See coastal tourism
Data	Ibid

- ***Fisheries and aquaculture***

HR skills	There are three major challenges for the fisheries sector in terms of fisheries development. Firstly, is providing traditional small-scale fishermen with the skills in their business and to diversify their activities firstly by moving off-reef and to start

	using deeper water fishing gear such as longlines but also by other sources of income (e.g. eco-tourism). A second is the new emphasis on aquaculture development, which has a need for specialist skills for stock husbandry, hatchery operation, farm and health ([27]) management which are not necessarily transferable form traditional fishing practises. A third, and more pervasive problem is persuading young people to take up fishing or fish farming, which are often regarded as high-risk, lower status occupations.
Infrastructure	For fishing, the move of off-reef fishing and development of longer chill chains means a requirement for larger, more sophisticated vessels. These need safe and convenient moorings, as well as on-going maintenance support. This can be a particular challenge in areas where there is an expanding recreational vessel fleet competing for space in small harbours and marinas.
Research	Good fisheries research is essential for sustainable management of these limited resources. The region benefits form a high level of often EU funded research into tuna stocks and fisheries, but lacks a similar emphasis on non-tuna bycatch species (e.g. mahi mahi) and reef fish species. Again, with the emergence of aquaculture activities, there is little local capacity for research in environmental carrying capacity needs, nor the impacts of development on the local marine environment, which is itself essential for supporting other maritime economic activities such as coastal tourism.
Data	Data on tuna catches and landings in the region is good and well organise at a regional level. However, like research, there is a need for better data on non-tuna catches to support stock assessment and fisheries management planning. For instance, there is still relatively little known about the demersal fish resources of the Mascareigne banks

- ***Ports***

HR skills	Port operation and management requires a wide range of skills, from engineering to facilities management to financial management. Mauritius, with its relatively large population and education facilities is best placed to respond, and also benefits from a well-resourced ports authority. The Seychelles, with its small population pool, has more of a challenge, but has good commercial linkages with external expertise from the Middle East, Africa and Europe. Toamasina Autonomous Port is Madagascar's primary cargo port is governed by a public body, but the operation of the port has been contracted to Madagascar International Container Terminal Services (MICTS), a subsidiary of Filipino company International Container Terminal Services Inc..
Infrastructure	Self-evidently good infrastructure is essential for port development and growth. In the Seychelles, this is constrained by the very narrow coastal strip, meaning intense competition for land with other coastal activities, including urban development, and has thus responded to land reclamation which is expensive and has associated environmental costs. Port Louis is also hemmed in by the own itself to the east, but does have land to the north and south where it can expend if necessary. Madagascar benefits form an extensive coastline with a number of natural harbours, but can be constrained by poor connecting infrastructure. The Comores is also restricted by poor communications infrastructure, but despite its small size, has reasonable land areas for expansion. Even if Réunion and Mayotte have port capacities, the unbalanced cost of the work is a structural limitation.
Research	There is little research capability in this region associated with port development. Mauritius has built a considerable IT research capacity, but this is mainly aimed at financial and other secondary markets.

([27]) Including local veterinary skills.

Data	In Mauritius, the Mauritius Ports Authority has a considerable statistical data collection facility operating in Port Louis. Similarly, in the Screeches, the Seychelles Ports Authority has an 'Information Centre', but its capacity is not known to the consultants.

- ***Maritime transport***

HR skills	As island nations, maritime transport skills are essential in this region. However, the sector in constantly evolving, with new propulsion technologies, the introduction of emission restrictions and ever-changing international obligations under IMO, ILO etc. This creates challenges in terms of skills development. Secondly, it has become increasingly difficult to encourage young people to enter the maritime sector, especially if it means spending long periods away from home.
Infrastructure	See ports –regional maritime short sea shipping (IOC): still weak but a long term regional willingness to develop a viable short sea shipping activity to facilitate the connectivity of the regional markets (IOC, 2015 ([28]))
Research	See ports
Data	See ports

- ***Shipbuilding***

HR skills	There are some specialist skills (e.g. for fishing vessels operating in the region) being developed in the regions for vessel building and repair, especially in Mauritius, but the main challenge is for scaling this up and challenging established yards, esp. in Asia. It is likely that these small island states, with their limited population pool and need to import all raw materials will never be able to compete for mainstream ship-building. However, they may be able to expand the current niche build and repair business to satisfy regional needs e.g. for fishing vessels, small patrol craft and inter-island transport needs. This will require additional skill sets.
Infrastructure	The large areas, and deep-water access needed for ship-building and repair are in short supply, both in Mauritius and the Seychelles. For instance, CNOI in Mauritius is struggling to expand as completion for space within the existing main part of Port Louis is fierce. Madagascar potentially has candidate areas, as well as access to raw materials, but lacks many of the skills required to take advantage.
Research	If a strategy to expand ship building and maintenance services for regional maritime assets develops, there will be a need for specialist research to ensure competitiveness with yards from outside the region.
Data	No major data gaps known.

([28]) The Case for a Regional Maritime Service, Facts and Figures: http://commissionoceanindien.org/activites/cabotage-maritime/the-case-for-a-regional-maritime-service-facts-and-figures-january-2014/ , access: 7 Sept. 2017. 39 p.

- ***Ship repair***

HR skills	See shipbuilding (the two sectors being strongly interconnected at the sea basin level)
Infrastructure	In Réunion and Mayotte, high costs of work are a brake for heavy ship repair services development. Light repair services complementary to repair services in Mauritius and Seychelles is under investigation to maintain these key services for ships calling or mooring in Réunion
Research	-
Data	-

- ***Non-economic activity — maritime surveillance, a key public driven activity for the Indian Ocean's blue economy***

HR skills	Maritime surveillance is a rapidly evolving sector, with increasing dependence on either autonomous vehicles or remote sensing to cover the large sea areas involved. This will inevitably require additional, highly skilled personnel to ensure regional capacity in this area. Whilst Mauritius has a rapidly developing IT sector, this is lacking from the other three states.
Infrastructure	Whilst there are good port facilities, and a number of traditional surface vessel assets, there is a general lack of facilities to take advantage of newer technologies in remote sensing in the region. Different actions are led at different levels (national, regional (IOC))
Research	Most maritime surveillance research is based in either South Africa or Reunion. Strategic alliances need to be built on a regional basis to reduce the overall cost burden and to develop compatible surveillance systems that can work together.
Data	Whilst there has been considerable progress made in the monitoring, control and surveillance of commercial tuna fishing, there are still challenges in obtaining comprehensive data to allow management of all maritime risks and threats in the region.

4.1 Synoptic view on the gaps identified

The following table reports a synoptic view of main gaps identified grouping them by 'legislative', 'non-legislative' and 'financial' gaps.

ACTIVITY	LEGISLATIVE	NON-LEGISLATIVE	FINANCIAL
Coastal tourism		• Language skills limitation (e.g. in Réunion and Mayotte) and training resources non-diversified enough to respond to the growing of quality workers in the sector (e.g. in Mauritius) and English-speaking customers' demand • Weak integrated approach of the tourism development at national and sea basin level • Tourism observatories, such as in Mayotte, may require continuous or additional support to collect data in collaboration national statistical Institutes (INSEE in Mayotte and Réunion). • Remark: high end opportunities such as ecotourism may require some degree of site-	

ACTIVITY	LEGISLATIVE	NON-LEGISLATIVE	FINANCIAL
		specific R&D on technologies e.g. to provide water / electricity supplies and responsible sewage treatment in remote, off-grid areas, all of them must take into account cyclonic risks • Lack of good infrastructure such as roads, energy, and accommodation in rural areas in Madagascar, Mayotte, and Comoros to boost coastal tourism	
Cruise tourism		• Ports in the sea basin still be adapted for a better answer to the signs of a growing demand of cruise tourism back in the sea basin • Still a weak integrated approach of developing coastal and cruise tourism at the sea basin level	
Fisheries and aquaculture		• Small-scale fishermen lacking skills to run their business more efficiently and to diversify their fishing activities • Lacking local specialist skills for stock husbandry, hatchery operation and farm management • Weak knowledge of fish stocks exploited by small-scale fisheries (including tuna associated species caught by artisanal fishermen in the sea basin) and non-tuna fish stocks in general • Unattractiveness of the sector for young people • Safe and convenient moorings, as well as on-going maintenance support without conflicting with other maritime sectors (such as the recreational fleets) • Little local capacity for research in environmental carrying capacity needs and in the impacts of aquaculture development on the local marine environment (essential to support other maritime economic activities (e.g. coastal tourism)	
Ports		• Poor connecting infrastructure in Madagascar and Comoros • Little research capability in this region associated with port development	
Maritime transport		• Still weak regional short sea shipping to facilitate exchange between the sea basin countries (regional maritime links between islands) • Difficulty to encourage young people to enter the maritime sector, especially if it means spending long periods away from home	
Shipbuilding		• Impossibility to compete for mainstream ship-building, especially with Asia and South Africa, due to the limited population pool and the need to import all raw materials for ship building (however possibility of scaling up for regional needs) • Short supply of deep water access (Mauritius for instance) • Lack of many shipbuilding skills in Madagascar, at least in the short term	
Ship repair		• High costs of work are a brake for heavy ship repair services development in Réunion and Mayotte compared to neighbour countries	

Mapping actors and maritime strategies

5. Mapping public authorities in the South West Indian Ocean sea basin

Table 4 — Public authorities surveyed in the South West Indian Ocean sea basin

Maritime sector(s)	Name of the authority	Main roles and responsibilities	Address	website
Cross-cutting	African Union	Intergovernmental organisation fostering the development of and the implementation of Panafrican policies by sectors between its country members	African Union Headquarters P.O. Box 3243 | Roosevelt Street (Old Airport Area) | W21K19 | Addis Ababa, Ethiopia	http://www.au.int/maritime
	Indian Ocean Commission	Intergovernmental organisation fostering the development of and the implementation of Panafrican policies by sectors between its country members (Indian Ocean countries	Blue Tower — 3ème étage Rue de l'Institut Ebène Mauritius	http://www.commissionoceanindien.org
	COMESA — Common Market for Eastern and Southern Africa	Intergovernmental economic organisations strengthening regional market integration (main organisations relevant to the study)	COMESA Centre Ben Bella Road P. O. Box 30051 Lusaka, Zambia	http://www.comesa.int/
	IORA — Indian Ocean Rim Association		3rd Floor | Tower I | NeXTeracom Building Cybercity | Ebene | Mauritius	http://www.iora.net
	SADC — Southern African Development Community	Regional integration of the SADC members and poverty eradication within Southern Africa through economic development and ensuring peace and security. 15 Member States; Angola, Botswana, Democratic Republic of Congo, Lesotho, Madagascar, Malawi, Mauritius, Mozambique, Namibia, Seychelles, South Africa, Swaziland, Tanzania, Zambia and Zimbabwe.	SADC House Plot No. 54385 Central Business District Private Bag 0095 Gaborone Botswana	www.sadc.int Cf. Fiche of Programmes de coopération au titre de l'objectif 'coopération territoriale européenne' 2014 -2023

Maritime sector(s)	Name of the authority	Main roles and responsibilities	Address	website
	UNECA — United Nations Economic Commission for Africa	Mainstreaming Blue Economy into national development plans of African countries (see UNECA, 2016. Africa's Blue Economy: A policy handbook)	Economic Commission for Africa P.O. Box 3001 Addis Ababa, Ethiopia	www.uneca.org
	Comité de Concertation Régionale (Comité de Suivi du programme de coopération)	Indian Ocean Consultation Committee — monitoring EU cooperation programmes in the Indian Ocean	Management authority: Conseil Régional de la Réunion	See also the Indian Ocean Commission website ([29])
	Conseil maritime ultramarin du bassin sud océan Indien	Overseas Maritime Council of the Southern Indian Ocean basin (French Council)	Council under the co-presidence of the Préfets ([30]) of Réunion, Mayotte and Terres Australes et Antarctiques françaises (TAAF)	See for instance, the Préfecture of Réunion ([31])
	Ministère des Outre-Mer, France	Ministry of French overseas territories and French outermost regions	27, rue Oudinot 75007 Paris	www.outre-mer.gouv.fr
	ONML — Observatoire national de la mer et du littoral	French observatory of maritime and coastal activities	Service de l'Observation et des statistiques (SOeS) Commissariat général au développement durable, 5, route d'Olivet, 45100 Orléans	http://www.onml.fr/accueil/
	Ifremer — Institute français de recherche pour l'exploitation de la mer	French Marine Research Institute	Siège social — 155, rue J. J. Rousseau — 92138 Issy Les Moulineaux Cedex	www.ifremer.fr
Fisheries	Indian Ocean Tuna Commission (IOTC	Management of tunas and fish associated to tuna catches	OTC Secretariat Le Chantier Mall (2nd floor)	www.iotc.org

([29]) http://commissionoceanindien.org/activites/programme-regional-de-surveillance-des-peches/activites/comite-de-concertation-regionale-le-role-remarquable-de-la-coi-en-faveur-de-lintegration-regionale-souligne-28-fev-14-reunion/ , access: 23 November 2016.
([30]) High public representative of the French State in each of these French territories.
([31]) http://www.reunion.gouv.fr/installation-du-conseil-maritime-ultramarin-du-a1374.html , access: 23 November 2016.

Maritime sector(s)	Name of the authority	Main roles and responsibilities	Address	website
			PO Box 1011 Victoria Mahé — Seychelles	
	Ifremer	French Marine Research Institute involved in marine research and fisheries research (including data collection)	Ifemer — Centre de Bretagne Unité d'économie maritime — UMR CNRS 6308 AMURE CS10070 29280 Plouzané — France	http://sih.ifremer.fr/Publications/Synthes es/Outre-Mer
Ports (including dredging)	PMAESA — Port Management Association of Eastern and Southern Africa	Non-profit, inter-governmental organization made up of Port Operators, Government Line Ministries, Logistics and Maritime Service Providers and other port and shipping stakeholders from the Eastern, Western and Southern African and Indian Ocean regions Note: France, Réunion and Mayotte are not members	KPA Building Asset No 480038, Kaunda Ave, Mombasa, Kenya P O Box 99209 — 80107 Mombasa — Kenya	www.pmaesa.org
Maritime surveillance (including fisheries surveillance)	DMSOI — Direction de la Mer Sud océan Indien French Maritime Directorate in the southern Indian Ocean	DMSOI's global missions: - Maritime surveillance and maritime safety a) in the sea basin on behalf of France and other countries in the Indian Ocean and b) in the French territories in the Indian Ocean; - Participate in the implementation of the sea and coastal national strategy in the French territories in the Indian Ocean within an integrated regulatory approach between the different maritime sectors based on the French blue book in the Indian Ocean	11, rue de la Compagnie — 97487 Saint Denis Cedex	http://www.dm.sud-ocean-indien.developpement-durable.gouv.fr/ See also: http://www.dm.sud-ocean-indien.developpement-durable.gouv.fr/plaquette-de-presentation-a185.html

6. Summary of existing maritime strategies

This Chapter lists some of the blue economy or blue economy related regional and national strategies developed within the Indian sea basin. Africa's Blue Economy and/or Blue Growth ([32]) framework has recently become spotlighted in the Africa's Union's Agenda 2063, the African Integrated Maritime Strategy (AIM Strategy 2050) and the recent initiative of the United Nations Economic Commission for Africa (UNECA) on Harnessing the Blue Economy for Africa's development. Agenda 2063 states for instance that 'Africa's Blue economy', which covers three times the size of its landmass, shall be a major contributor to continental transformation and growth. In this regard, the 2050 AIM Strategy provides a broad framework for the protection and sustainable exploitation of the African Maritime Domain (AMD) for wealth creation while the recent UNECA initiative provides a step-by-step guide on how to mainstream the Blue Economy into continental, sub-regional, and national policies, plans, laws, regulations, and practices for the development of African Sustainable Blue Economy strategies. Alongside these Blue Economy strategies, the Indian Ocean Commission and the Nairobi Convention play a coordinating role in implementing a series of intervention projects in the South West Indian Ocean sea basin including maritime surveillance. Also, the Convention projects aim to stem any further degradation of the ecosystem and to reverse the degradation and destruction of critical habitats. To end, France has developed a National Law on the Blue economy development and national strategies by maritime sectors for instance a national strategy for the sea and the coasts and a port national strategy. The national strategies are planned to be adapted to the specific context and development by French sea basins including the Indian Ocean sea basin where Mayotte and Réunion are located.

6.1 African Union

6.1.1 Africa's Union's Agenda 2063

- **Overview**

On the occasion of the Golden Jubilee of the Organization of African Unity (OAU), the African Union (AU) Summit tasked the African Union Commission (AUC), supported by the New Partnership for Africa's Development (NEPAD) Planning and Coordinating Agency (NPCA), the African Development Bank (AfDB) and the UN Economic Commission for Africa (UNECA), to prepare a 50-year continental agenda, the Agenda 2063. The extensive consultative development process of Agenda 2063 resulted in the Aspirations of the African People, the driver of Agenda 2063. In preparation for this agenda, were also realized an extensive review of African development experiences, analysis of challenges and opportunities of today, as well as a review of national plans, regional and continental frameworks and technical studies, including drawing upon publications and research materials from many institutions and organizations. Agenda 2063 will be first introduced through a ten-year implementation plan, the framework of which outlines the goals, priority areas and targets, indicative strategies and milestones for each of the 7 aspirations.

- **Geographical level (local/national/sea basin)**

Whole Africa. National and sea basins. *Note that France, Mayotte, Réunion and the EU are not members of the African Union, being non-African States (or non-African States representative for the EU).*

([32]) The Blue Economy incorporates technologies that are mature or post-mature stage such as oil & gas, tourism, shipping, and fishing. Blue Growth is the long-term strategy to support sustainable growth within the Blue Economy incorporating life-cycle assessment. It encompasses technologies that have high growth and job potential and by their nature they are early stage and novel (definition taken from the EU Research project MARIBE, see: https://maribe.eu/ , access: August 2017.

- **Maritime sectors involved**

All sectors.

- **Objectives**
  - Blue economy for accelerated economic growth
  - Environmentally sustainable and climate resilient economies and communities

- **Implementing tools**

This strategy uses many tools, including:

  - Implementing the African Integrated Maritime Strategy
  - Developing/implementing policies and programmes for sustainable utilization of marine resources to increase their contribution to GDP
  - Providing policies / incentives and positive regulatory environment for the creation new businesses with platforms based on: (i) deep ocean water applications (ii) marine hydrocarbon and mineral exploration and exploitation (iv) marine biotechnology and off (vi) aquaculture development
  - Developing/ implementing policies to support the application of marine spatial planning and integrated adaptive oceans policy/governance for Exclusive Economic Zones (EEZs)
  - Developing/implementing policies for the growth of port operations and marine transport
  - Promoting the sustainable use and management of coastal zones and marine resources to build climate resilient and sustainable communities
  - Creating representative marine protected areas for resilience, sustainability and conservation of aquatic biodiversity ⎫ Establish bank of genetic marine resources to restore degraded eco-systems and vulnerable / threatened species

- **Actors involved and beneficiaries**

All maritime actors. Beneficiaries: all stakeholders

- **Possible contribution to the fulfilment of BG objectives**

Fostering blue growth.

- **Website address**

http://www.au.int/en/agenda2063

6.1.2 Africa's Integrated Maritime Strategy 2050

- **Overview**

The 2050 Africa's Integrated Maritime Strategy (2050 AIM Strategy) was published by the African Union in 2012 as a regional approach covering both inland and maritime waters.

- **Geographical level (local/national/sea basin)**

Whole Africa (African States). All Sea basins.

- **Maritime sectors involved**

Sectors involved:

  - Fisheries and aquaculture
  - Marine tourism, maritime governance, maritime security, marine spatial planning and environmental monitoring

- **Objectives**
  - Establish a Combined Exclusive Maritime Zone of Africa (CEMZA)
  - Engage civil society and all other stakeholders to improve awareness on maritime issues

- Enhance political will at community, national, regional and continental levels
- Enhance wealth creation, and regional and international trade performance through maritime-centric capacity and capability building
- Ensure security and safety of maritime transportation systems
- Minimize environmental damage and expedite recovery from catastrophic events
- Prevent hostile and criminal acts at sea, and coordinate/harmonize the prosecution of the offenders
- Protect populations, including AMD heritage, assets and critical infrastructure from maritime pollution and dumping of toxic and nuclear waste
- Improve Integrated Coastal Zone/Area Management in Africa
- Promote the ratification, domestication and implementation of international legal instruments
- Ensure synergies and coherence between sectoral policies within and between the RECs/RMs
- Protect the right of access to sea and freedom of transit of goods for land-locked States.

- **Implementing tools**
  - Capacity building
  - A series of out-reach initiatives including an annual Maritime Security and Development Conference and representation of African maritime interests in international fora

- **Actors involved and beneficiaries**

Key stakeholders: AU Member States, local communities, specialized regional institutions and associations, the African maritime private sector, strategic development partners and the international community as a whole

- **Possible contribution to the fulfilment of BG objectives**

Pan African development strategy of maritime sectors in African countries.

- **Website address**

http://www.au.int/maritime

6.2 The Policy Framework and Reform Strategy for Fisheries and Aquaculture in Africa

- **Overview**

The Policy Framework and Reform Strategy for Fisheries and Aquaculture was created in 2011 to endorse the recommendations of the first Conference of African Ministers of Fisheries and Aquaculture (CAMFA I) held in September 2010. This Policy Framework and Reform Strategy is structured around three main entities: (1) regional specific priorities of common interest to all or most of the countries in each of the five regions of Africa, (2) supporting and delivering mechanisms to assist and facilitate implementation of agreed strategies and (3) suggesting indicators to measure success.

- **Geographical level (local/national/sea basin)**

Whole Africa sea basins. Both Mayotte and Réunion are out of the AU's jurisdiction (see section hereabove).

- **Maritime sectors involved**

Fisheries and aquaculture

- **Objectives**
  - Improve the productivity of fisheries and aquaculture
  - Enhance the contribution of fish to sustainable food and nutritional security and economic growth within the Comprehensive Africa Agriculture Development Programme (CAADP) framework

- **Implementing tools**

— Economic, ecosystems and welfare approaches to reform fisheries
— Fiscal reforms

- **Actors involved and beneficiaries**

African governments, actors of the fisheries and aquaculture sectors

- **Possible contribution to the fulfilment of BG objectives**

Fostering blue growth

- **Website address**

http://www.au.int/en/documents/30266/policy-framework-and-reform-strategy-fisheries-and-aquaculture-africa and the African Union — Interafrican Bureau for Animal Resources (AU-IBAR) coordinating the implementation of the Strategy.

6.3 The Nairobi Convention

- **Overview**

The Convention for the Protection, Management and Development of the Marine and Coastal Environment of the Eastern African Region (the Nairobi Convention) and associated protocol underpins the Eastern African region, in particular South Africa, Mozambique, Tanzania, Kenya, Somalia, Seychelles, Mauritius, Comoros, Madagascar and France. The Convention is composed of three protocols: (1) Protocol for land-based sources of pollution and (2) Protocol for protected wild fauna and flora, and (3) Protocol on cooperation in cases of pollution emergencies, and a forth protocol on Integrated Coastal Zone Management, which is under discussion. Adding to these protocols, there are a Climate Change Strategy that covers the coastal terrestrial areas and marine extent, and a Strategic Action Programme for land based sources of pollution, including river basins. The action plan for the Nairobi Convention details objectives for the protection, management, and development of the marine and coastal environment of the eastern African region.

- **Geographical level (local/national/sea basin)**

Indian Ocean Sea basin (Somalia down to south Africa). *Both Mayotte and Réunion are represented by France in this Convention*.

- **Maritime sectors involved**

All sectors

- **Objectives**

To obtain a prosperous Western Indian Ocean Region with healthy rivers, coasts and oceans

- **Implementing tools**

Providing a mechanism for regional cooperation, coordination and collaborative actions

- **Actors involved and beneficiaries**

All actors

- **Possible contribution to the fulfilment of BG objectives**

Fostering blue growth

- **Website address**

http://web.unep.org/nairobiconvention/

6.4 The Indian Ocean Commission — IOC's initiatives regarding the blue economy strategy

- **Overview**

The Indian Ocean Commission (IOC) is an intergovernmental organization created in 1982 in Port Louis (Mauritius) and institutionalized in 1984 by the Victoria Agreement (Seychelles). It brings together five countries in the Indian Ocean region: Comoros, Reunion, Madagascar, Mauritius, Seychelles. In 2002, the IOC, along with the Common Market for Eastern and Southern Africa (COMESA), East African Community (EAC) and Intergovernmental Authority on Development (IGAD), created an International Coordination Committee (IRCC), which was later mandated by the states of the ESA-IO.

The IOC has two specific tasks:

— it is a preferred vehicle construction and development of all the Indian Ocean area, in which the movements and physical and immaterial connections must be able to perform effectively;
— it then provides a framework for its member states to conduct external collective actions of integration and cooperation with the Eastern and Southern African region, the African continent (AU), the South (South-South cooperation) and the rest of the world, as well as multilateral institutions and various donors.

IOC operates where its action brings added value compared to actions of its members and / or regional organisations of broader initiatives (COMESA, SADC Tripartite), to continental institutions (African Union) and / or multilateral (UN, WTO, etc.). While continuing to respect the principles of coordination, complementarity and subsidiarity, IOC plays a supporting role, or pulse, in fields where its members need an increased specific support and where broader regional organizations adequately taken into account their specific island.

IOC leads different projects or actions or studies in accordance with its mandate.

The IOC has a pole of blue and green economy growth with the objective to sustainable and integrated development of the blue economy in the IOC area of competence.

6.4.1 Elaboration of the blue economy strategy of the IOC

- **Overview**

During the meetings of the governing body of the IOC, several politicians highlighted the potential of the blue economy and the need for the IO countries to take advantage of the ocean in their durable development approach. With these considerations in mind, the IOC intends to develop a regional strategy on blue economy.

- **Geographical level (local/national/sea basin)**

Both national and Indian Ocean sea basin. Both Mayotte and Réunion are represented by France in the IOC.

- **Maritime sectors involved**

Main sectors involved:

— Fisheries and aquaculture
— Tourism
— R&D
— Technological innovation
— Marine biotechnology
— Renewable energies
— Maritime transport
— Exploration and exploitation of mineral resources at sea

- **Objectives**
  — A secure regional space of intelligent economic growth

  - A resilient and durable insular and oceanic environment
  - Promote Indianoceania's identity

- **Implementing tools**

To elaborate strategies and national action plans concerning blue economy through sharing of experiences better practices and local valuations, the IOC has organized a regional conference on blue economy. This conference will identify opportunities and restrictions of each of the main sectors concerned by blue economy.

- **Actors involved and beneficiaries**

Public and private actors

- **Possible contribution to the fulfilment of BG objectives**

Blue economy is the central concept of this strategy

- **Website address**

http://commissionoceanindien.org/axes-strategiques/poles-de-croissance-regionale-bleue-et-verte-specialisations-et-valorisation-economique/

In December 2016, the Indian Commission organised a two-day workshop for the Indian Ocean blue growth. Located in Réunion, it was coordinated by the local public authorities of the EU outermost region: the local representation of the State — Préfecture — and the Regional council — Conseil Régional). The outcomes of the workshop are presented in the Chapter 3 above.

6.4.2 Regional fishing and aquaculture strategy if the Indian Ocean Commission

- **Geographical level (local/national/sea basin)**

National and Indian Ocean sea basin. Both Mayotte and Réunion are represented by France in the IOC.

- **Maritime sectors involved**

Fisheries and aquaculture

- **Objectives**
  - Provide guidance and set priorities to the IOC member states for the management and conservation of fisheries resources and sustainable development of aquaculture
  - Strengthen the cooperation to enable the IOC member states to improve the management of the fisheries sector, especially in the case of shared resources or common interest, and realize the potential for growth and jobs related to sustainable development

- **Implementing tools**

The Regional Strategy defines a comprehensive program of regional cooperation among countries in the areas of conservation, management and the responsible and sustainable exploitation of fisheries resources and the sustainable development of aquaculture.

- **Actors involved and beneficiaries**

All actors

- **Possible contribution to the fulfilment of BG objectives**

Promoting blue growth

- **Website address**

http://commissionoceanindien.org/axes-strategiques/poles-de-croissance-regionale-bleue-et-verte-specialisations-et-valorisation-economique/

6.4.3 SmartFish Programme (EU funded)

- **Geographical level (local/national/sea basin)**

Both National and Indian Ocean sea basin. Both Mayotte and Réunion are represented by France in the IOC.

- **Maritime sectors involved**
  - Marine and inland fisheries
  - Aquaculture

- **Objectives**

The sustainable exploitation of marine and lake fisheries resources in order to improve level of social, economic and environmental development and deepen regional integration in the ESA-IO

- **Implementing tools**

Actions are implemented in five areas: sustainable fisheries management, governance, monitoring, control and surveillance, trade ([33]) in fish and fisheries by-products and food security

- **Actors involved and beneficiaries**

Actors involved:

  - The European Union
  - The Indian Ocean Commission
  - The FAO

- **Possible contribution to the fulfilment of BG objectives**

Fostering blue growth

- **Website address**

http://commissionoceanindien.org/activites/smartfish/

6.4.4 Maritime surveillance

- **Geographical level (local/national/sea basin)**

Both national and Indian Ocean sea basin. Both Mayotte and Réunion are represented by France in the IOC.

- **Maritime sectors involved**

All sectors

- **Objectives**

Implementation of medium and long term regional strategies against piracy and promoting maritime safety

- **Implementing tools**
  - An action plan for Somalia focused on the roots of the problem;
  - Strengthening national and regional capacities regarding the arrest, transfer, detention and prosecution of pirates;
  - Strengthening regional capacity to fight against illicit financial flows linked to piracy and deal with the economic impact of piracy in the region;
  - Building national and regional capacity for maritime safety;
  - The establishment of a network for information exchange and coordination.

([33]) Including market studies for aquaculture and fishery products. Available online on the provided website.

- **Actors involved and beneficiaries**

Regional organizations ESA-IO, COMESA, EAC and IGAD

- **Possible contribution to the fulfilment of BG objectives**

Improvement of maritime safety

- **Website address**

For instance, for fisheries, the regional fisheries surveillance programme funded by the EU that ended in 2014 and part of the activities are now being implemented by the EU funded programme Smartfish: http://commissionoceanindien.org/activites/programme-regional-de-surveillance-des-peches/a-propos/

6.4.5 BIODIVERSITY project (EU funded)

- **Geographical level (local/national/sea basin)**

Indian Ocean sea basin and National. Both Mayotte and Réunion are represented by France in the IOC.

- **Maritime sectors involved**

All sectors.

- **Objectives**

Develop and strengthen the national and regional capacities for sustainable participatory management of coastal, marine and island specific biodiversity during a period of five years. Five main results are expected:

— Improved and harmonised policies and legal and institutional frameworks across the region for the sustainable use of biodiversity
— Enhanced education, sensitisation, communication and information tools for the management of the use of biodiversity, in support of decision makers at regional, national and community levels
— Improved systems for networking and exchange of data, statistics, and other biodiversity related information.
— Biodiversity Thematic Centres created or enhanced for the exchange of information, experiences and best practices in the sustainable use of the biodiversity.
— Enhanced contribution of biodiversity to sustainable economic development and sustainable livelihoods across the region

- **Implementing tools**

Activities include:

— Capacity building
— Sensitisation and information-sharing of biodiversity data and best practises
— A Call for Proposals scheme giving opportunity to Non-State Actors, including local communities, to develop and implement small biodiversity management projects, including revenue-generating projects such as small-scale fisheries, aquaculture and farming

The project started in 2014 and will end in 2018.

- **Actors involved and beneficiaries**

Actors of the ESA-IO region (Comoros, Madagascar, Mauritius, Seychelles, Kenya and Tanzania) and the Secretariat of the Nairobi Convention

- **Possible contribution to the fulfilment of BG objectives**

Fostering blue growth

- **Website address**

http://commissionoceanindien.org/fileadmin/resources/Biodiversite/others/Depliant_VA_final.compressed.pdf

6.4.6 ISLANDS Programme (EU funded)

The programme started in 2011.

- **Geographical level (local/national/sea basin)**

Both national and Indian Ocean sea basin. Both Mayotte and Réunion are represented by France in the IOC.

- **Maritime sectors involved**

All sectors

- **Objectives**

An increased level of social, economic and environmental development and deeper regional integration through the sustainable development of SIDS in the ESA-IO region.

- **Implementing tools**
  - Promote Sustainable Development;
  - Address common challenges such as Coral Reef degradation, Natural Disaster, and Climate Change;
  - A Monitoring and Evaluation System to track countries progress in implementing the Mauritius Strategy.

- **Actors involved and beneficiaries**

Actors involved:

- IOC member States;
- The Secretariat of the Pacific Regional Environment Programme (SPREP);
- The Caribbean Community Climate Change Centre (5Cs);
- The EU;
- The United Nations Department of Economic and Social Affairs (DESA).

Beneficiaries: Comoros, Madagascar, Mauritius, Seychelles and Zanzibar of the United Republic of Tanzania.

- **Possible contribution to the fulfilment of BG objectives**

Fostering blue growth.

- **Website address**

http://commissionoceanindien.org/activites/islands/le-programme/ and its brochure ([34]).

6.5 Seychelles: Regional Fusion & Law Enforcement Centre for Safety & Security at Sea (REFLECS$_3$)

- **Overview**

The Regional Fusion and Law Enforcement Centre for Safety and Security at Sea (REFLECS$_3$) was originally known as the Regional Anti-Piracy Prosecution and Intelligence Co-ordination Centre (RAPPICC), and was established in 2012 as a joint initiative by the UK and the Seychelles governments. The centre has strengthened regional cooperation and established a joint intelligence task force within the eastern African region, consisting

([34]) http://31.222.186.27/moodle/pluginfile.php/792/mod_resource/content/3/brochure_ISLANDS_en_finale_13.pdf , access: 23 November 2016.

of Seychelles, Tanzania and Kenya. The centre has been successful in developing cases for the prosecution of pirates.

- **Geographical level (local/national/sea basin)**

Eastern African Sea basin. Both Mayotte and Réunion are not part of this strategy.

- **Maritime sectors involved**

Maritime security.

- **Objectives**

Create a multi-disciplinary and multinational centre for law enforcement cooperation in partnership with Seychelles, wider Indian Ocean nations and international partners to combat the threat from regional piracy and maritime linked transnational organized crime.

- **Implementing tools**

Create sustainable regional capability and capability to undermine the piracy business model by bringing pirate leaders, financiers and enablers to justice and depriving them of the proceeds and benefits of their crimes through:

  — Intelligence development
  — Investigation and precautions
  — Capacity and capability-building in the region.

- **Actors involved and beneficiaries**

Main actors involved:

  — The REFLECS3 Management Board
  — The REFLECS3 Steering Group
  — The Contact Group on Piracy off the Coast of Somalia (CGPCS).

- **Possible contribution to the fulfilment of BG objectives**

Helps to improve an integrated maritime surveillance in Seychelles.

- **Website address**

http://www.rappicc.sc/index.html

6.6 The Blue Economy: Seychelles' Vision for Sustainable Development in the Indian Ocean

- **Overview**

The Seychelles government adopted the National Development Strategy (NDS) 2015-2019 in November 2015, with the concept of the 'Blue Economy' as its centrepiece. The concept emphasises the economic potential of Seychelles' vast Exclusive Economic Zone (EEZ), a marine area of 1 374 000 km² (the second largest in Africa), for inclusive growth. The vision reflects that the Seychelles, like many small island developing states, as well as many African coastal states, lack the capacity to manage their oceanic territory and exclusive economic zones, and that it is important that this capacity is increased through regional and international cooperation, and sharing of resources (HE Chatham House, 2014).

Also, the Commonwealth, a voluntary association of 52 independent and equal sovereign states is assisting the government of Seychelles, member of the Association, in its Blue Economy Strategic Roadmap and Implementation from 2016 to 2018. ([35])

([35]) http://thecommonwealth.org/project/seychelles-blue-economy-strategic-roadmap-and-implementation , November 2016.

- **Geographical level (local/national/sea basin)**

Seychelles' Sea basin. Both Mayotte and Réunion are not part of this strategy.

- **Maritime sectors involved**

All sectors

- **Objectives**

Change the relationship with the ocean from one of extraction to one of development conservation and sustainable use of ocean resources, oil and mineral wealth extraction, bio-prospecting, sustainable energy production and marine transport, as well as branding Seychelles a 'blue' tourism destination

- **Implementing tools**

Not yet a formal strategy but the concept integrates:

  — Conservation and sustainable use of ocean resources
  — Oil and mineral wealth extraction
  — Bio-prospecting
  — Sustainable energy production and marine transport
  — Branding Seychelles a 'blue' tourism destination

- **Actors involved and beneficiaries**

All actors.

- **Possible contribution to the fulfilment of BG objectives**

'Blue Economy' is a central concept of this strategy

- **Website address**

Office of the President of the Republic of Seychelles to follow the Government Blue economy strategy and its roadmap development: http://www.statehouse.gov.sc/

6.7 Mauritius — blue economy strategy

- **Geographical level (local/national/sea basin)**

National.

- **Maritime sectors involved**

All sectors.

- **Objectives**

General objective: develop an oceanic economy to diversify the Mauritian economy.

- **Implementing tools**
  — New ministry created with the Institute of Oceanography and all maritime activities;
  — Promoting of sea prospections for oil and mineral resources for their sustainable exploitation (Speech of the President of Mauritius in January 2015) and
  — Development of a master plan to develop the ocean economy in Mauritius. To draft it, the Ministry would have signed an agreement with the Commonwealth Secretariat to develop a master plan on a National Ocean Policy Paper covering the next five years according to one press media (DefiMediaGroup) ([36]). The situation of the plan is unknown (absence of information on the website of the Ministry).

([36]) 30 March 2016 http://defimedia.info/ocean-economy-roadmap-burgeoning-sector , access: October 2016.

- **Actors involved and beneficiaries**

All maritime actors.

- **Possible contribution to the fulfilment of BG objectives**

Fostering blue growth.

- **Website address**

Ministry of Ocean Economy, Marine Resources, Fisheries, Shipping and Outer Island ([37])

6.8 France — Loi sur l'économie bleue — 2016 (French Law on Blue Economy)

- **Geographical level (local/national/sea basin)**

National. Both Mayotte and Réunion are represented by France.

- **Maritime sectors involved**

All sectors.

- **Objectives**

General objective: Foster new perspectives of development and attractiveness for all maritime actors.

- **Implementing tools**
  - Development by sea basin strategies
  - Simplify the regulation on maritime activities
  - Reinforce the competitiveness of the French flag
  - Reinforce the maritime and in ports security
  - Foster the evolution of port activities
  - Modernise the fisheries and aquaculture sector
  - Development of the yachting sector
  - Better integration of the maritime responsibility

- **Actors involved and beneficiaries**

All maritime actors.

- **Possible contribution to the fulfilment of BG objectives**

Fostering blue growth.

- **Website address**

Portal of French legislation: Legifrance — loi pour l'économie bleue ([38])

See also, the recently approved National stragety — SNML — to develop blue growth published in the French Official Journal ([39]): Décret n° 2017-222 du 23 février 2017 Stratégie nationale pour la mer et le littoral

Additional details in the 'Réunion' and 'Mayotte' annexes)

([37]) http://defimedia.info/ocean-economy-roadmap-burgeoning-sector , access: 7 Sept. 2017
([38]) https://www.legifrance.gouv.fr/affichTexte.do?cidTexte=JORFTEXT000032728685&categorieLien=id , access: 7 Sept. 2017
([39]) www.legifrance.gouv.fr

6.9 France — Livre Bleu de l'Océan indien (2012) (Blue Book for the Indian Ocean)

- **Geographical level (local/national/sea basin)**

Indian Ocean basin: Mayotte, Réunion, TAAF

- **Maritime sectors involved**

Sectors involved: all maritime sectors

— Maritime transport
— Port
— Tourism
— Fisheries
— Aquaculture
— Marine Renewable energies.

- **Objectives**

Regional implementation of the Integrated Maritime Policy (IMP) for blue growth: French Blue Book (Stratégie nationale pour la mer et les oceans — SNML).

- **Implementing tools**

Definition of objectives for IMP implementation not only for French territories but also with a regional basin implication (see annexes 'Réunion' and 'Mayotte' for additional details on the implementation of the Indian Ocean Blue Book and the SNML)

Road map for Governance, Blue economy, protection and sustainable management of ecosystem, research and innovation, State action on Sea.

- **Actors involved and beneficiaries**

Main actors involved:

— Ministère de l'Environnement, de l'Énergie et de la Mer ;
— Ministère des outre-mer;
— Local stakeholders.

- **Possible contribution to the fulfilment of BG objectives**

Full compliance with BG Objectives as it is an IMP.

- **Website**

State services in Réunion (préfecture de la Réunion): http://www.reunion.gouv.fr/le-livre-bleu-sud-ocean-indien-a122.html

7. Mapping main stakeholders in the South West Indian Ocean sea basin

Table 5 — Main stakeholders surveyed in the South West Indian Ocean sea basin

Maritime sector(s)	Name of the stakeholders	Main roles and responsibilities	Address	website
Cross-cutting sector	EURODOM	Association representing the French outermost regions and its private stakeholders in all sectors in Brussels (among other activities)	251, Avenue Louise– Boîte 5 — B-1050 Bruxelles Or Eurodom Paris, 25, rue du Général Foy — F-75008 Paris	http://www.eurodom.org/
	Pôle de compétitivité Mer Bretagne Atlantique	French association grouping public and private organisations in Brittany and on the French Atlantic coast to foster research, development and innovation in the maritime sectors		http://www.pole-mer.fr/ www.pole-mer-bretagne-atlantique.com
	Pôle de compétitivité Mer Méditerranée	French association grouping public and private organisations on the French Mediterranean Coast to foster research, development and innovation in the maritime sectors	93, Forum de la Méditerranée, Technopôle de la Mer 83 190 Ollioules France	http://www.polemermediterranee.com
Fisheries and aquaculture	Ferme marine de Mahébourg	Red drum fish farming	Royal Road, Pointe aux Feuilles, Mauritius	www.fmm.mu
Extraction of salt	Compagnie salinière de Madagascar	Salt extraction company in Diego Suarez (80 000 t produced in 2015 — 25 000 t exported	Propriété Salines Plion, Antsahampano — BP. 29, Antsiranana, 201 Madagascar	http://www.csm.mg
	Sel Gemme de Madagascar	Salt extraction company in Tulear	Not identified (note: salt production is about to close or is closed now)	-
	Mont Calme Ltd, Mauritius	Salt extraction (closing – land to be used for real estate development)	Tamarin salt ponds, Mauritius	-
	Fleur de sel tropicale	Salt trader importing from the Indian Ocean	-	-
Ports (including dredging)	Association of Ports of the Indian Ocean Islands (APIOI)	The aim of the association is to provide a basic platform for member countries - through cooperation, collaboration, competition, dialogue/information sharing and, to some extent, specialisation - to face mounting challenges regionally and internationally in the liner shipping, cargo transportation, handling and other maritime/port-related industries. Port	OC headquarters	www.apioi.com

Maritime sector(s)	Name of the stakeholders	Main roles and responsibilities	Address	website
		authorities of Madagascar, Mauritius, Mayotte, Reunion and Seychelles are all members		
	Port Management Authorities in Seychelles, Madagascar, Mauritius and South Africa (Cape Town)	-	-	-
	Transnet National Ports Authority	Provides port infrastructure and marine services at the eight South African commercial seaports	P O Box 32696 Braamfontein 2017 South Africa Port of Durban: Box 1027 Durban 4000 South Africa	https://www.transnetnationalportsauthority.net.
Shipbuilding	CNOI — Chantier naval de l'océan Indien	Shipbuilding and ship repair company	Mer Rouge Port-Louis Republic of Mauritius	www.cnoi.info
	SECREN — Société d'études, de construction et de réparation navales	Shipbuilding and ship repair company	1, Rue Lavigerie — BP 135 Antsiranana, Madagascar	www.secren.mg
Ship repair	See CNOI and SECREN above	-	-	-
Coastal tourism	Tourism agencies in Seychelles, Mauritius and Madagascar	-	-	-
Cruise tourism	CIOA — Cruise Indian Ocean Association	Promoting cruise tourism in the Indian Ocean. Executive committee of the CIOA: Kenya, Tanzania, South Africa, Mauritius, Reunion Island (France) Note: Countries are represented by the ports authorities and their tourism bodies within this circuit region	P O Box 99209 — 80107, Mombasa — Kenya	www.cruiseindianocean.org

Mapping Blue Growth and assessing the potential

8. Growth drivers and barriers

Growth drivers and barriers are analysed for existing maritime sectors and maritime sectors with potential in terms of R&D, Access to finance, Infrastructure, Education, training and skills, Public engagement, and Local cooperation ([40]). The oil and gas sector is not analysed due to its current non-existence although presented in the summary fiche for the Indian Ocean sea basin.

	Fisheries and aquaculture	
	Drivers to growth	**Barriers for growth**
R&D	IOC support with external aid such as the EU: research and development of the fisheries and aquaculture sector is a strategic axis of the Indian Ocean Commission Direct external of the EU and EU Member States to third countries IOTC support with funds from compulsory and non-compulsory contributions ([41]) Support by NGOs (civil society)	Fisheries knowledge in terms of stock status, in particular for non-tuna species and stocks exploited by small-scale fisheries
Access to finance	Several development projects support the sector in the sea basin	Heavy investment requiring robust operators to renew industrial fishing vessels, which are ageing (attraction of foreign direct investment necessary)
Infrastructure	A sustainable port network to land (or tranship) and process high value species in the sea basin: good infrastructures for landing and tuna canning in Seychelles, Mauritius and Madagascar and for toothfish landed in Réunion	Sea basin far from high value market (except for tropical tuna and Antarctic toothfish, see left) and having costs to import marine fish farm feed and trade farmed fish seeds at large scale production (at least in the short term) Poor landing infrastructure in coastal rural areas (e.g. Madagascar) Cyclonic conditions: costs of cyclonic resistant infrastructure to farm marine fish in open sea
Education, training & skills	Availability of fisheries training and research centres in the sea basin (especially in Réunion and South Africa, smaller but operating research units in other sea basin States	Professionalism of small-scale fishermen in the sea basin States
Public engagement	IOC, EU and EU Member States (such as France), World Bank and NGO engagement in the sea basin EU driver for environmental sustainability: EU public fund not allowed to build new EU fishing fleet capacity (Common Fisheries Policy) to avoid stimulating overcapacity and overfishing Tuna management through IOTC: internationally binding conservation and management measures improving for sustainable tuna fishing Creation of a regional fisheries	Risk of unfair competition if fleet development rules and fishing management rules are different between fishing States targeting the same fish stocks in the sea basin

([40]) i.e. synergies and conflicts at local level with other activities, be they maritime or non-maritime.
([41]) For further details on the EU voluntary contributions to the IOTC, see the 2015 Ex post evaluation on Union financial measures for the implementation of the Common Fisheries Policy and in the area of the Law of the Sea 2007-2013, https://bookshop.europa.eu/en/ex-post-evaluation-on-union-financial-measures-for-the-implementation-of-the-common-fisheries-policy-and-in-the-area-of-the-law-of-the-sea-2007-2013-pbKL0415586/?AllPersonalAuthorNames=true , available to the public since February 2017.

	Fisheries and aquaculture	
	Drivers to growth	**Barriers for growth**
	management organisation to manage non-tuna species in the high seas in the Indian Ocean (South Indian Ocean Fisheries Agreement — SIOFA) with a secretariat in Réunion (FAO) ([42])	
Local cooperation	Synergies with tourism	Competition with other maritime sectors (cruise tourism, goods landed by containers), especially in ports

	Blue biotechnology	
	Drivers to growth	**Barriers for growth**
R&D	Availability of R&D facilities in Réunion	Still absent or low presence of blue biotechnology R&D in the sea basin with the exception of Réunion
Access to finance	Existence of EU and IOC project funding, for instance in Réunion	Emerging projects — business validity (that is economic viability of the private sector for investment)
Infrastructure	French research and development laboratories in Réunion	-
Education, training & skills	Education and skills in EU outermost regions (possibility of knowledge and skill transfer in the sea basin)	High skill availability almost absent (except in Réunion to some extent)
Public engagement	IOC and State (France) support	Additional EU support would be welcome from other EU funding mechanism than ERDF (stakeholder consultation) while project ideas exist
Local cooperation	Knowledge share between stakeholders within different maritime sectors (Réunion)	-

	Extraction of salt	
	Drivers to growth	**Barriers for growth**
R&D	Learning experience from international research and use of the technology of salt extraction by evaporation	-
Access to finance	-	Salt extraction by solar evaporation: a maritime sector with lower visibility than others for public support (see also local cooperation below)
Infrastructure	-	Except Madagascar, other sites in the sea basin are now either going to close (Mauritius), run for eco-tourism only or just started as a small commercial production (Réunion)
Education, training & skills	-	See access to finance above
Public engagement	-	See access to finance above
Local cooperation	-	Competition with other coastal use (e.g. real estate development, for instance recently in Mauritius)

([42]) http://www.fao.org/fishery/rfb/siofa/en , latest access: 15 March 2017.

	Desalination	
	Drivers to growth	**Barriers for growth**
R&D	Lessons learned from Canary Islands desalination (using inshore wind energy in the production process) and at international level (for instance in Middle East)	-
Access to finance	Water request and water loss in the distribution networks: public service interest in the medium term for new technologies to produce freshwater	Production costs for the moment (
Infrastructure	Existing solutions (Mayotte, Seychelles, Mauritius and worldwide)	
Education, training & skills	Education and skills in EU outermost regions	Not identified
Public engagement	State and local authorities support	Priority axes of local public investment (e.g. in Mayotte, it used to be a second priority to develop a second desalination unit until a several freshwater shortages in Dec. 2016 — Feb. 2017 had the effect for the local public authorities to modify their priorities for public support to water access and plan, among other actions, the development of the second unit in 18 months
Local cooperation	Growing population may be a medium-term driver to develop desalination units in territories with demographic growth and seasonal freshwater shortages (e.g. Mayotte)	The operations shall minimise their impacts on the environment (consumption of energy and discharge of salt)

	Maritime transport	
	Drivers to growth	**Barriers for growth**
R&D	Worldwide and EU focus on technology development to replace partly fossil fuel for ship propulsion and generating electricity on board and on marine pollution by ships	-
Access to finance	Needs by Indian Ocean Islands for imports first	Private investors to ensure the financial viability of regional trade by short sea shipping (objective of public authorities but slow progress for several years) Regional market will limit the volumes of maritime transport in the sea basin in the medium to long term (not an issue in the short term yet)
Infrastructure	Good port network in the sea basin (several islands on maritime roads to ship goods to Asia, Southern Africa and Middle East)	Port competition to provide similar services to some extent (therefore complementary services to foster by sea basin public support)
Education, training & skills	Education and skills in EU outermost regions	-
Public engagement	Through development of ports in the different Indian Ocean countries including EU outermost regions (see ports) R&D development (see R&D above)	-
Local cooperation	The Indian Ocean sea basin is not having traffic issue at sea (compared to other sea basins like in the North Eastern Atlantic)	Competition with other maritime sectors at port (space management required at ports)

Maritime transport	
Drivers to growth	**Barriers for growth**
Maritime safety at sea: piracy attacks has been reduced by maritime surveillance	

	Ports (including dredging)	
	Drivers to growth	**Barriers for growth**
R&D	Not identified	Environmental sustainability
Access to finance	State and local authorities support	Port competition
Infrastructure	State and local authorities support	-
Education, training & skills	Not identified	Deep knowledge of the specific soil conditions: (skills to be developed at the sea basin) to adapt, if necessary and legally possible, the local regulation on environmental impact assessment and monitoring against international regulations (case in Réunion, see the specific annex)
Public engagement	State and local authorities support	Not identified
Local cooperation	Use of port infrastructure by several maritime sectors to land goods and people (tourism, fisheries, maritime transport for instance)	Competition of users from different maritime sectors at port (space and service management required at ports) especially for emerging ones (cruise tourism)

	Shipbuilding	
	Drivers to growth	**Barriers for growth**
R&D	Transfer of French expertise (main land to local shipbuilding) — ship of the future (low carbon)	-
Access to finance	See 'infrastructure' below	Investors to ensure the financial viability of the shipyards to maintain the activity
Infrastructure	Existing infrastructure (Mauritius, South Africa, Madagascar) to build ships (fishing vessels, patrol vessels, small maritime transport vessels)	Asian competition for large (but also small) ship building; for instance, workforce cost in Reunion or Mayotte higher than in other neighbour countries preventing growth of the sector in these two EU outermost regions
Education, training & skills	Not identified	Not identified
Public engagement	Globally support for maritime surveillance, maritime transport* ship building (*: for passenger traffic and national transport) and for industrial fishing vessels	
Local cooperation	Not identified (potential for building leisure boats at the sea basin level to investigate)	Space at port (competition of port users to manage and land availability to develop in some ports)

	Ship repair	
	Drivers to growth	**Barriers for growth**
R&D	Not identified	Not identified
Access to finance	Insular geography and natural coastal heritage– favour maritime lines to transport goods and leisure boating hence ship repair services at the sea basin level	Not identified
Infrastructure	Existing capacities for each type of vessel	Workforce cost in Réunion and Mayotte (see ship building above)
Education, training	Not identified	Risk of loss of local skills in the medium

& skills		term if ship repair localised in only a few locations in the sea basin
Public engagement	Usually, a public support through development of ancillary services when ports are developed (e.g. in Mayotte)	-
Local cooperation	A key ancillary service to sustain several maritime sectors using ports (cruise tourism, maritime transport and marine fishing)	-

	Coastal tourism	
	Drivers to growth	**Barriers for growth**
R&D	Presence of tourism boards in several countries in the sea basin (especially Indian ocean islands) fostering R&D in the sector	-
Access to finance	State and local authorities support	Logistics to reach the sea basin: costs compared to other sea basin with similar attractiveness
Infrastructure	State and local authorities support	Business validity for hotel upgrades and services delivery (viability of coastal tourism projects if meeting the market demand)
Education, training & skills	Not identified	Language skills: English and French speaking (at least these two languages) to respond to a global offer of tourisms. Other languages is recommended to be developed by employees in the sector
Public engagement	State and local authorities support	-
Local cooperation	When coastal tourism is present and grow in an area, its rollout effects is to naturally develop the global economy of this area by integrated development	-

	Cruise tourism	
	Drivers to growth	**Barriers for growth**
R&D	Not identified	Not identified
Access to finance	International cruise operators' willingness to develop an offer in the sea basin State and local public authorities support	Administrative burden — a concern by stakeholders interviewed: slow decision to obtain urgent public funds especially for SMEs (cruise tourism is having potential in the sea basin)
Infrastructure	State and local authorities support and coordination In EU outermost regions: presence of health services and waste management at port at international standards (an asset for cruise agencies and for international tourism at the sea basin level)	Regional competition Still a need to adapt infrastructure and services in ports in the sea basin to respond to the potential offer seen by international cruise operators
Education, training & skills	Not identified	Need of increasing the number of employees speaking several international languages to attract new customers
Public engagement	State and local authorities support	Public organisations should be encouraged to develop tourism as an economic sector (integrated approach as a supply chain) Coherence of public support within the sea basin requested by EU outermost region stakeholders when developing the sector
Local cooperation	Remark: complementarity of landscapes and biodiversity in the sea basin:	Not identified

Cruise tourism	
Drivers to growth	**Barriers for growth**
potential for an integrated offer at the sea basin level	

	Other marine renewable energies (MRE)	
	Drivers to growth	**Barriers for growth**
R&D	IOC support strategic axis 4 — Réunion experimental platform Presence of an internationally renowned laboratory on atmosphere and cyclones in Réunion (LACY ([43])) however fund required to run more projects and to communicate on the competency of the laboratory	R&D funding availability to continue (and if possible to increase) Lack of knowledge on situation within extreme weather conditions (characteristics of waves and currents within these conditions)
Access to finance	Low or free carbon energy production	SWAC technology cost for the time being but the technology is improving (see infrastructure)
Infrastructure	Some potential for SWAC use in the sea basin and to some extent for ocean thermal energy conversion (OTEC) after gaining experience on using MREs in the sea basin	Several technologies are at R&D stage (pilot projects) only
Education, training & skills	Education and skills available to some extent in the EU Outermost regions	Lack of high skills in the basin
Public engagement	Strong public engagement due to international policies to reduce use of oil and gas to respond to climate change	Production costs of energy generated from MREs is still high compared to terrestrial renewable energies and several project failures in the sea basin, which are a barrier to full public support
Local cooperation	Climate change adaptation and the need to reduce the dependence on fossil energy — being a limited resource — fostering use of MREs in the economies of all the sea basin countries	-

([43]) http://lacy.univ-reunion.fr/ , access: October 2016.

8.1.1 Synoptic view of legislative, non-legislative and financial growth drivers and barriers

The following table reports a synoptic view of main growth drivers and barriers grouping them under legislative, non-legislative and financial frames

Growth drivers			
ACTIVITY	LEGISLATIVE	NON-LEGISLATIVE	FINANCIAL
Fisheries and aquaculture	• Tuna fisheries management through IOTC: internationally binding conservation and management measures improving for sustainable tuna fishing • Driver for environmental sustainability: EU public fund not allowed to build new EU fishing vessels (Common Fisheries Policy rule) to avoid stimulating overcapacity and overfishing and favour sustainable fishing where the EU fleets operate	• Good infrastructure for tuna landing and processing in the sea basin (especially in Seychelles, Mauritius but also Madagascar) and for toothfish and associated species in Réunion • Availability of fisheries training and research centres in the sea basin (especially in Réunion and South Africa, smaller research units in other sea basin States) • Synergies with tourism (eco-tourism)	• Research and development: public engagement by the IOC, regional fisheries management organisations (IOTC and SIOFA), the EU and other development organisations (e.g. World Bank, French Development Agencies) • Public engagement by the civil society
Blue biotechnology		Availability of R&D facilities in Réunion: possibility of knowledge and skill transfer in the sea basin	Existence of EU and IOC project funding, for instance in Réunion
Extraction of salt		• Learning experience from international research and use of the technology of salt extraction by evaporation • Local market: Madagascar and potential one as a niche for locally produced salt in Réunion	
Desalination		• Existing desalination units in Mayotte (one) but mainly in Seychelles and Mauritius for touristic demand with drinkable waters (hotels) • Demographic population in the medium term in territories with demographic growth and seasonal freshwater shortages (e.g. Mayotte)	Public and private engagement in Seychelles, Mauritius and Mayotte
Maritime transport		• Several islands on maritime roads to ship goods to Asia, Southern Africa and Middle East • Growth in the sea basin to meet imports and transhipment needs first • Public engagement through port development and maritime surveillance	
Ports (including dredging)		• Growth in the sea basin to meet imports and transhipment needs first (see ports above) • Use of port infrastructure by several maritime sectors to land goods and people	Public engagement

Growth drivers			
ACTIVITY	LEGISLATIVE	NON-LEGISLATIVE	FINANCIAL
Shipbuilding	Transfer of French expertise (main land to local shipbuilding) — ship of the future (low carbon)	• Existing infrastructure (Mauritius, South Africa, Madagascar) to build ships (fishing vessels, patrol vessels, small maritime transport vessels) mostly • Worldwide and EU focus on technology development to replace partly fossil fuel for ship propulsion and generating electricity on board and on marine pollution by ships	
Ship repair		• Insular geography and natural coastal heritage enabling ship repair services to be present at the sea basin level • Existing repair capacities for each type of vessel at the sea basin level • A key ancillary service to sustain several maritime sectors using ports (cruise tourism, maritime transport and marine fishing) • Usually, a public support through development of ancillary services when ports are developed (e.g. in Mayotte)	
Coastal tourism		• Presence of tourism boards in several countries in the sea basin (especially Indian ocean islands) fostering R&D in the sector with public support • Rollout effects by naturally developing the global economy of an area where coastal tourism grows	
Cruise tourism		• International cruise operators' willingness to develop an offer in the sea basin • State and local public authorities support • In EU outermost regions: presence of health services and waste management at port at international standards (an asset for cruise agencies and for international tourism at the sea basin level) • Complementarity of landscapes and biodiversity in the sea basin	
Other marine renewable energies		• R&D: IOC support strategic axis 4 — Réunion as an experimental platform • Some potential for SWAC use for cooling in the sea basin • Climate change adaptation and the need to reduce the dependence on fossil energy — being a limited resource — fostering use of MREs in the economies of all the sea basin countries • Presence of an internationally renowned laboratory on atmosphere and cyclones in Réunion (LACY) however fund required to run more projects and to communicate on the competency of the laboratory • Education and skills available to some extent in the EU Outermost regions (the potential is still to be supported for growth)	• Climate change adaptation and the need to reduce the dependence on fossil energy — being a limited resource — fostering use of MREs in the economies of all the sea basin countries with a strong public engagement • SWAC technology: costly for the time being

Barriers to growth			
ACTIVITY	LEGISLATIVE	NON-LEGISLATIVE	FINANCIAL
Fisheries and aquaculture	Fisheries: • Risk of unfair competition if fleet development rules and fishing management rules are different between fishing States targeting the same fish stocks in the sea basin • In case of poor integrated management of the different maritime sectors: competition with other maritime sectors (cruise tourism, goods landed by containers), especially in ports	Fisheries: • Fisheries knowledge in terms of stock status, in particular for non-tuna species and stocks exploited by small-scale fisheries • Poor landing infrastructure in coastal rural areas to land wild fish (e.g. Madagascar) Aquaculture: • Sea basin far from high value market impacting the development of large scale marine aquaculture production (at least in the short term) • Cyclonic conditions: costs of cyclonic resistant infrastructure to farm marine fish in open sea	Fisheries: • Heavy investment requiring robust operators to renew industrial fishing vessels, which are ageing (attraction of foreign direct investment necessary)
Blue biotechnology		• Still absent or low presence of blue biotechnology R&D in the sea basin with the exception of Réunion • High skill availability almost absent (except in Réunion to some extent)	
Extraction of salt		• Sea basin market of salt met by imports of salt whose origin is outside the sea basin: except Madagascar, other sites in the sea basin are now either going to close (Mauritius), run for eco-tourism only or just started as a small commercial production (Réunion) • Competition with other coastal use (e.g. real estate development, for instance recently in Mauritius)	Salt extraction by solar evaporation: a maritime sector with lower visibility than others for public support
Desalination	For an environmentally sustainable production, the desalination units shall minimise their impacts on the environment (consumption of energy and discharge of salt)		• Priority axes of local public investment • Dependency on public support for large scale commercial production (for the moment)
Maritime transport		Competition between countries for transhipment	Private investors to ensure the financial viability of regional trade by short sea shipping (objective of several public authorities but slow progress for several years)
Ports (including dredging)	Lack of knowledge of the specific soil and tropical conditions to develop ports (skills to be developed at the sea basin): to adapt, if necessary and legally	Competition of users from different maritime sectors at port (space and service management required at ports)	

Barriers to growth			
ACTIVITY	LEGISLATIVE	NON-LEGISLATIVE	FINANCIAL
	possible, the local regulation on environmental impact assessment and monitoring against international regulations (case in Réunion, see the specific annex)		
Shipbuilding	Globally support for maritime surveillance, maritime transport* ship building (*: for passenger traffic and national transport) and for industrial fishing vessels	• Asian and South African competitions for large (but also small) ship building • Competition at the sea basin level based on workforce cost difference • Space at port (competition of port users to manage and land availability to develop in some ports)	Investors to ensure the financial viability of the shipyards to maintain the activity
Ship repair		• Workforce cost in Réunion and Mayotte • Risk of loss of local skills in the medium term if ship repair localised in only a few locations in the sea basin	
Coastal tourism		• Language skills: employees in the sector shall speak at least both English and French to respond to a touristic global offer • Logistics to reach the sea basin: costs compared to other sea basin with similar attractiveness	
Cruise tourism		• A need to adapt infrastructure and services to cruise ships in ports in the sea basin • Public organisations should be encouraged to develop tourism as an economic sector (integrated approach as a supply chain) • Lack of coherence of public support within the sea basin (requested by EU outermost region stakeholders when developing the sector) • Low number of local employees speaking several international languages to attract new customers	Slow decision to obtain urgent public funds especially for SMEs (cruise tourism is having potential in the sea basin)
Other marine renewable energies (than offshore marine wind farm)		• Lack of knowledge on situation within extreme weather conditions (characteristics of waves and currents within these conditions) • Several technologies are at R&D stage (pilot projects) • Lack of high skills in the basin	Production costs of energy generated from MREs is still high compared to terrestrial renewable energies and several project failures, which is a brake to concrete public support

9. Future potential for mature and pre-developing activities

9.1 Fisheries and aquaculture

Marine fisheries

- **Potential for employment:**
  - Tuna fishing: commercial tuna, apart from yellowfin tuna, are currently not subject to overfishing and not overfished (IOTC) ([44]). Several countries and territories have fleet development plans (for instance Mayotte). Fleet development is possible to some extent provided that rigorous anticipation and monitoring of their effect on stocks and local communities fishing the same species be applied (for instance in Comoros). There is therefore some potential for additional jobs in the sector in the medium-long term.
  - Tooth fish fishing: currently the sector is mature and potential for employment is likely to be low. Some investigations to exploit new fishing grounds near the TAA waters are running (see annex 'Mayotte');
  - Other stocks: still some potential of employment to diversify sources of fish for current operators in the sea basin

- **Potential for value added:** low to medium (medium for value-addition in the post-harvest supply chain)

- **Potential for competitiveness**:

Availability of human resources adequately skilled.	Artisanal fishermen to be professionalised (situation in almost all countries with an existing marine fishing sector).
Potential for attracting FDI.	To some extent (investors knowing the sector and prone to invest when new opportunities arise – for instance targeting new stocks in the sea basin).
Availability of public funds.	Availability (EU and the World Bank for instance) for sustainable development of the sector.
Availability of support sectors / facilities (e.g. Research, infrastructures, etc.).	Landing infrastructure and sites usually require improvements a) to respond to national and international standards in terms of seafood hygiene and b) to minimise fish waste.
Potential in terms of positioning in the international market.	The sector is globally competitive Potential is foreseen on national and sea basin markets rather than focusing first on international markets (requiring international seafood safety standards, quality production to meet niche markets such as eco-label products or large-scale volumes to compete).

Marine aquaculture

- **Potential for employment:**
  - Low in the short term, due to technology and production costs required to produce high volume of farmed fish, competition with imported products and wild fish caught by local fishermen at competitive price for the time being and at least in the short term.

([44]) http://lacy.univ-reunion.fr/ , access: October 2016. http://www.iotc.org/science/status-summary-species-tuna-and-tuna-species-under-iotc-mandate-well-other-species-impacted-iotc , access: 25 November 2016.

- Some potential for current farmed fishing (Mauritius): to meet the sea basin demand and sell products at premium price as a niche outside the sea basin (red drum production)
- Maintenance of employment: currently the situation in shrimp farming in Madagascar being in competition with farmed shrimp from South-Eastern Asia (Vietnam and Thailand for instance).

- **Potential for value added:**

Low (similar situation as potential for employment). Value-addition is currently in place on shrimp. Value-addition is alike employment to be maintained in an internationally competitive sector

- **Potential for competitiveness**:

Availability of human resources adequately skilled.	Marine aquaculture technicians are high skilled meaning that import of specific human resources is needed at least in the short term then maintenance and transfer of these skills, at least part of them, to local resources are necessary within a continuous and long-term approach.
Potential for attracting FDI.	Low for the time being.
Availability of public funds.	Yes. The focus on research, development and innovation (RDI) is however highly relevant to develop marine aquaculture in the sea basin.
Availability of support sectors / facilities (e.g. Research, infrastructures, etc.).	Availability of research facilities in Réunion and in South Africa with a network for these research centres for collaboration worldwide.

9.2 Blue biotechnology

- **Potential for employment:** medium**,** high potential regarding the biodiversity but the sector is only at research and development level in the sea basin. The volume of local employment is likely to remain non-substantial compared to maritime sectors like tourism, fisheries and the ports

- **Potential for value added:** medium regarding the targeted sectors (cosmetics, medicine, green energy). Local value-addition is the objective of developing this sector but part of the blue-biotechnology raw material produced locally may continue to increase their value after being exported to processing hubs of biotechnology companies based in the US, Asia and Europe and where the international markets are.

- **Potential for competitiveness**: medium.

Availability of human resources adequately skilled.	Low but existing research structure in Réunion for instance.
Potential for attracting FDI.	High regarding the targeted sectors (cosmetics, medicine, etc.).
Availability of public funds.	Medium for R&D
Availability of support sectors / facilities (e.g. Research, infrastructures, etc.).	Existing research structure in Réunion, better structuration at regional level.

Potential in terms of positioning in the international market.	High regarding the biodiversity of the sea basin.

9.3 Extraction of salt

- **Potential for employment:** low to mature sector declining due to land competition with other sectors such as real estate development (for instance in Mauritius).
- **Potential for value added:** low (only created by indirect value-addition).
- **Potential for competitiveness**:

Availability of human resources adequately skilled.	Availability of human resources in current salt pond sites.
Potential for attracting FDI.	Low.
Availability of public funds.	Low except for maintaining traditional operations for ecotourism (for instance in Réunion and Mayotte).
Availability of support sectors / facilities (e.g. Research, infrastructures, etc.).	International research is not focusing on this sector generally.
Potential in terms of positioning in the international market.	Current commercial productions targeting national demand.

9.4 Seabed mining

- **Potential for employment:** low.
- **Potential for value added:** low to medium.
- **Potential for competitiveness**: not identified.

Availability of human resources adequately skilled.	Low.
Potential for attracting FDI.	Low to medium (highly dependent of minerals and metals market costs).
Availability of public funds.	Low to medium.
Availability of support sectors / facilities (e.g. Research, infrastructures, etc.).	Low (however a relevant research institute in the basin: IFREMER in Réunion).
Potential in terms of positioning in the international market.	Low.

9.5 Desalination

- **Potential for employment:** low in operation- medium for building. The technology requires a small workforce to operate and is still costly for large-scale development. In the short term, other green energy is expected to be developed in priority with public support in the sea basin except to respond partly to need of freshwater during severe seasonal draughts where countries or the customers will

have the 'purchasing power' to produce and operate desalinated water (e.g. in Mayotte, see the specific annex or Seychelles). In the medium term, the technology may be less costly to operate and is likely to meet the needs of a growing population on islands with a very limited supply of clean drinking water therefore providing local employments.

- **Potential for value added:** low-medium. The value-addition for the countries developing such a technology would be only indirect by purchasing some equipment locally and using local services

- **Potential for competitiveness**: low (local production with existing solutions).

Availability of human resources adequately skilled.	Low: imported technologies and manpower for building.
Potential for attracting FDI.	Medium: tourism support.
Availability of public funds.	High in the medium term: growing and young population may increase demand for such technology, which may also be available at low operation costs.
Availability of support sectors / facilities (e.g. Research, infrastructures, etc.).	Few operators in the sea basin (one desalination site in Mayotte for instance, none in Réunion).
Potential in terms of positioning in the international market.	Low, meeting local demand for drinking water.

9.6 Maritime transport

- **Potential for employment**: medium regarding the insular geography and the sea basin population increase that naturally creates a demand for goods then for maritime transport for goods within the basin. However, part of the crew is recruited outside the sea basin (especially for international maritime links). The potential for employment could increase by attracting maritime transport companies to recruit seaman located in the basin.

- **Potential for value added:** medium — the sector provides indirect value-addition to the sea basin countries by using ports facilities and services.

- **Potential for competitiveness**: medium.

Availability of human resources adequately skilled.	Medium.
Potential for attracting FDI.	Medium/high (CMA CGM for instance in Réunion in terms of non-local investor).
Availability of public funds.	Availability related to port development in the sea basin and regional short sea shipping (the basin market for goods exchanges is to be improved, short sea shipping in the sea basin.
Availability of support sectors / facilities (e.g. Research, infrastructures, etc.).	Cf. Ports below.
Potential in terms of positioning in the international market.	Medium/high regarding the regional market and its location on the Asian — African maritime transport road.

9.7 Ports (including dredging)

- **Potential for employment:** medium/high. Port activities are having the potential for growth by regional and international trade of goods with private and public engagements. Low (stevedores) to high skilled employments will be needed including the need to attract educated specialists in environmental impact assessment and monitoring of tropical ports with a robust knowledge in EU and international legislations and the local environment (preferably remaining in the sea basin).

- **Potential for value added:** medium. Increase the value of imported for re-export may increase port activities but this potential is still under investigation by setting free trade areas near ports (e.g. in Réunion and Mayotte).

- **Potential for competitiveness**: medium.

Availability of human resources adequately skilled.	Medium.
Potential for attracting FDI.	Low (public infrastructures).
Availability of public funds.	Medium (supporting action for attractiveness).
Availability of support sectors / facilities (e.g. Research, infrastructures, etc.).	Medium/high: port infrastructure development is in the high agenda for public organisations such as EU, IOC and UNECA.
Potential in terms of positioning in the international market.	In accordance with maritime transport trend (see 'maritime transport' above).

9.8 Ship building

- **Potential for employment:** low-medium in the basin, employment is meeting the local demand to build fishing vessels (mostly for fleet renewal in the sea basin), patrol vessels (ibid) and small maritime transport ships first and the potential is expected to remain. Large shipyards (for super-tankers and cruise ships) are located outside the sea basin. Potential for employments to build leisure boat in the sea basin is to be investigated however the ship building sector is not expected to grow substantially.

- **Potential for value added:** low to medium**,** value-addition by ancillary services and purchase of equipments locally and in the basin to operate ship building sites

- **Potential for competitiveness**: low regarding the growing Asian market for large ships, medium for other ships and vessels.

availability of human resources adequately skilled.	Medium: in South Africa and Madagascar for instance.
potential for attracting FDI.	low — regional market validity.
availability of public funds.	Availability (for instance, recently, an EU support to build two new ferries for the public transport company in Mayotte — one has been recently delivered).
availability of support sectors / facilities (e.g. research, infrastructures, etc.).	To some extent availability in some countries (e.g. in Madagascar when maritime transport ships have been recently built to respond to a demand in Mayotte).
potential in terms of positioning in the international market.	Low for large ship building (medium for other).

9.9 Ship repair

- **Potential for employment:** low to medium regarding the insular geography of the sea basin countries and the need of maintaining a minimum level of ship repair services in all ports in the sea basin. The volume of additional employees with different skills will increase at the sea basin level to meet the growth of repair activities for maritime transport but the increase is not expected to be substantial as heavy repair and maintenance are achieved systematically in the same ports in the sea basin (or beyond).

- **Potential for value added:** low-medium. The value-addition for the territories is and is likely to continue to be based mainly on indirect value-addition (VA of repair services using local equipment and services for instance) however, due to the insular geography and the remote location of the countries in the sea basin, parts of the equipment ought to be imported (equipment purchased outside the sea basin).

- **Potential for competitiveness**: low at global scale- medium for regional market

Availability of human resources adequately skilled.	Medium.
Potential for attracting FDI.	Medium market dependent and competition within the sea basin.
Availability of public funds.	Low (mainly growing by private sector initiatives first). However, light ship repair services are important and the public sector is aware of this permanent need (see annexes 'Réunion' and 'Mayotte').
Availability of support sectors / facilities (e.g. Research, infrastructures, etc.).	Medium (port infrastructure with different types of capacities).
Potential in terms of positioning in the international market.	Low to medium in accordance with workforce cost.

9.10 Coastal tourism

- **Potential for employment:** medium/high Coastal tourism provides an important amount of employments in the sea basin in the sea basin. The development of employment is to be expected as there is still some potential for growth to professionalise the sector and by an integrated development approach of the tourism sector at national and sea basin level.

- **Potential for value added:** medium/high; Value-addition can increase by improving the quality and diversity of the offers in the sea basin in terms of infrastructure, logistics, accommodation and staff skills and providing diverse and complementary coastal tourism activities within the sea basin

- **Potential for competitiveness**: medium

Availability of human resources adequately skilled.	Medium to high (low in Comoros and Mayotte).
Potential for attracting FDI.	Medium to high (to be mitigated for security and safety reasons and tourists' origins).
Availability of public funds.	Low (private activities) mainly on port facilities
Availability of support sectors / facilities (e.g. Research, infrastructures, etc.).	Medium.
Potential in terms of positioning in the international market.	Medium to high (regarding global touristic offer to be balanced by travel cost.

9.11 Cruise tourism

- **Potential for employment:** medium for indirect employment should there be activities at cruise stops (in the Indian Ocean countries where ships stops) with port management authorities and other port users be willing to cooperate to develop the sector showing optimistic signs of growth

- **Potential for value added:** low to medium with global touristic offer of services and long stops at port calls (integrated development of the economic sector)

- **Potential for competitiveness**: low/medium regarding global market

Availability of human resources adequately skilled.	Low to medium.
Potential for attracting FDI.	Medium to high (touristic destination) -signs of increasing interest of some cruise companies to provide cruises in the sea basin.
Availability of public funds.	Low to medium (cf. Port) some countries show extreme willingness to develop cruise tourism, conflict of port users for port facilities availability is to be managed to develop the sector harmoniously.
Availability of support sectors / facilities (e.g. Research, infrastructures, etc.).	Low to medium (cf. Port).
Potential in terms of positioning in the international market.	Low to medium regarding cruise costs compared to other international offers.

9.12 Marine wind energy

In the short term, marine wind energy has low potential due to cyclonic risks in the basin: the technology is currently not adapted to such conditions for offshore wind farming (with the exception by land reclamation in Mauritius and public engagement in Seychelles). Offshore marine wind farming may have higher potential in the islands sea basin in the medium term when technology would have been supported to build wind farms supporting cyclones and by a robust maritime spatial planning to avoid conflict users including the marine biodiversity protection.

- **Potential for employment:** low in the short term (see para. Above), and medium in a long-term perspective regarding technologies level of readiness for commercial applications therefore creating some local jobs for local management, maintenance, repair services and local RDI. It will be medium and high-skilled jobs with long term employment. However, the automation of the MRE technologies will not require a large number of local employment at the sea basin level

- **Potential for value added:** low-medium, the MRE technologies will be developed outside the sea basin level (at least a perspective that may not change in the medium term). Some value-addition by local management, maintenance and repair services and RDI can be expected in the future.

- **Potential for competitiveness**: low (cyclonic risk versus technologies)

Availability of human resources adequately skilled.	Low (emerging sector).
Potential for attracting FDI.	Low to medium (public support required regarding energy price and attracting FDI).
Availability of public funds.	Medium to high (low carbon and climate change adaptation policies in several countries and at the sea basin level).
Availability of support sectors / facilities (e.g. Research, infrastructures, etc.).	Low: research facilities in Réunion mainly Infrastructure: terrestrial wind energy in the sea basin mostly such as in Réunion.
Potential in terms of positioning in the international market.	Low-medium: some opportunities to develop international research on MREs in extreme weather conditions (cyclones) however the potential is low in terms of industrial development of MREs to meet the international demand for commercial technologies with competitive costs.

9.13 Other marine renewable energies than marine wind farming

- **Potential for employment:** medium. Some technologies seeing potential for commercial production (SWAC). There a medium potential in a medium-term perspective by increase of a diversity of MRE technologies financially suitable for commercial applications therefore creating some local jobs for local management, maintenance, repair services and local RDI. It will be medium and high-skilled jobs with long term employment. However, the automation of the MRE technologies will not require a large number of local employment at the sea basin level.

- **Potential for value added:** low-medium, the MRE technologies will be developed outside the sea basin level (at least a perspective that may not change in the

medium term). Some value-addition by local management, maintenance and repair services and RDI can be expected in the future.

- **Potential for competitiveness**: low (medium for RDI) in accordance with regional resources

Availability of human resources adequately skilled;	Low (emerging sector)
Potential for attracting FDI;	Low to medium (public support required regarding energy price and attracting FDI)
Availability of public funds;	Medium to high (Low carbon and climate change adaptation policies in several countries and at the sea basin level)
Availability of support sectors / facilities (e.g. Research, infrastructures, etc.)	Low: research facilities in Réunion mainly
Potential in terms of positioning in the international market	Low tomedium: some opportunities to develop international research on MREs in extreme weather conditions however the potential is low in terms of industrial development of MREs to meet the international demand for commercial technologies

9.14 Synoptic view of the future potential by activity

The following table reports a synoptic view of main future potential for each activity grouping them by potential for 'employment', 'value added' and 'competitiveness' gaps.

ACTIVITY	POTENTIAL FOR EMPLOYMENT	POTENTIAL FOR VALUE ADDED	POTENTIAL FOR COMPETITIVENESS
Fisheries and aquaculture	Fisheries: Medium • Several countries and territories in the sea basin have tuna fishing fleet development plans (for instance Mayotte). Fleet development is possible to some extent provided that rigorous anticipation and monitoring of their effect on stocks and local communities fishing the same species be applied; • Diversification of wild species caught to meet the market demand is under investigation Aquaculture: low to medium • The insular situation of the countries in the sea basin and the weather conditions prevent the possibility to develop large scale marine aquaculture and producing low cost products • Some potential for some additional employment by a steady growth in Mauritius (red drum production) • Some potential may exist to	Fisheries and aquaculture: medium • Exponential growth of the processing sector to add value to the raw products is not expected but improving local value addition is likely to be an objective of operators catching fish then selling fish at national level and within the sea basin	Fisheries: medium: • Wild fish are a limited resource and the local industrial harvest sector is therefore competitive • The concrete development of the tuna fishing fleet will be dependent on the financial capacity and commitment of private investors • Public funding is committed to support the sector to be environmentally sustainable Aquaculture • The sector is highly competitive to meet the international market and to provide fish at suitable price for the local market (except for premium products as niche market) • Public funds are committed to support the sector by providing local research and development to develop independent productions, which are financially and environmentally sustainable

ACTIVITY	POTENTIAL FOR EMPLOYMENT	POTENTIAL FOR VALUE ADDED	POTENTIAL FOR COMPETITIVENESS
	provide employments by producing green algae (as biofuel) and spirulina (as micronutrients) in Réunion • Offshore aquaculture is still costly to operate and provide fish at competitive market price		
Blue biotechnology	Medium • High potential regarding the biodiversity but the sector is only at research and development level in the sea basin. • The volume of local employment (high skilled jobs) may grow but not providing a large amount of employments compared to maritime sectors like tourism, fisheries and the ports.	Medium • Local value-addition is the objective but part of the blue-biotechnology raw material produced locally may continue to increase their value after being exported to processing hubs of biotechnology companies based in the US, Asia and Europe and where the international markets are.	Medium • There is high potential due to the marine biodiversity of the sea basin • However: ○ The sector is only at a RDI stage in the sea basin and the RDI is based in Réunion mostly; ○ Structuration of the sector to be competitive for RDI and, as a second objective, for commercial applications is necessary to foster the full biodiversity potential of the sea basin
Extraction of salt — salt evaporation pond production	Low: • A mature sector declining due to land competition with other sectors such as real estate development (for instance in Mauritius).	Low • Only created by indirect value-addition	Low: • The sector is mostly maintained by public support for eco-tourism in Réunion and Mayotte • Public engagement prioritise supports to other maritime sectors
Seabed mining[23]		Low: • Non-existent or marginal sector • However, some local research on sea bed mining exploitation (Ifremer in Réunion).	
Desalination	Low-medium: • A technology requiring few employees to run and some local employments to build • Large scale commercial production not expected in the sea basin, at least in the short term: developed to respond to seasonal water shortages in countries having the purchasing power to run a technology which is still costly to build and operate (for instance in Mayotte, Seychelles and Mauritius)	Low-medium: • Indirect-value addition only (distribution of freshwater and use of local services and equipment)	Low: • Developing the technology in the sea basin is to provide waters locally by purchasing the technical equipment outside the sea basin

ACTIVITY	POTENTIAL FOR EMPLOYMENT	POTENTIAL FOR VALUE ADDED	POTENTIAL FOR COMPETITIVENESS
Maritime transport	Medium: • Potential of the sector for growth by population increase in the sea basin and by transhipment activities for ports being the international maritime roads for transport of goods • Part of the crew is recruited outside the sea basin (especially for international maritime links). The potential for local employment could increase by attracting maritime transport companies to recruit seaman located in the basin.	Medium: • The sector provides indirect value-addition to the sea basin countries by using ports facilities and services in the sea basin	Medium: • Location of some countries in the sea basin on the international maritime roads for transport of goods favouring transhipment activities (e.g. Mayotte, Réunion, Mauritius) • Public engagement through port development and short sea shipping development • Trade within countries in the sea basin is still to be improved (short sea shipping)
Ports (including dredging)	Medium-high: • By the growth potential of the port activities in the sea basin (import and transhipment of goods first)	Medium • Indirect value-addition to the countries • Adding-value to products in sea basin countries may increase port activities by this potential is still under investigation.	Medium-high: • Port infrastructure development is in the high agenda for public organisations such as the EU.
Shipbuilding	Low-medium: • The potential is expected to remain in the sea basin to build fishing vessels, patrol vessels and small transport ships (renewing them) • Potential for employments to build leisure boat in the sea basin • Nevertheless, the ship building sector is not expected to grow substantially in the sea basin.	Low-medium: • The value-addition for the territories is likely to continue to be based on indirect value-addition (sector with low direct value-addition)	Low-medium • Competitive within the sea basin by presence of several shipyards to build ships however Not competitive for large ships
Ship repair	Low-medium: • A slight increase of employments to expect by the progressive growth of maritime transport activities in the sea basin • By a need to maintaining a minimum level of ship repair services in all ports in the sea basin and the presence of 'regional hub' for ship repair facilities in the sea basin	Low-medium: • The value-addition for the territories is likely to continue to be based on indirect value-addition (sector with low direct value-addition)	Low-medium: • A competitive market at the sea basin level with regional hub to provide heavy repair services (absence of potential at international level) • Sector growing by private sector initiatives mostly (with some exceptions)
Coastal tourism	Medium-high: • Coastal tourism provides an important amount of employments in the sea basin. The development of employment is to be	Medium-high: • VA can be increase by improving the quality of the offer in the sea	Medium: • In the short term, security and safety issues in countries has had negative impact on attracting tourists (for instance in Madagascar

ACTIVITY	POTENTIAL FOR EMPLOYMENT	POTENTIAL FOR VALUE ADDED	POTENTIAL FOR COMPETITIVENESS
	expected as there is still some potential for growth to professionalise the sector and by an integrated development approach of the tourism sector at national and sea basin level.	basin and providing diverse and complementary coastal tourism activities within the sea basin through an integrated development of the tourism economy at the sea basin and the national levels	and in Réunion (sharks)) • Travelling to the sea basin is still costly compared to other destinations with similar geographical conditions; however public engagements are developing a competitive logistical offer to reduce this barrier for growth
Cruise tourism	Medium: • Indirect local employment that could benefit from cruise ship calls. Conditions: local cooperation at the ports to develop cruise tourism	Low to medium: • Medium with global touristic offer of services and long stops at port calls (with an integrated development of the economic sector)	Low to medium: • Competitive market at international level however public and private interest to develop the sector in the Indian Ocean sea basin
Marine wind energy	Low in the short term: • The technology is to be adapted to extreme weather conditions in the sea basin (in general, priority is given by the public sector in developing other MREs apart from Seychelles and Mauritius (however for the latter by land reclamation)	Low in the short term: • The MRE technologies are developed outside the sea basin level. Some value-addition by MRE local management, maintenance and repair services of the MRE tools can be expected in the medium-long term by an adapted technology to the sea basin level.	Low (medium for RDI): • A potential to develop international research on marine wind energy in extreme weather conditions however the potential is low in terms of local industrial development of MREs to meet an international demand • Public engagement to attract private investors (e.g. Mauritius) to develop MREs at commercial level at the sea basin level to reduce the carbon energy dependency, adapt to climate changes
Other marine renewable energies	Medium: • Some technologies seeing potential for financially suitable commercial production (SWAC) and with the perspective of a diverse panel of improved and locally-adapted technologies to be financially suitable for local large scale commercial applications therefore creating some local jobs (medium and high-skilled long-term employments) for local management, maintenance and repair services and local RDI	Low-medium: • The MRE technologies are developed outside the sea basin level. Some value-addition by MRE local management, maintenance and repair services of the MRE tools can be expected in the future.	Low (medium for RDI): • A potential to develop international research on MREs in extreme weather conditions however the potential is low in terms of industrial development of MREs to meet the international demand for commercial technologies • Public engagement to develop commercial MREs at the sea basin level to reduce the carbon energy dependency, adapt to climate changes by attracting private investors

10. Investments opportunities

At sea basin scale, investment opportunities are showing potential mainly for:

- costal and Cruise tourism with a global, diversified and integrated offer to discover the sea basin heritage (with longer stops of cruise ships at ports to develop coastal tourism further);
- port development to sustain the progressive growing demand of goods trade in the next ten years and to accompany other maritime sectors to grow with synergies in port facilities and services definition and uses such as for cruise tourism;
- blue technologies: for research, development and innovation (RDI) at least in the medium term;
- fisheries and aquaculture to respond to local and international ([45]) demand for food proteins with potential opportunities to diversify the panel of species to produce and with synergies with other sectors blue technologies for instance;
- marine renewable energy:
  - for RDI in adapting technologies to tropical and extreme weather conditions and for commercial applications adapted to the conditions of the sea basin (cyclones, reefs and marine biodiversity protection);
  - by an integrated development approach with non-marine renewable energies and other sectors (SWAC — Blue technologies);
- ship repair as an ancillary service for several maritime sectors (tourism, port activities and fisheries mostly).

11. Most relevant maritime activities at sea basin level

<u>The selected ([46]) maritime sectors are:</u>

1. fisheries and aquaculture;
2. blue biotechnology: blue biotechnology has been selected for its possible future potential while other relevant sectors could have been analysed too such as salt extraction, oil and gas extraction and seabed mining;
3. ports (including dredging): maritime transport is indirectly analysed through this maritime sector;
4. shipbuilding;
5. ship repair;
6. coastal tourism;
7. cruise tourism;
8. other MREs than marine wind farms: although commercial potential for marine wind farms are investigated by Seychelles and Mauritius. In the EU outermost regions, further oceanographic data will be required to develop offshore wind farms resistant to extreme weather conditions during cyclone occurrence in Réunion (other MRE technology are developed for commercial use). In Mayotte, managing users' conflict must be investigated too for the Mayotte waters are a marine protected area (see annexes 'Réunion' and 'Mayotte'). The Chapter focus here on other MREs then.

([45]) For tuna and associated species mostly but also toothfish.
([46]) Selection approach: 8 'relevant' maritime activities have been identified based <u>first</u> on the EU outermost regions[46] commercial interests and needs or potential ones in the selected maritime sectors in terms of trade, provision of services and works and second on potential for third country socio-economic development.

11.1 Fisheries and aquaculture

- **Workforce characteristics:** at present the main industrial fisheries are dominated by overseas crew. For instance, most of the EU purse seiners have French or Spanish crew (depending upon the flag), supplemented by largely West African crew. Even the large (12 vessel) Seychelles-flagged purse seine fleet is mainly Spanish and West African crew. Whilst skill gaps are an important element of this (there is limited nautical training capacity in the region, the main reason for the lack of regional crew is a combination of cultural and socio-economic reasons. There are better paid and more comfortable employment options in local tourism and associated economic activities. One area where skill development is potentially worth pursuing is encouraging artisanal fishers to diversify their activities, both into 'pescatourism' as well as off-reef fishing. The latter has challenges e.g. the use of larger vessels and more complex gear, but has the potential to provide good livelihood opportunities, especially when focusing on high value species using low impact gear e.g. mechanised handlines. For aquaculture, where every different skill sets are needed compared to wild capture fisheries, there is a real lack of capacity within the region, especially for the more skilled work e.g. in hatcheries and laboratories. Given the current focus on offshore aquaculture development in Mauritius and efforts to restart Mayotte's drum production, investment into capacity-building in aquaculture is a real option.
- **Administrative burden**: fisheries are increasingly heavily regulated at both national and regional levels, both to improve sustainability (e.g. reduce effort on pressure stocks) and combat illegal, unregulated, and underreported (IUU) fishing. However, there are no major administrative barriers to fisheries development.
- **Restrictive practices:** no restrictive barriers have been identified that might seriously harm the development of the sector. As mentioned above, industrial fishing practises are increasingly regulated, with limits on Fish Aggregating Device (FAD) use, very limited bottom trawling (which can be environmentally destructive) in the region (apart from the west coast of Madagascar). IUU fishing — one a major issue in the region, is now less prevalent due to strong action taken at regional level on monitoring, control and surveillance.
- **Sources of finance:** Finance is a major issue across the sector. Fishing requires considerable investment, both in capital equipment (boats, vessels, etc.) as well as running costs (crew salaries, fuel, etc.). The longline feet in Reunion is now ageing, and is seeking both private and public sector support to do so. However pubic funding from the EMFF is designed not to increase fishing capacity, and thus is difficult to apply to vessel renewal.

11.2 Blue biotechnology

- **Workforce characteristics:** This is relativity new sector, not only in the sea basin, but also in general. It is a complex and varied field, requiring new and advanced skills in areas like biochemistry, genetics and marine ecology. Many of these skills are simply not available in the region, although there are developing academic and commercial research nodes in Reunion and Mauritius. It is likely that this industry will be dependent upon imported specialists for some time to come, especially if it remains a niche industry. There could be an opportunity to develop education and training for high skills at regional basin (that is transfer knowledge) based on Réunion R&D network.
- **Administrative burden**: no specific administrative burden has been identified. Given this is a niche small-scale but high value industry,

administrative burdens will be limited to restrictions on ethical testing and use of biological material, esp. biocides, if these even exist.

- **Restrictive practices:** not identified
- **Sources of finance:** genuine new initiatives in blue biotechnology are likely to be attractive to investors, so far as there has been a degree of robust business-planning and risk assessment. Some funds, such as the EU Horizon 2020 may also have opportunities for blue biotechnology research and start-ups. However, businesses seem to struggle during the second and third rounds of funding, lowering their growth potential. Some SMEs struggle to reach a certain critical mass, making them less attractive for investment by large established companies of their sector (Ecorys et al., 2012).

11.3 Ports (including dredging)

- **Workforce characteristics:** Ports have existed in the sea basin for a long time and there is now considerable local competence in most aspects of planning, operations and management. The Mauritius Port Authority is a good example of this, running a success full and strategically planned integrated port facility in Port Louis. Thus, most of the requisite skills are available. However, most pots are still dependent upon imported labour, inc. Port Louis and Port Victoria.

  With an improvement of skills, the heterogeneous costs of workforce with different costs increase the concurrency in the area for regional and international market.
- **Administrative burden**: EU regulations and standards harmonisation in Réunion and Mayotte are an administrative burden in the short term although in the medium term having positive effects to attract investors and customers (transport companies and cruise tourists, for instance). Indian Ocean Memorandum of Understanding (MoU) ([47]) for Port state control has been adopted by all bordering countries (latest revision in 2016).
- **Restrictive practices:** Environmental impacts of practices and exploitation, especially when land reclamation or dredge spoil dumping is involved.
- **Sources of finance:** Mainly based on public resources (State and local authorities).

11.4 Shipbuilding

- **Workforce characteristics:** the availability of workforce with different costs provides with a high level of concurrency in the area. Therefore, it is not developed except in South Africa and Mauritius. It also requires specialist skills, some of which are not available in the region. But there are specialist shipyards who have developed a high level of competence for locally important vessels e.g. small, specialist fishing and patrol vessels.
- **Administrative burden**: not identified. In Mauritius, there is very limited space for expansion of ship-building in Port Louis, and the MPA is unwilling to facilitate expansion, even into low use areas.
- **Restrictive practices:** Environmental impacts of the activities.
- **Sources of finance:** mainly based on private sector for maritime transport.

([47]) http://www.iomou.org/moumain.htm , access: August 2017.

11.5 Ship repair

- **Workforce characteristics:** the availability of workforce with different costs provides with a high level of concurrency in the area.
- **Administrative burden**: not identified.
- **Restrictive practices:** Environmental impacts of the activities (pollution risks).
- **Sources of finance:** Mainly from private sector but also from public sector (French Navy in Réunion for instance but also for maritime passenger transport).

11.6 Coastal tourism

- **Workforce characteristics:** Large availability in the sea basin with a large range of services. Again, has been developed of a long period with considerable state support and in skill development. But skill development will need to keep pace with industry development e.g. into eco-tourism, high-end niche tourism, etc. Short term efforts are on-going and should be enhanced in French mother tongue countries on foreign language (at least English) education to be in accordance with tourist's origins
- **Administrative burden**: EU legislation such as the Water Framework directive (Waste water treatment in Mayotte) or marine protected areas regulation
- **Restrictive practices:** Sustainability of activity and associated practices for marine environment protection, esp. considering freshwater use and sewage disposal. There is also the risk of poorly planned expansion into environmentally or culturally sensitive areas.
- **Sources of finance:**
  - Public sector for education, environmental issues, transport network, etc.
  - Private sector as it is a commercial activity.

11.7 Cruise tourism

- **Workforce characteristics:** In connection with port workforce and coastal tourism for touristic welcoming.
- **Administrative burden**: see coastal tourism.
- **Restrictive practices:** Sustainability of activity and associated practices for marine environment protection.
- **Sources of finance:** public and private partnership.

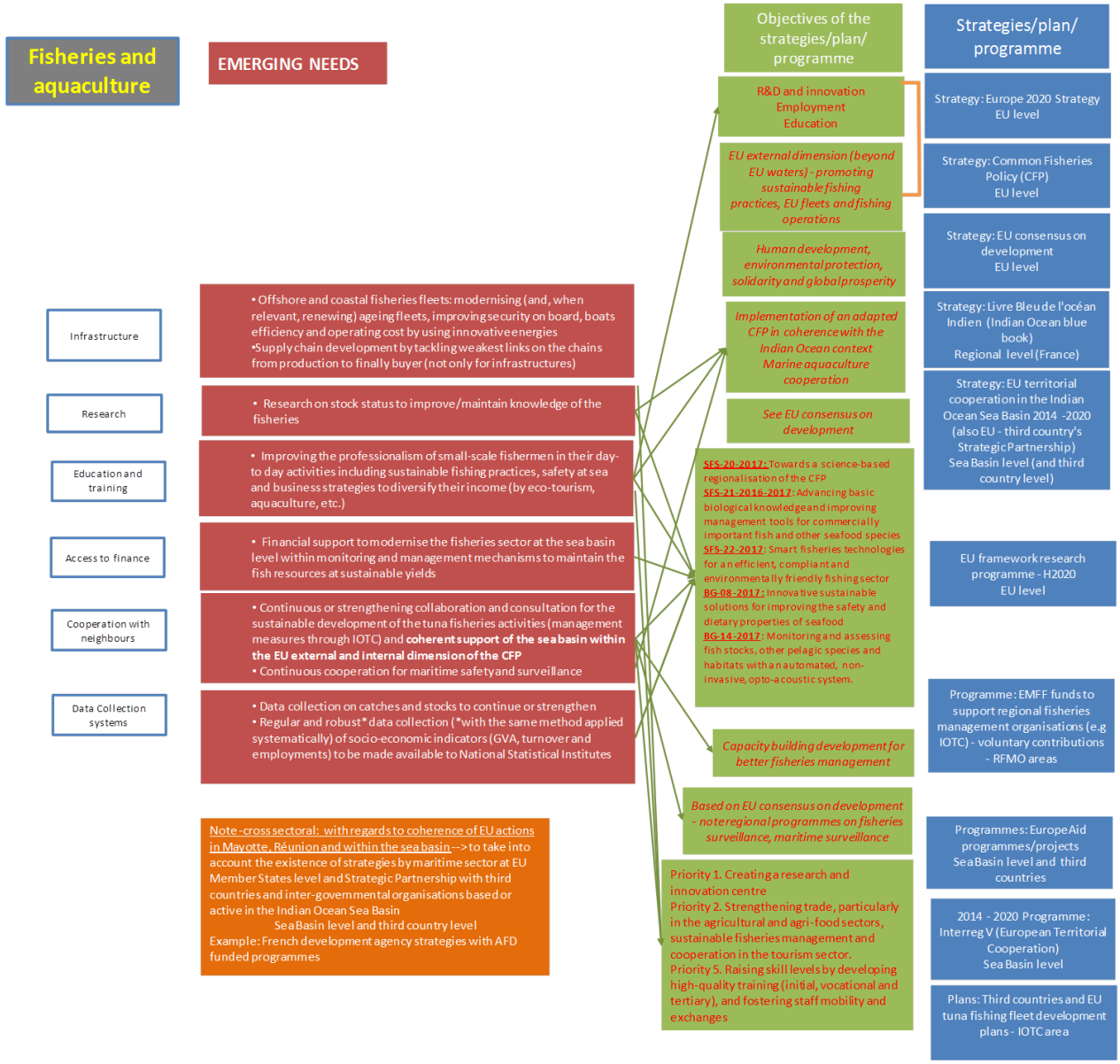
11.8 Other marine renewable energies than marine wind energy

- **Workforce characteristics:** Lack of high skilled workforce for the time being as it is an emerging sector. R& D availability in Réunion for experimentation.
- **Administrative burden**: Environmental impact assessment laws for settlement and production but also for cable for energy delivery.
- **Restrictive practices:** Environmental impact assessment laws and other maritime activities' compatibility.
- **Sources of finance:** Mainly public investment in accordance with public policies (COP); low carbon energy production and O&G independency.

Identifying public and private intervention

12. Potential financial, legislative and non-legislative measures for stimulating growth

Figure 2 — Logical diagram for fishery and aquaculture

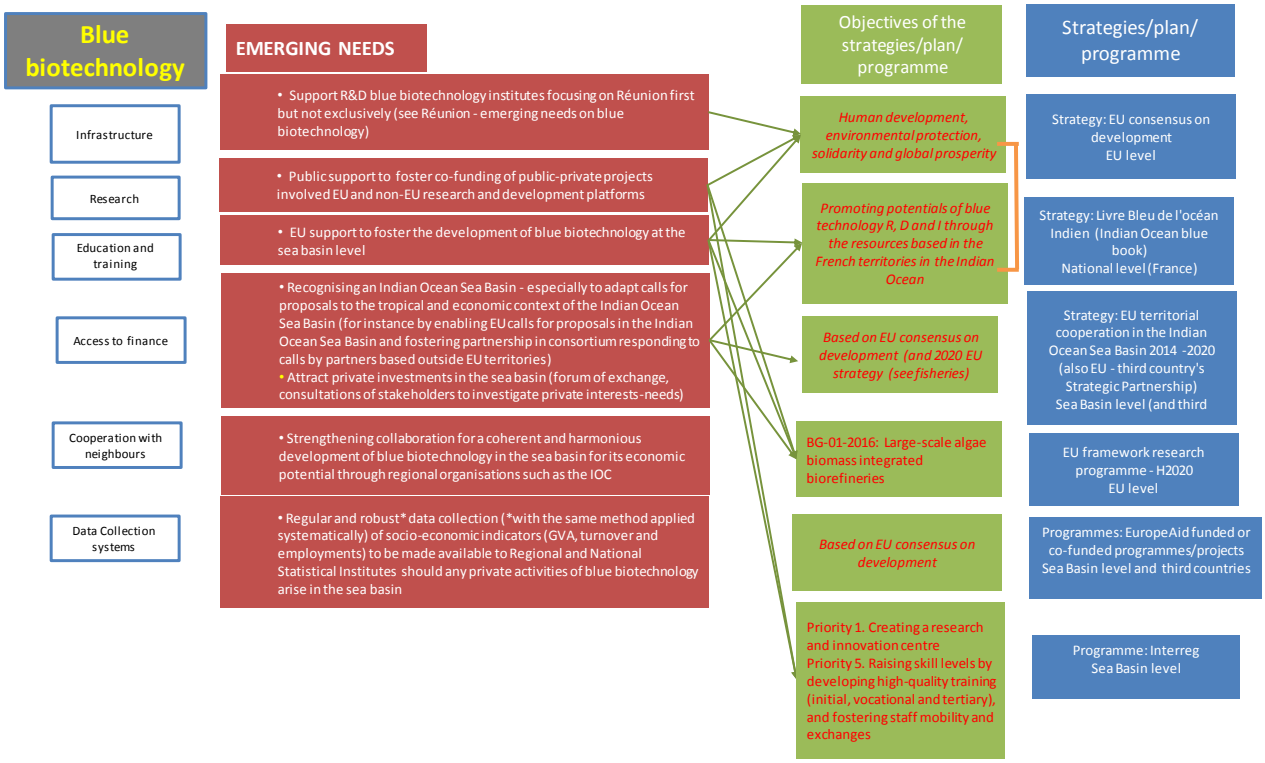


As the fishing fleet modernises, with implications for the shore-support the newer vessels will need, this is likely to required development of shore side infrastructure, such as quays and on-shore processing and marketing. Whilst most fisheries research in tuna fisheries are coordinated through IOTC, there is a need to develop **local centres of excellence for fisheries and aquaculture research**, especially for local stocks. As fishing fleets evolve and modernise, with greater emphasis on off-reef fishing and high value catches, this will require considerable upskilling of both crews as well as their on-shore counterparts. Investment into both capture fisheries and aquaculture is often regarded as high risk, and thus investors need to develop long-term, **sustainable business plans for vessel / fishing gear replacement**, as well as investment into new aquaculture technologies such as cage farm units, recirculation systems and feed mills.

The region's tuna and billfish purse seine longline fisheries target highly migratory stocks that are under the management remit of the Indian Ocean Tuna Commission. The EU has a strong role to play in supporting research and development in these fisheries, and as

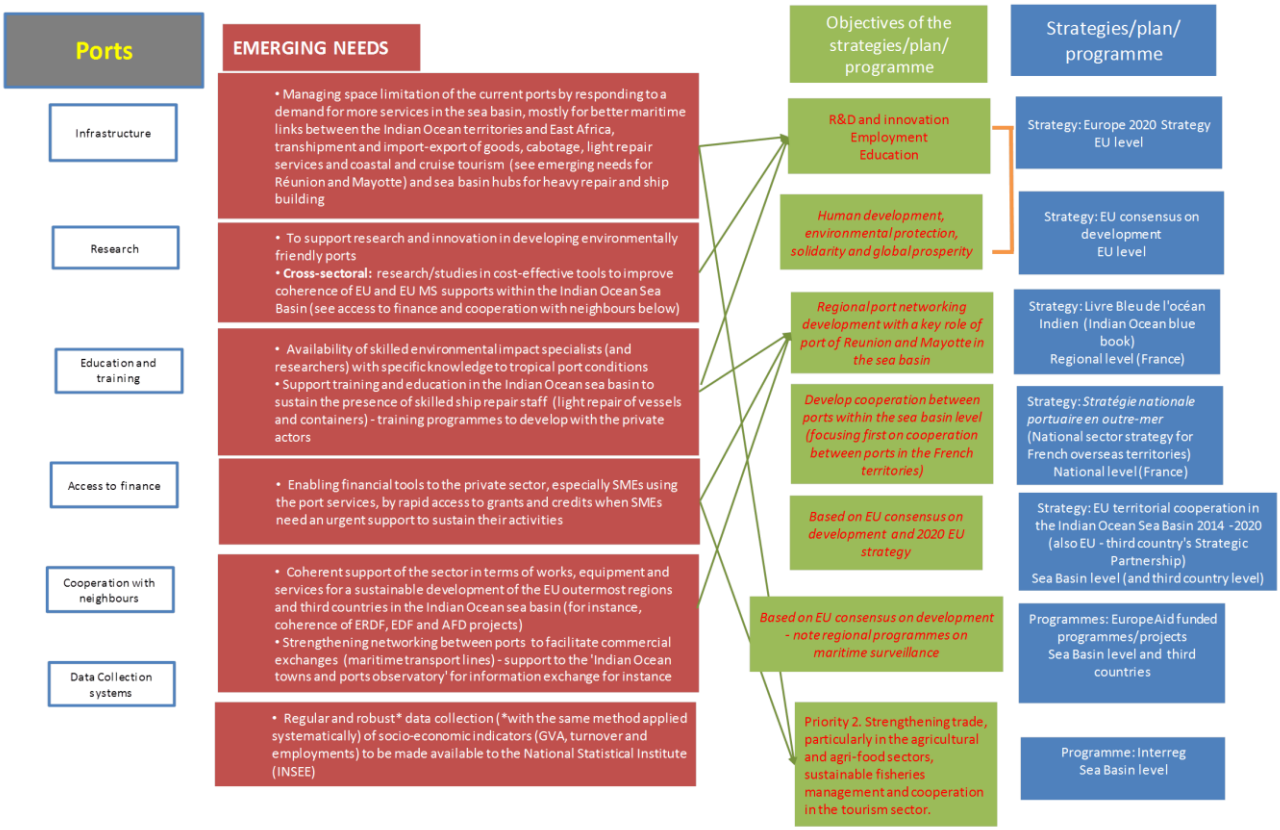
noted above, could provide a regional centre of excellence in fisheries science and management. Finally, the region can learn from EU expertise developed under the EU Data Collection Framework (DCF) to support and strengthen new forms of fisheries and related marine environmental data collection in the region.

Figure 3 — Logical diagram for blue biotechnology



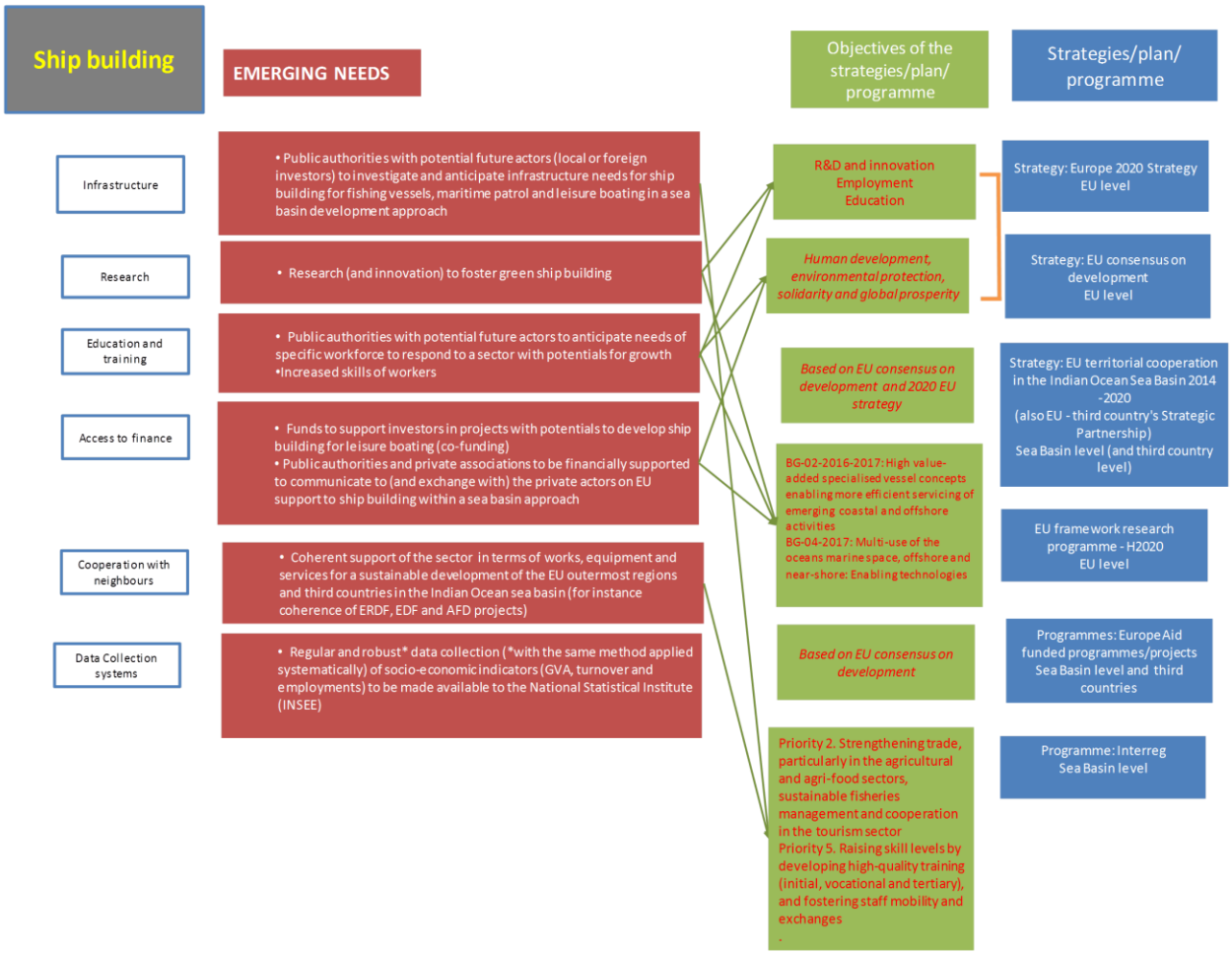
The development of blue technology will require **critical infrastructure in the form of laboratories** and industrial units for commercialising pilot developments. Much of this initial research and development will come from public funding, with partnerships (both with EU and non-EU research organisations) a key factor in this. Whilst these partnerships will bring new technologies to the region, they should also focus on developing regional capacity through education and training. These partnerships will also provide leverage for new financing, attracted through access to the unique range of tropical and sub-tropical bio-resources in the region. It is important that, with this highly specialised sector, that regional development is coordinated in order to **reduce the potential for local competition**, and to allow for an **integrated network of specialist skills in biotechnology** to develop over the region. In order to assist and facilitate development of this small but growing blue sector, it is important that growth is carefully monitored through appropriate data collection, allowing further interventions where needed.

Figure 4 — Logical diagram for ports



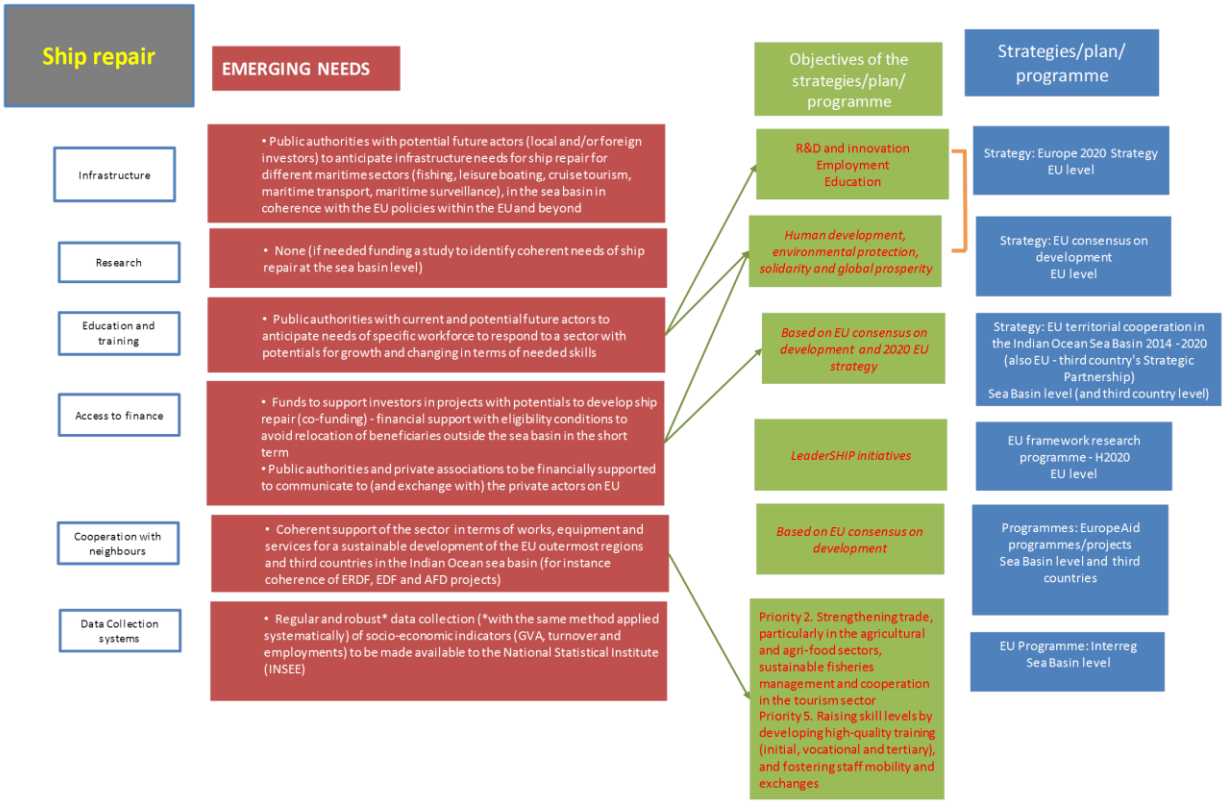
To allow seaborne connectivity to grow, the **development of port infrastructure** to accommodate both new and diversified vessel traffic is essential. This includes both **harbour structures** as well as **on-land port service support**. Research and development will mainly be focused on ensuring that ports operate sustainably and efficiently, thus reducing both environmental as well as economic costs. With the development of these ports, and the introduction of new management and information systems that enable this sustainable development, education and training will be essential. Similarly, capacity development to enable the diversification of port activities will also be important, with the private sector playing an essential role in this. Access to finance is also an important consideration, both for the major investments by key partners, as well as by the multitude of SMEs that both support and contribute to modern port development. Again, a **coordinated approach to regional development** is needed, with an **integrated sea transport development strategy** to assist cooperation between neighbouring island states. This development will need to be supported by **good information on vessel traffic**, **port operations**, cargo **throughput** and efficiencies, all requiring novel forms of data collection.

Figure 5 — Logical diagram for ship building



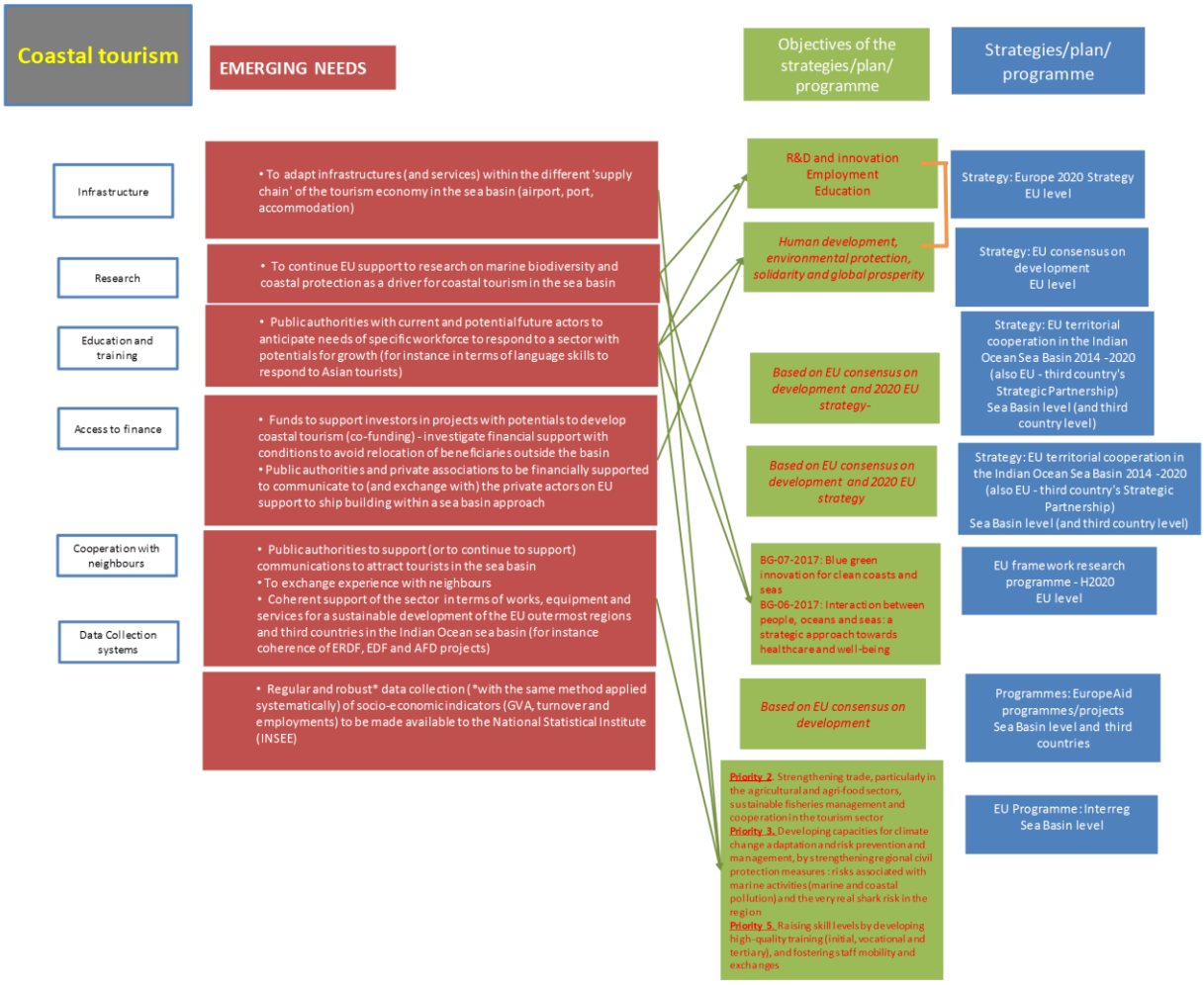
With the anticipated increase in sea vessel traffic, as well as the development of fisheries, maritime surveillance, maritime leisure and marine energy, there will need to be the associated development of **niche ship-building capability and infrastructure** to service regional needs. This is likely to be accompanied by some targeted research to ensure that appropriate construction techniques are developed that reduce environmental and cost demands and so that **ship-building is well suited to the physical demands** of this sea basin. As new ship-building technologies are introduced, these will need to be accompanied by education and training to ensure that efficiency and quality can be maintained in a highly competitive sector. With the aforementioned diversification e.g. into supporting maritime surveillance and marine leisure development, there will need to be investment into new facilities and services — this will need to be accompanied by access to finance and funding support. Given the need to maintain connectivity with neighbouring countries, cooperation with neighbours will be essential to ensure that networks are built within the region. Again, this development will need to be accompanied by data collection, focusing on maintaining regional competitiveness with ship-building operations from outside of this Indian Ocean sub-basin.

Figure 6 — Logical diagram for ship repair



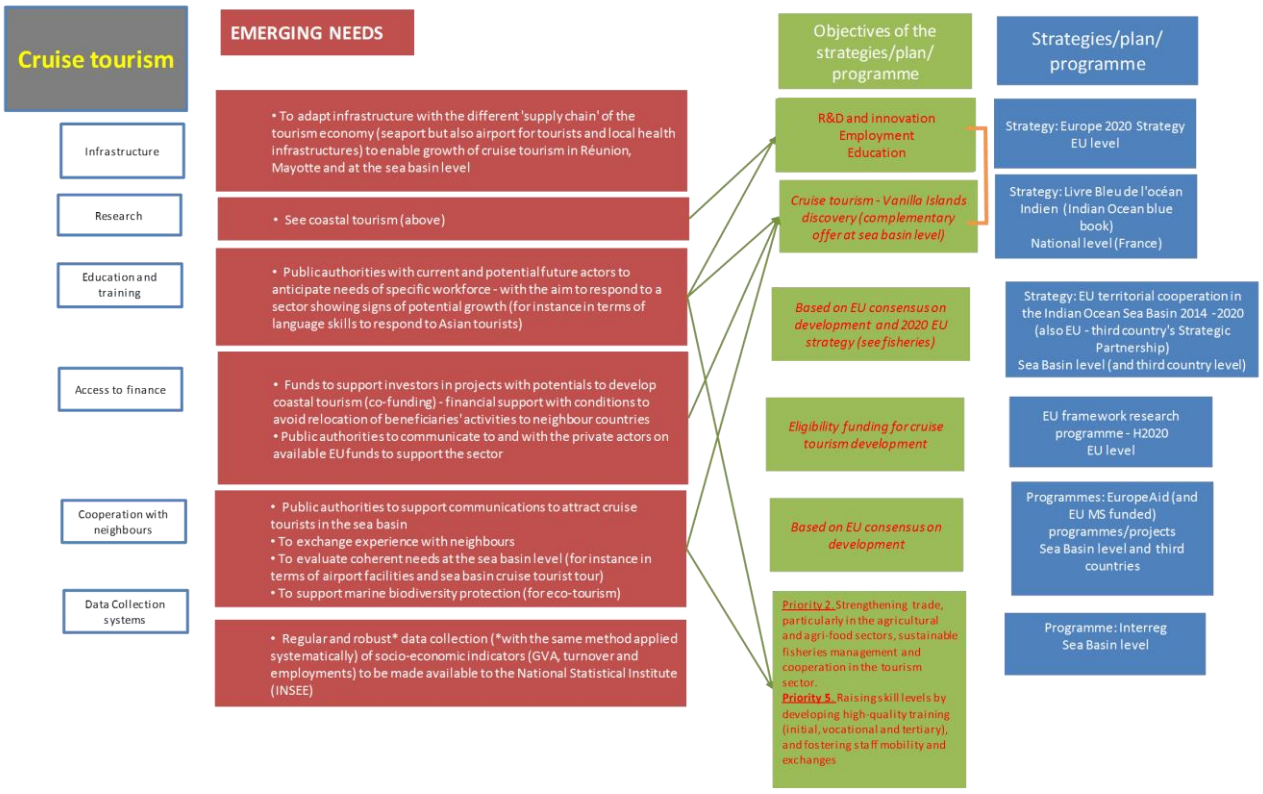
As with ship-building, diversification into different marine economic activities, such as fisheries and marine recreation, will add new opportunities for the repair and servicing of vessels. Whilst some capacity already exists, such as CNOI in Mauritius, this will expand, exerting demand for **new infrastructure and land area** to accommodate berthed, dry-docked vessels, workshops and administrative space. Some degree of research will also be required, to ensure that local development barriers are removed and that sustainability and efficiency can be enhanced. This will inevitably require new capacity-building in the form of education and training, especially as the work load diversifies and expands. Access to finance will also need to fund new capital expenditure, as well as enable businesses to survive what will inevitably be a variable demand. Again, it is important that the **region cooperates to build the overall regional capacity** and does not introduce competition between neighbours. Finally, data collection need to be improved to support this developing sector.

Figure 7 — Logical diagram for coastal tourism



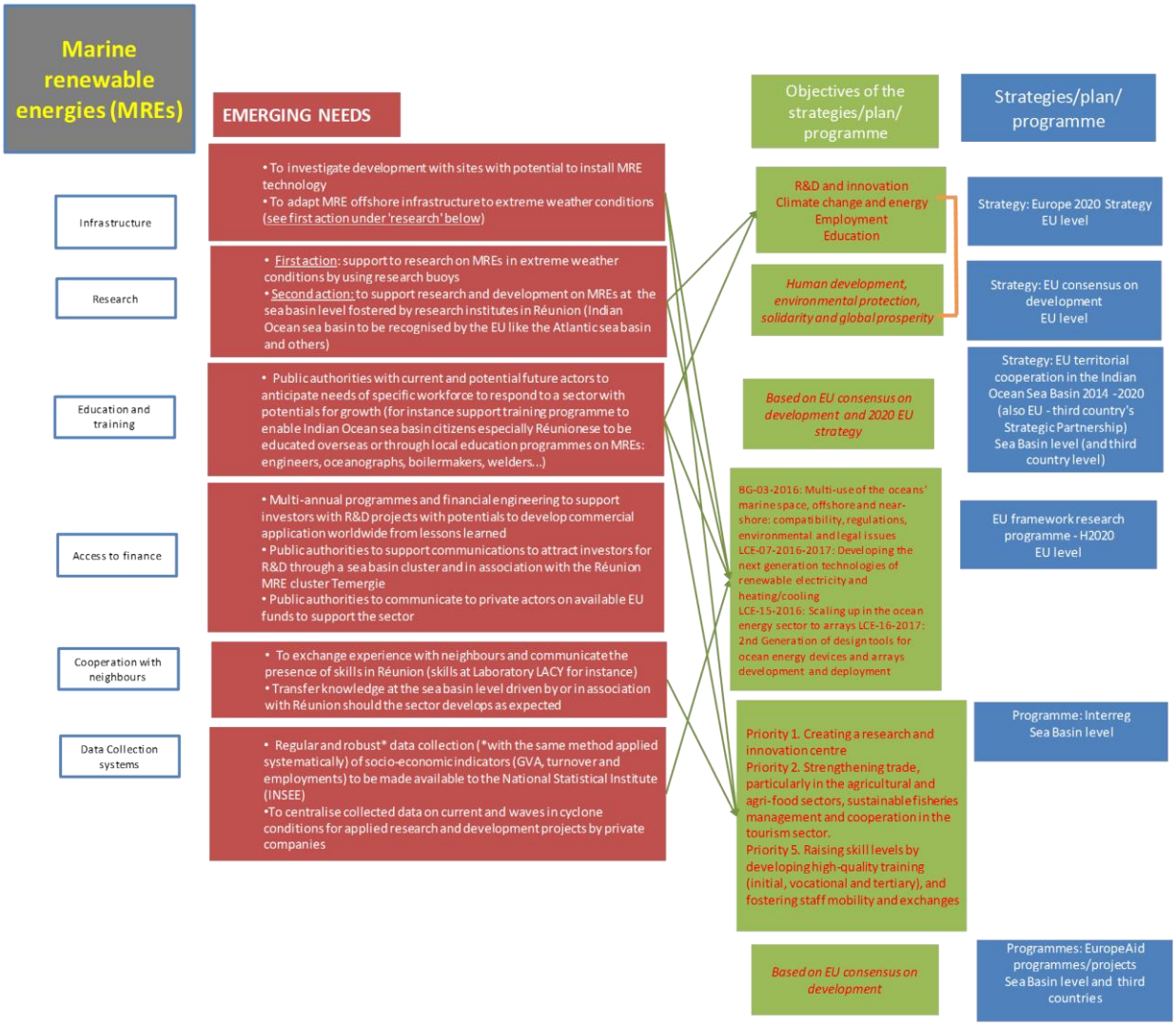
Whilst **coastal tourism is a well-developed sector in much of the region**, it is both evolving and expanding to new locations. With the latter, this often required the development of **new infrastructure** such as roads, power supplies and **water supply / treatment**, as well as the **guest and support facilities** themselves. Research will also be required, especially to accompany **niche tourism development such as eco-tourism**, as well as wide marine environmental research to assist conserve the marine biodiversity assets on which coastal tourism so much depends, especially in the face of climate change. This expansion and diversification will require a combination of new skills and local knowledge, thus imposing a particular blend of education and training challenges. All these three aspects will require access to finance, especially in developing support infrastructure and facilities. The region — which is highly diverse in terms of its physical and cultural assets — needs to promote **coordination between neighbours** in developing a coherent regional coastal tourism strategy. As before, public and private sector support to development is very much dependent upon good information, and thus data collection on the different aspects of coastal tourism development is essential.

Figure 8 — Logical diagram for cruise tourism



Cruise tourism requires development to be driven under an integrated supply chain approach to provide an attractive **sea basin cruise tourism package** with diverse activities in the sea basin at each stop. This requirement is relevant for infrastructure at ports, airports and accommodation but also for education and training for the sector to have available a **skilled workforce including language skills** to respond to new tourists coming from Asia and the US. In terms of access to finance and cooperation with neighbours, EU is suggested to support communication programmes to promote the sea basin and to help the sea basin territories to exchange experience for an integrated and coherent development of the sector in the sea basin (for instance through regional associations identified during the study). Protecting the marine biodiversity is also crucial for the long-term presence of cruise tourism in the sea basin. The marine fauna and flora create leisure activities for tourists such as marine mammal watching and snorkeling.

Figure 9 — Logical diagram for marine renewable energies



Marine renewable energy development in the sea basin is at a burgeoning stage at the sea basin level with some commercial applications only in Réunion and Seychelles. The sector requires to be supported for **research, development and innovation** driven by the EU to develop a hub in the EU outermost region Réunion for dissemination of knowledge and education within the sea basin level.

Also, research and studies at the sea basin level should focus on understanding how MRE technology can behave in extreme weather conditions and identifying sites with potential to install suitable MRE technologies. The EU outermost region in the South West Indian Ocean sea basin has the potential to drive those studies.

Alike in Réunion, development of MREs is suggested to be **steered by multi-annual programmes** with relevant financial engineering by private and public partnerships involving human resources in the EU outermost regions and in other territories to share knowledge and develop the sector harmoniously. Platforms of exchange between private and public stakeholders at the sea basin level should be supported by the EU too. These platforms could be coordinated or should involve the MRE cluster in Réunion, Témergie to foster the development of a R&D and training hub of MREs in Réunion (cooperation with neighbours).

13. Position of the EU outermost regions in the sea-basin

Over the next page is presented a SWOT analysis of the position of Mayotte and Réunion in the South Western Indian Ocean sea basin. While Réunion and Mayotte have 'weaknesses' due to the costs of their workforce being EU outermost regions and French territories binding them to specific standards for work such as working taxes, infrastructures standards, both EU outermost regions have for strengths a working population being skilled or having the possibility to use the EU and the French education, research and training infrastructures to provide quality services compared to their neighbours (for instance to develop their tourism). It is on those strengths that Réunion and Mayotte will be able and shall develop their blue economy within the sea basin.

Table 6 — SWOT analysis of the position of the ORs in the South Western Indian Ocean sea basin

Strengths	Weaknesses
Blue economy competitiveness	
Presence of internationally recognised port infrastructure, services, ancillary services in Réunion (ISPS certified and Mayotte harbour planning to become ISPS certified) Both Mayotte and Réunion on maritime roads for maritime transport of goods	Administrative burden: several layers of public services supportive or able to support the same maritime economic activities: OR (region — and department councils when operating in the outermost regions —, national and EU levels) compared to other independent territories in Caribbean-Amazonia area.
Labour market in the maritime sector	
Recognised quality and productivity of the workforce in the EU outermost region in general	Low skilled workforce yet high associated costs resulting from stringent EU and French employment standards Difficulty to authorise skilled non-EU workforce to carry out short term business travels, for instance for training or urgent equipment maintenance or repair, in the EU outermost regions in the sea basin due to strong visa application procedures
Maritime research and innovation	
Presence of research infrastructure in Réunion with a strong international network of researchers (IFREMER, IRD for instance)	Non-possibility to respond to call for EU proposals on blue growth (the Indian Ocean Sea basin is not recognised in the calls) Blue biotechnology and marine renewable energies in Réunion and in the medium term in Mayotte: coherent and long-term public vision to support maritime research and innovation a) to be complementary and competitive at the sea basin, supporting local investment (see threat below) and, to some extent, in the global market and b) to foster sea basin innovation in the sea basin

Opportunities	Threats
Blue economy competitiveness	
Maritime transport and ports: demographic increase at least in the next two decades will provide growth to the sector in the sea basin	Fisheries and ports: environmentally unsustainable growth in the sea basin creating unfair competition for the ports of the EU outermost region and increasing the risk of over-exploitation of pelagic and non-pelagic fisheries resources. Governance issues in Mayotte: risk of social tensions (social context: high unemployment rates in the EU outermost regions especially for the younger population and in Mayotte)
Labour market in the maritime sector	
Réunion with a potential to retain part of the skilled workforce in the sea basin being a EU territory with political stability, good research and education capacities up to universities. Regional training and capacity-building with equivalent industries in Mauritius, Seychelles and South Africa.	Non-availability of specific skills locally Inability to retain locally skilled workforce willing to work in other international places (Mayotte and Réunion being remote islands away from the key international markets for employment)
Maritime research and innovation	
Availability of EU Funds for research and innovation at the EU level Partnerships with EU research institutions and service providers Potential to collaborate with other R&D service providers in Southern and Eastern Africa	Uncontrolled entry of private investors in research, development and innovation

14. Conclusions on concrete measures by type and timeframe

By analysing the Logical diagrams above, and considering all findings reported in the previous sections of the report, the following 'legislative', 'non-legislative' and 'financial' conclusions emerged. For each of them, the following details are provided:

- possible time frame for fulfilling the potential (short/medium term or long term);
- stakeholders that should activate the measure;
- possible best practices experimented in other territories (i.e. cross-fertilisation possibilities);
- measures for achieving/implementing them.

- ***Fisheries and aquaculture***

	Conclusion	Time frame	Stakeholders that should activate the measure	Possible best practice/flagship project	Measures for achieving them
LEGISLATIVE	No legislative conclusion identified				
NON-LEGISLATIVE	**The coherence of public support within the sea basin should be improved or better recognised and understood by stakeholders located in the EU outermost regions.** Local stakeholders in Réunion and Mayotte expressed their concern that public support from the EU and EU Member States to neighbour countries might conflict with — or otherwise undermine — efforts to develop the EU outermost regions in the Indian Ocean (cross sectoral issue) In the recent years, there have been increasing numbers of consultations with stakeholders to obtain their views of defining coherent EU funding strategies by evaluating whether public funds coming from the EU (and other organisations within the EU) are coherent with the external and internal dimensions of the CFP. However, there is still room for improvement.	Long term approach	European Commission and EU Member States including France	Consultations to Mayotte and Réunion stakeholders on sustainable fisheries partnership agreements between the EU and non-EU countries (SFPA) in the sea basin is to become more systematic during the evaluation process, a legally binding step to negotiate or renegotiate SFPAs	Recommended measures to define more coherent strategies to develop the fishery sector in the sea basin by: • Regular round tables and information exchange between EC services (for instance DG MARE and EuropeAid) and EU and EU MS developing agencies for project design and monitoring. • Stakeholder consultations including stakeholders in the EU Outermost Regions during the project cycle especially during the project design. • Enhanced communications and exchange to stakeholders on the coherence of public supports to neighbour countries and to EU outermost regions.

	Conclusion	Time frame	Stakeholders that should activate the measure	Possible best practice/flagship project	Measures for achieving them
	Several maritime sectors expressed their concern about the need for a coherent approach to developing the sea basin either within their maritime sector or between maritime sectors present or having the potential to develop in the Indian Ocean sea basin. Also, the EU outermost regions in the sea basin are French territories only. EU private stakeholders may have an interest to be represented by and exchange within an EU Maritime Cluster covering the Indian Ocean sea basin	Short term actions for a long-term approach	Indian Ocean Commission (coordination) Sea basin countries: public authorities in association with representatives of their maritime sectors Maritime Cluster in Réunion	Maritime Clusters in France European network of maritime clusters ([48])	Studies: • Development of a blue economy platform of exchange at the sea basin level between public and private stakeholders with an open public access and restricted access to specific information. • Creation of a sea basin network of national maritime clusters • Pilot development of a Maritime Cluster based in Réunion with EU stakeholders interested in investing or operating business within the Indian Ocean sea basin. Such cluster could be an EU projected cluster based on the current Maritime Cluster in Réunion. The study will analyse the potential, needs (including private and public support) and gaps to create the platform and the network above.
	Fish stock status are still unknown and should continue to be monitored to support sustainable fishing, covering both tuna and non-tuna species (esp. straddling stocks)	-	For tuna species through IOTC For non-tuna species, through SIOFA at sea basin level	Not available	The EU should continue to co-fund research on stock status (through voluntary contributions to RFMOs).
FINANCIAL	No financial conclusion identified				

([48]) www.enmc.eu , latest access: 16 March 2017.

- ***Blue biotechnology***

	Conclusion	Time frame	Stakeholders that should activate the measure	Possible best practice/flagship project	Measures for achieving them
LEGISLATIVE	No legislative conclusion identified				
NON-LEGISLATIVE	**Blue biotechnology in the Indian Ocean has potential by developing the already existent structures of research, training, education and development in Réunion with a sea basin leadership**, however public supports to blue biotechnology projects should get more diversified by extending supports beyond ERDF, by enabling R&D with partners outside EU territories and by being adapted to R&D in tropical areas	Short term and medium term	Through IOC and in association with other international cooperation agencies such as the World Bank Local and local public authorities from Réunion such as the Conseil Régional in association with the cluster Qualitropic and Cyroi	Not available	EU to co-fund or fund local, regional and international workshops between public and private stakeholders to foster co-funding of blue biotechnology projects at the sea basin level. Continuous input and proactive initiatives from the blue growth industries in preparing and regularly modifying education programmes supported by the European Structural and Investment Funds.

	Conclusion	Time frame	Stakeholders that should activate the measure	Possible best practice/flagship project	Measures for achieving them
	Several maritime sectors expressed their concern about the weak coherence of developing the sea basin either within their maritime sector or between maritime sectors. Also, the EU outermost regions in the sea basin are French territories only. EU private stakeholders may have an interest to be represented by and exchange within an EU Maritime Cluster covering the Indian Ocean sea basin	Short term actions for a long-term approach	Indian Ocean Commission (coordination) Sea basin countries: public authorities in association with representatives of their maritime sectors Maritime Cluster in Réunion	Maritime Clusters in France European network of maritime clusters ([49])	Studies: • Development of a blue economy platform of exchange at the sea basin level between public and private stakeholders with an open public access and restricted access to specific information. • Creation of a sea basin network of national maritime clusters. • Pilot development of a Maritime Cluster based in Réunion with EU stakeholders interested in investing or operating business within the Indian Ocean sea basin. Such cluster could be an EU projected cluster based on the current Maritime Cluster in Réunion. The study will analyse the potential, needs (including private and public support) and gaps to create the platform and the network above.
FINANCIAL	No financial conclusion identified				

([49]) www.enmc.eu, latest access: 16 March 2017.

- ***Ports***

	Conclusion	Time frame	Stakeholders that should activate the measure	Possible best practice/flagship project	Measures for achieving them
LEGISLATIVE	See remark in the line below.				
NON-LEGISLATIVE	**Coherence of public supports within the sea basin should be improved or better perceived by stakeholders located in the EU outermost regions**. Indeed, local stakeholders in the visited EU outermost regions — Réunion and Mayotte — expressed their concern about public supports from the EU and EU Member States to neighbour countries conflicting with the development of the EU outermost regions in the Indian Ocean (cross sectoral issue)	Long term approach for the EU (and EU MS) to participate to the development of the sea basin	European Commission and EU Member States including France	In the recent years, consultations of stakeholders have increased to obtain their views on defining EU fund strategies and evaluating whether public funds coming from the EU or from EU Member State are coherent with the EU external and internal dimensions of the CFP. However, there is room for improvements.	Recommended measures to define more coherent strategies to develop ports in the sea basin by: • Regular round tables and information exchange between EC services (for instance DG MARE and EuropeAid) and EU and EU MS developing agencies for project design and monitoring. • Public and private stakeholder consultations including stakeholders in the EU Outermost Regions over the entire project cycle and especially during the project design. • Enhanced communications to stakeholders on the coherence of public supports to neighbour countries and to EU outermost regions. Remark: exchange of goods and transport of passengers by sea within the sea basin should be facilitated by coherent trade, maritime transport and labour legislation between neighbour countries too (based on World Trade Organisation rules).

	Conclusion	Time frame	Stakeholders that should activate the measure	Possible best practice/flagship project	Measures for achieving them
	Several maritime sectors expressed their concern about the **weak coherence of developing the sea basin either within their maritime sector or between maritime sectors**. Also, the EU outermost regions in the sea basin are French territories only. EU private stakeholders may have an interest to be represented by and exchange within an EU Maritime Cluster covering the Indian Ocean sea basin	Short term actions for a long-term approach	Indian Ocean Commission (coordination) Sea basin countries: public authorities in association with representatives of their maritime sectors Maritime Cluster in Réunion	Maritime Clusters in France European network of maritime clusters ([50])	Studies: • Development of a blue economy platform of exchange at the sea basin level between public and private stakeholders with an open public access and restricted access to specific information. • Creation of a sea basin network of national maritime clusters. • Pilot development of a Maritime Cluster based in Réunion with EU stakeholders interested in investing or operating business within the Indian Ocean sea basin. Such cluster could be an EU projected cluster based on the current Maritime Cluster in Réunion. The study will analyse the potential, needs (including private and public support) and gaps to create the platform and the network above.
FINANCIAL	No financial conclusion identified				

([50]) www.enmc.eu , latest access: 16 March 2017.

- ***Ship building***

	Conclusion	Time frame	Stakeholders that should activate the measure	Possible best practice/flagship project	Measures for achieving them
LEGISLATIVE	No legislative conclusion identified				
NON-LEGISLATIVE	With the anticipated increase in sea vessel traffic and the development of fisheries, maritime surveillance, maritime leisure and marine energy, **regional needs for sustainable shipyards are present**.	Medium to long term measures	Ports of the Indian Ocean in association with shipyard owners at the sea basin level Potential coordination: Observatoires Villes Ports de l'océan Indien (based in Réunion)	Not available	EU to foster: • Regular networking of ports and shipyard owners at the sea basin level by co-funding a platform or platforms of exchange and regional workshops between private and public stakeholders to identify and update needs and sources of funding for heavy investments (being private and public). • Developments of new ship building technologies (green technologies) by co-funding innovative projects.

	Conclusion	Time frame	Stakeholders that should activate the measure	Possible best practice/flagship project	Measures for achieving them
	Several maritime sectors expressed their concern about the **weak coherence of developing the sea basin either within their maritime sector or between maritime sectors**. Also, the EU outermost regions in the sea basin are French territories only. EU private stakeholders may have an interest to be represented by and exchange within an EU Maritime Cluster covering the Indian Ocean sea basin	Short term actions for a long-term approach	Indian Ocean Commission (coordination) Sea basin countries: public authorities in association with representatives of their maritime sectors Maritime Cluster in Réunion	Maritime Clusters in France European network of maritime clusters ([51])	Studies: • Development of a blue economy platform of exchange at the sea basin level between public and private stakeholders with an open public access and restricted access to specific information. • Creation of a sea basin network of national maritime clusters. • Pilot development of a Maritime Cluster based in Réunion with EU stakeholders interested in investing or operating business within the Indian Ocean sea basin. Such cluster could be an EU projected cluster based on the current Maritime Cluster in Réunion. The study will analyse the potential, needs (including private and public support) and gaps to create the platform and the network above.
FINANCIAL	No financial conclusion identified				

([51]) www.enmc.eu , latest access: 16 March 2017.

- ***Ship repair***

	Conclusion	Time frame	Stakeholders that should activate the measure	Possible best practice/flagship project	Measures for achieving them
LEGISLATIVE.	No legislative conclusion identified				
NON-LEGISLATIVE	**Maritime sectors requiring ship repair services need the presence of skilled ship repair staff in several locations in the sea basin** (for fishing vessels, maritime patrol boats and container ships). For instance, the Grand Port Maritime de la Réunion is planning to identify repair needs at the outermost region level as complementary services to repair services already existing in other neighbour countries for a sustainable and competitive presence of these services	Short term (measure 1) Medium term (measure 2)	Ports of the Indian Ocean Potential coordination: Observatoires Villes Ports de l'océan Indien (based in Réunion)	Not available	• Market study to evaluate competitiveness and requested level of skills within the sea basin level. • To co-fund or fund training programmes to train technicians or new comers in the sector in different ports of the sea basin. • To evaluate or re-evaluate needs for specific workforce (and infrastructure) at the sea basin level on a regular basis to update training programme and infrastructure requirements.

	Conclusion	Time frame	Stakeholders that should activate the measure	Possible best practice/flagship project	Measures for achieving them
	Several maritime sectors expressed their concern about the weak coherence of developing the sea basin either within their maritime sector or between maritime sectors. Also, the EU outermost regions in the sea basin are French territories only. EU private stakeholders may have an interest to be represented by and exchange within an EU Maritime Cluster covering the Indian Ocean sea basin	Short term actions for a long-term approach	Indian Ocean Commission (coordination) Sea basin countries: public authorities in association with representatives of their maritime sectors Maritime Cluster in Réunion	Maritime Clusters in France European network of maritime clusters ([52])	Studies: • Development of a blue economy platform of exchange at the sea basin level between public and private stakeholders with an open public access and restricted access to specific information. • Creation of a sea basin network of national maritime clusters. • Pilot development of a Maritime Cluster based in Réunion with EU stakeholders interested in investing or operating business within the Indian Ocean sea basin. Such cluster could be an EU projected cluster based on the current Maritime Cluster in Réunion. The study will analyse the potential, needs (including private and public support) and gaps to create the platform and the network above.
FINANCIAL	See non-legislative measure item 3 above	-	-	-	-

([52]) www.enmc.eu , latest access: 16 March 2017.

- ***Coastal tourism***

	Conclusion	Time frame	Stakeholders that should activate the measure	Possible best practice/flagship project	Measures for achieving them
LEGISLATIVE	No legislative conclusion identified				
NON-LEGISLATIVE	Coastal tourism is sustainable should it be based on a **strong partnership between public and private organisations**, on a protected environment being marine or coastal and with **professional tourism staff**	Long term	Coordinated by the Indian Ocean Commission Ministries in charge of tourism in the countries of the sea basin in close relationship with ministries in charge of environment National tourism associations	Not available	Continuous support of the EU to • Identify and (co-)fund environmental protection initiatives having an impact on coastal tourism in the beneficiary countries; being legislative actions, development projects or research projects. • Co-fund platforms of exchanges or round tables enabling private and public organisations to meet at regional and national level and identify urgent and medium-term needs. • Co-fund training programme (for instance to learn international language skills) to tourism staff in the countries of the sea basin.

	Conclusion	Time frame	Stakeholders that should activate the measure	Possible best practice/flagship project	Measures for achieving them
NON-LEGISLATIVE	**Several maritime sectors expressed their concern about the weak coherence of developing the sea basin either within their maritime sector or between maritime sectors**. Also, the EU outermost regions in the sea basin are French territories only. EU private stakeholders may have an interest to be represented by and exchange within an EU Maritime Cluster covering the Indian Ocean sea basin	Short term actions for a long-term approach	Indian Ocean Commission (coordination) Sea basin: public authorities in association with representatives of their maritime sectors Maritime Cluster in Réunion	Maritime Clusters in France European network of maritime clusters ([53])	Studies: • Development of a blue economy platform of exchange at the sea basin level between public and private stakeholders with an open public access and restricted access to specific information. • Creation of a sea basin network of national maritime clusters. • Pilot development of a Maritime Cluster based in Réunion with EU stakeholders interested in investing or operating business within the Indian Ocean sea basin. Such cluster could be an EU projected cluster based on the current Maritime Cluster in Réunion. The study will analyse the potential, needs (including private and public support) and gaps to create the platform and the network above.
FINANCIAL	No financial conclusion identified				

([53]) www.enmc.eu , latest access: 16 March 2017.

- ***Cruise tourism***

	Conclusion	Time frame	Stakeholders that should activate the measure	Possible best practice/flagship project	Measures for achieving them
LEGISLATIVE	No legislative conclusion identified				
NON-LEGISLATIVE	**Development of cruise tourism in the sea basin requires an integrated and coherent approach**, that is a regional strategy in terms of infrastructure, services and supply chain using the diversity of offers from the different cruise destinations in the South Western Indian Ocean	Short term (measure 1) Short and medium term (measure 2)	National tourism associations and port authorities through a regional organisation representing them (for instance through the Cruise Indian Ocean Association, CIOA) Measure Through the Indian Ocean Commission — cruise tourism and, when appropriate, beyond (countries of the sea basin such as Kenya, Tanzania and South Africa are not members of the IOC)	Not available	• The EU should (continue to) co-fund platforms of exchange of information and views at the sea basin level through regional workshops and conferences and existing regional associations (or newly created ones if the existing ones fail to achieve the expected objectives of sharing information and views) to promote sustainable and safe tourism (all port users should be invited). • Using these platforms, urgent actions should be identified in parallel with. • Drafting of a specific regional strategy to develop cruise tourism at sea basin level (a strategy agreed between public and private stakeholders) in coherence with the regional blue economy strategy of the IOC expected to be published by the first quarter of the year 2017.

	Conclusion	Time frame	Stakeholders that should activate the measure	Possible best practice/flagship project	Measures for achieving them
NON-LEGISLATIVE	**Several maritime sectors expressed their concern about the weak coherence of developing the sea basin either within their maritime sector or between maritime sectors**. Also, the EU outermost regions in the sea basin are French territories only. EU private stakeholders may have an interest to be represented by and exchange within an EU Maritime Cluster covering the Indian Ocean sea basin	Short term actions for a long-term approach	Indian Ocean Commission (coordination) Public authorities in association with representatives of their maritime sectors Maritime Cluster in Réunion	Maritime Clusters in France European network of maritime clusters ([54])	Studies: • Development of a blue economy platform of exchange at the sea basin level between public and private stakeholders with an open public access and restricted access to specific information. • Creation of a sea basin network of national maritime clusters. • Pilot development of a Maritime Cluster based in Réunion with EU stakeholders interested in investing or operating business within the Indian Ocean sea basin. Such cluster could be an EU projected cluster based on the current Maritime Cluster in Réunion. • The study will analyse the potential, needs (including private and public support) and gaps to create the platform and the network above.
FINANCIAL	No financial conclusion identified				

- ***Marine renewable energies***

Conclusion	Time frame	Stakeholders that should activate the measure	Possible best practice/flagship project	Measures for achieving them

([54]) www.enmc.eu , latest access: 16 March 2017.

	Conclusion	Time frame	Stakeholders that should activate the measure	Possible best practice/flagship project	Measures for achieving them
LEGISLATIVE	No legislative conclusion identified				
NON-LEGISLATIVE	MREs in the sea basin is at a burgeoning stage at the sea basin level — at a research and development stage — and with few commercial applications in Réunion and Seychelles. Some of the latter having failed in Réunion. The sea basin countries should be fostered to **participate to research, training, education and development of MRE commercial applications** (with a medium-term strategy for Réunion to become a hub for research, development, and innovation at the sea basin level and beyond)	Short to medium term approach	Indian Ocean Commission in collaboration, amongst others, with the MRE cluster in Réunion Temergie	Not available	Through Interreg and/or EDF: • Development of a regional exchange platform on MRE; a platform with a restricted access to public-and private stakeholders being located or interested to develop MRE technologies in the sea basin. • Organisation of round tables to consult and facilitate exchange between public- and private stakeholders being located or interested to develop MRE technologies in the sea basin (activity funded by the EU). Organisation every two years for instance. • Support to site identifications for pilot projects that the EU and EU investors could co-fund; • Investigate the potential and needs to create a sea basin MRE cluster (interests of countries and private stakeholders, mechanisms to create and operate under a financially independent manner, sources of funding). Continuous input and proactive initiatives from the industry in preparing and regularly modifying education programmes supported by the European Structural and Investment Funds.

	Conclusion	Time frame	Stakeholders that should activate the measure	Possible best practice/flagship project	Measures for achieving them
NON-LEGISLATIVE	**Several maritime sectors expressed their concern about the weak coherence of developing the sea basin either within their maritime sector or between maritime sectors**. Also, the EU outermost regions in the sea basin are French territories only. EU private stakeholders may have an interest to be represented by and exchange within an EU Maritime Cluster covering the Indian Ocean sea basin	Short term actions for a long-term approach	Indian Ocean Commission (coordination) Sea basin countries: public authorities in association with representatives of their maritime sectors Maritime Cluster in Réunion	Maritime Clusters in France European network of maritime clusters ([55])	Studies: • Development of a blue economy platform of exchange at the sea basin level between public and private stakeholders with an open public access and restricted access to specific information. • Creation of a sea basin network of national maritime clusters. • Pilot development of a Maritime Cluster based in Réunion with EU stakeholders interested in investing or operating business within the Indian Ocean sea basin. Such cluster could be an EU projected cluster based on the current Maritime Cluster in Réunion. The study will analyse the potential, needs (including private and public support) and gaps to create the platform and the network above.
	The sea basin faces extreme weather conditions. **MRE technologies need to be adapted to these conditions for their commercial development at medium to large-scales**	Short to medium term approach	Through the Indian Ocean Commission in collaboration, amongst others, with the MRE cluster Temergie and the research institute Lacy based in Réunion	Not available	EU to co-fund research projects (through Interreg, EDF or H2020): • first short-term action, to obtain and collect data on extreme weather conditions of cyclones (current and wave data) by using research buoys (and to centralise data). • To develop adapted MRE technology providing energy (electricity and cool) at suitable costs compared to other sources of energy (by using preferably multi-annual programmes).
FINANCIAL	No financial conclusion identified				

([55]) www.enmc.eu , latest access: 16 March 2017.

List of interviewees

Organisation	Maritime activities concerned	Number of interviewees
Comité de Concertation Régionale (Comité de Suivi du programme de coopération)	Cross-cutting	1 (interviews of the Conseil Régional carried out in Réunion)
Conseil maritime ultramarin du bassin sud océan Indien		2 (indirect interviews carried out in Réunion and Mayotte — see Annexes 5 and 6)
Ministère des Outre-Mer, France		2
ONML — Observatoire national de la mer et du littoral, France		1
Ifremer — Institute français de recherche pour l'exploitation de la mer (France)		1
DMSOI — Direction de la Mer Sud océan Indien French Maritime Directorate in the southern Indian Ocean	Maritime surveillance (including fisheries surveillance)	5 (Interviews carried out in Réunion and Mayotte — see annexes 5 and 6)
Fleur de sel tropicale	Salt extraction (trader)	1

HOW TO OBTAIN EU PUBLICATIONS

Free publications:

- one copy:

via EU Bookshop (http://bookshop.europa.eu);

- more than one copy or posters/maps:
  from the European Union's representations (http://ec.europa.eu/represent_en.htm);
  from the delegations in non-EU countries
  (http://eeas.europa.eu/delegations/index_en.htm);
  by contacting the Europe Direct service (http://europa.eu/europedirect/index_en.htm)
  or calling 00 800 6 7 8 9 10 11 (freephone number from anywhere in the EU) (*).

  (*) The information given is free, as are most calls (though some operators, phone boxes or hotels may charge you).

Priced publications:

- via EU Bookshop (http://bookshop.europa.eu).

Priced subscriptions:

- via one of the sales agents of the Publications Office of the European Union
  (http://publications.europa.eu/others/agents/index_en.htm).

EA-04-17-826-EN-N

Publications Office

doi: 10.2826/941122